The ER Diagram for the scenario (Quiet Attic Films)

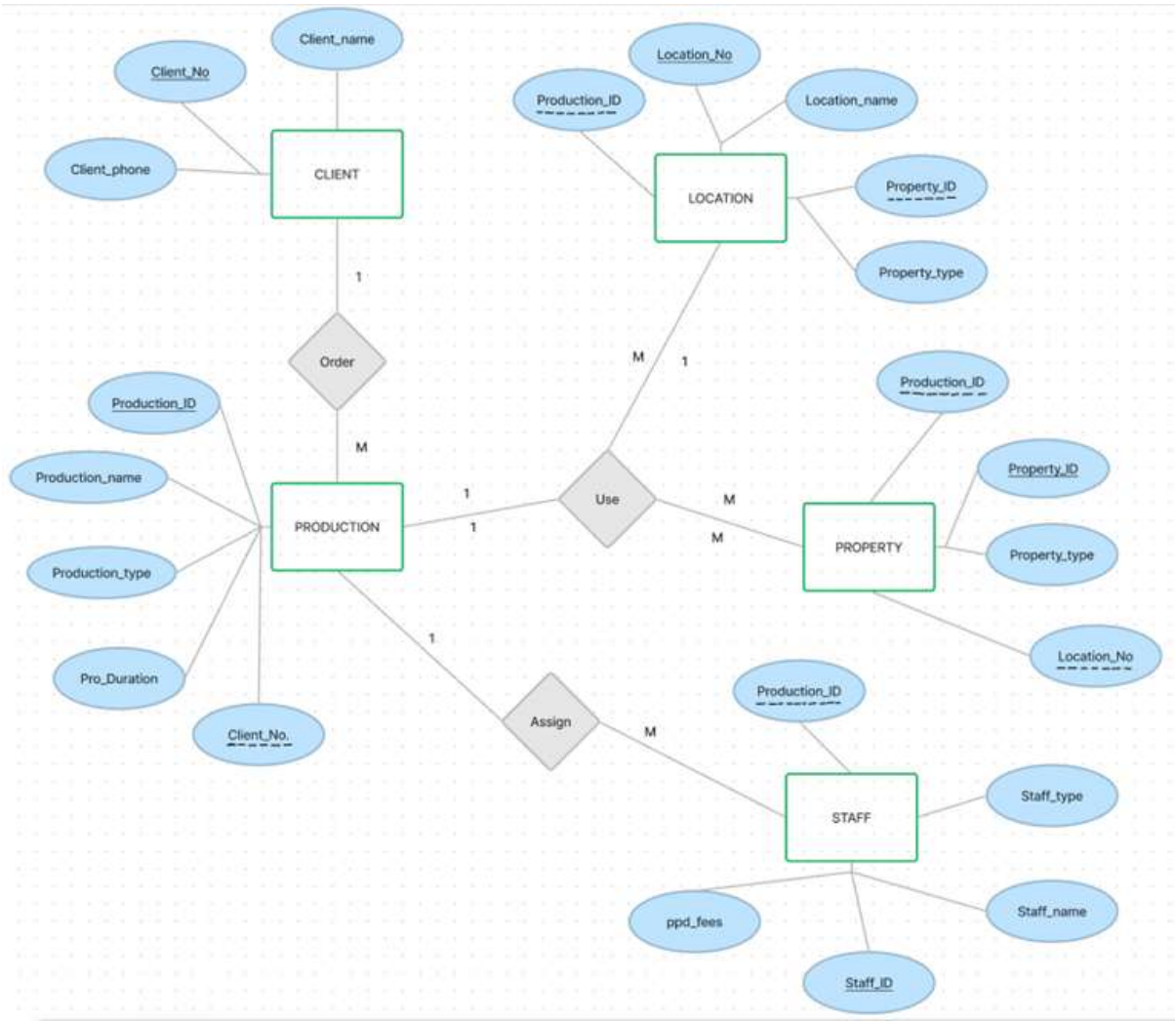

Figure 1- ERD for Quiet Attic films

Assumptions to the ERD

- One location can only have one production at a time, even though one production may use one or more locations.
- One location may have use on one or many properties.
- Many properties can belong to one production while one property won't be assigned in two productions at a time.

- One production may have one or more staff members while one staff member can only engage in one production at a time.

The logical database model to Quiet attic films

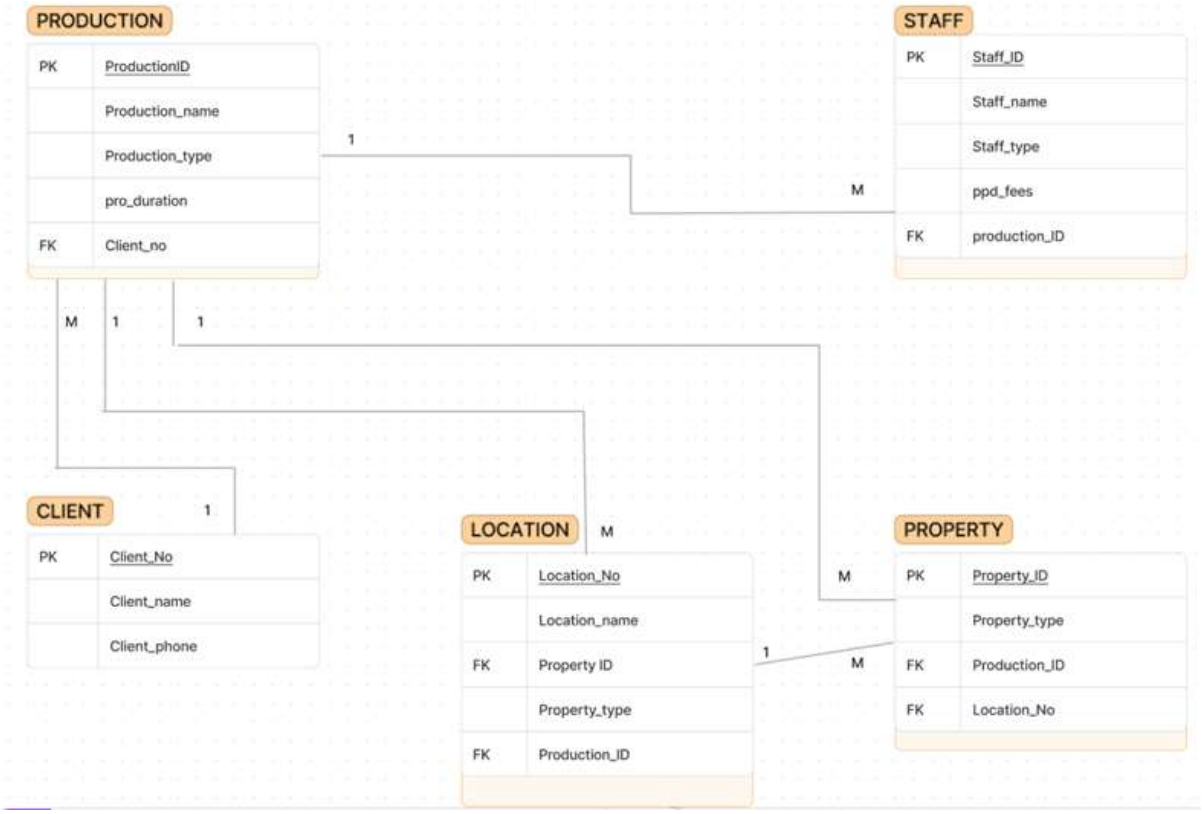

Figure 2- Logical database design for quiet attic films

Normalizing the database model of Quiet Attic Films

The above database model has no multivalued attributes or Composite attributes. Therefore, it is in the 1st Normal form.

As it has no partial Functional dependencies, it can be agreed that the database model is already in 2nd Normal form.

The Database for quiet attic films has a transitive dependency in the LOCATION entity. The normalization application for it are as follows.

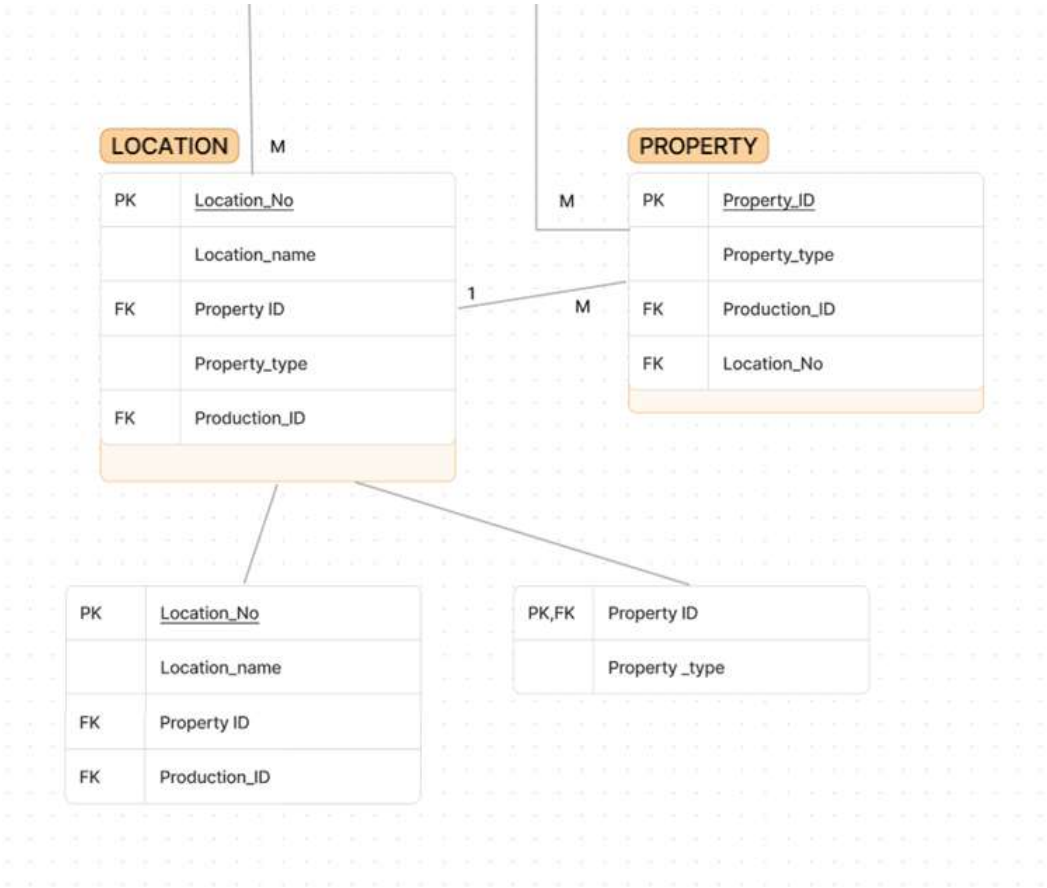

Figure 3- Normalization of the database design

User interface design of Quiet Attic Films (Wireframe Model)

An Interface design for the above scenario using the wireframe method is given below. This interface will be called “Quiet Attic Films -Records”

The login page for Quiet attic films- records

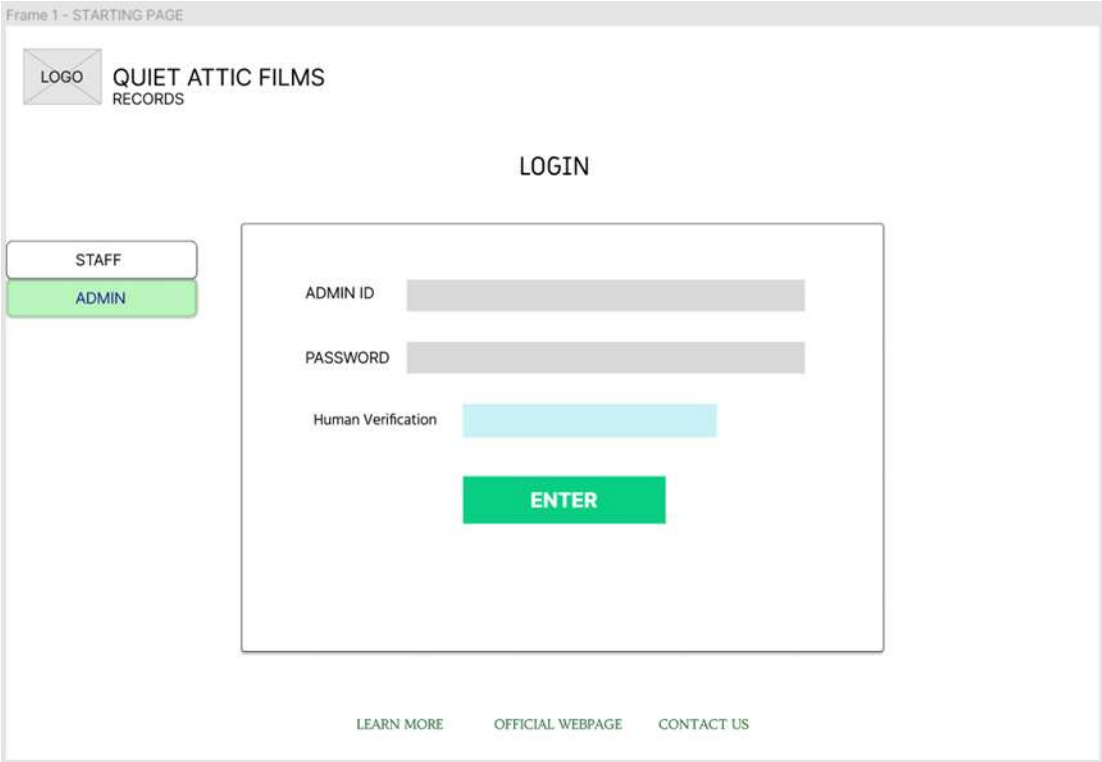

Figure 4- Login Page for Quiet Attic Films (Wireframe)

Here, the user can select and login as a staff member or Admin.

Selection Page

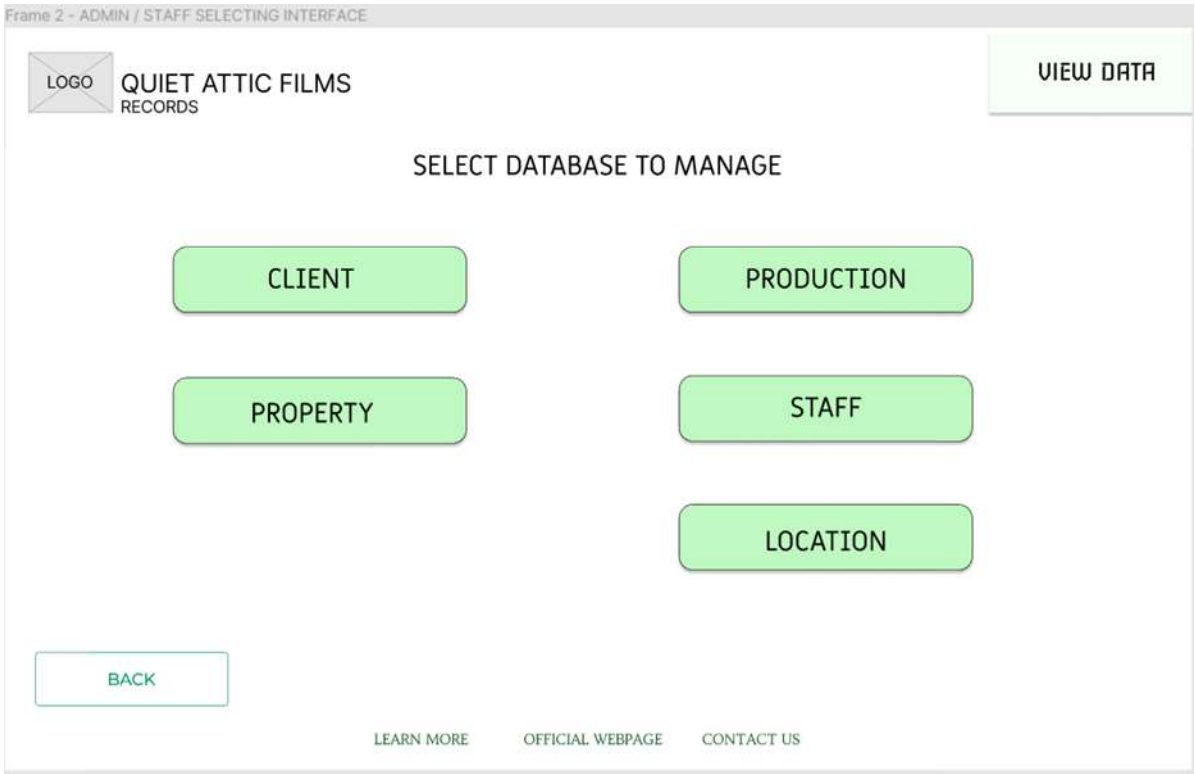

Figure 5-Selection page for quiet attic films (Wireframe)

Here the user can select which category (entity) of the database to manage.

View Data Page

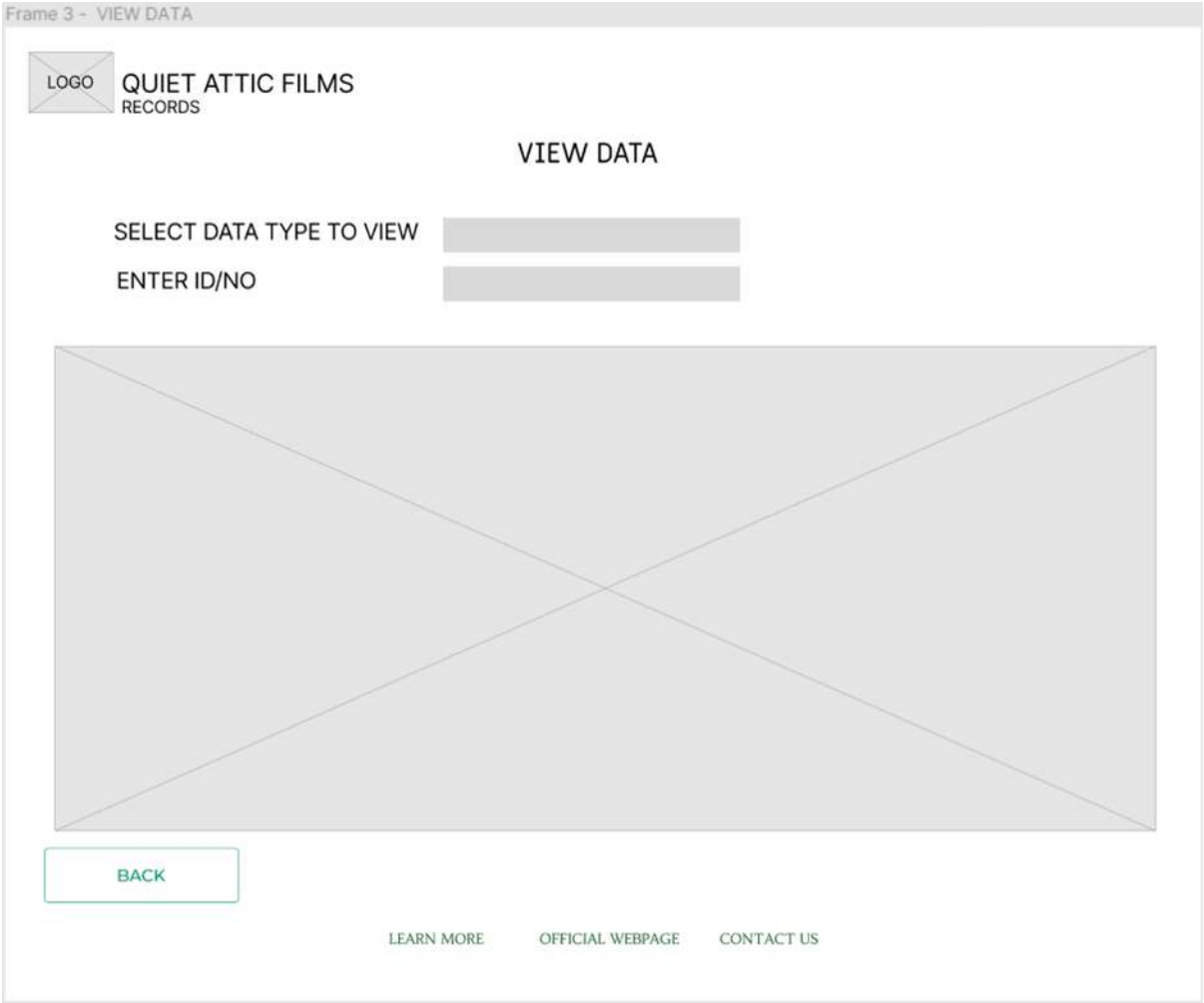

Figure 6- View data page for Quiet Attic films

This page will be useful to the user to view data. (Retrieve data from the database). The user will select the entity and enter the ID/ No. and get a copy of the data in the database below based on the record.

Manage Clients Interface (Wireframe map)

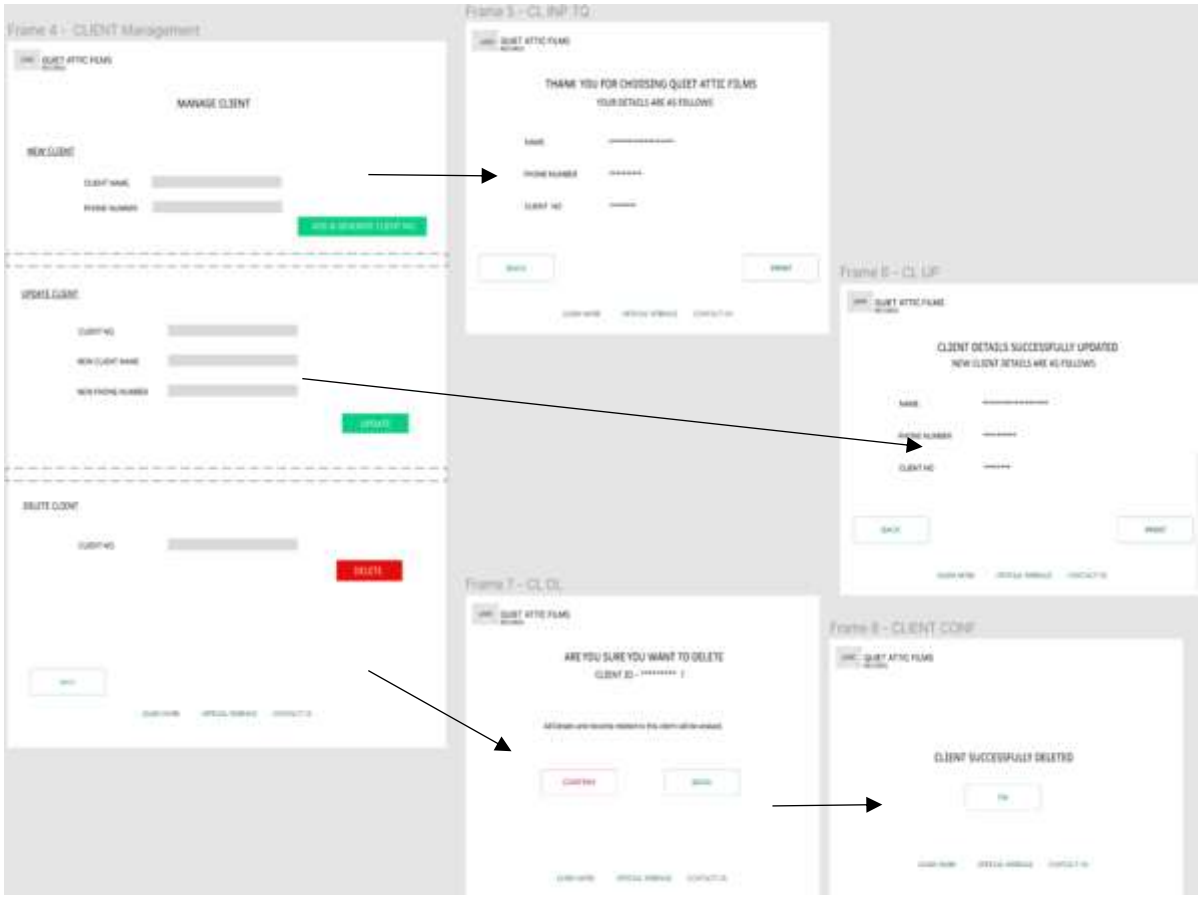

Figure 7-Manage clients page for quiet attic films (Wireframe map)

Shown above in figure 10 is the map of pages in the interface on managing the client data.

Insert Client Data Interface

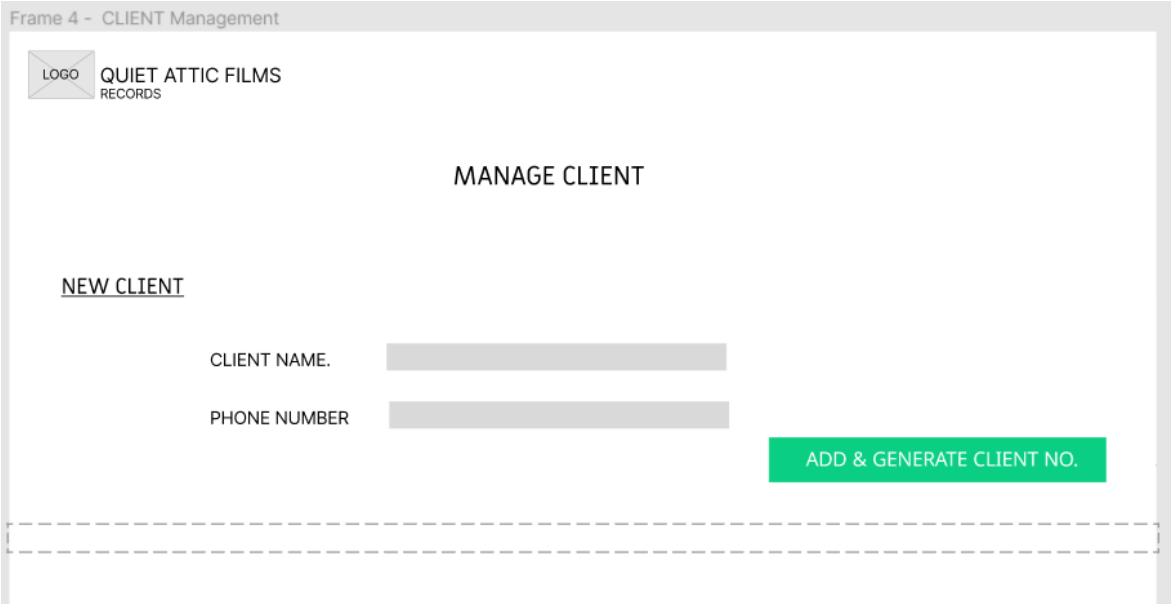

Figure 8-Insert clients for quiet attic films (Wireframe)

The user can insert client name and phone number and gain a generated client number which will be unique. The output page providing the user about the information of the client after inserting data is shown below.

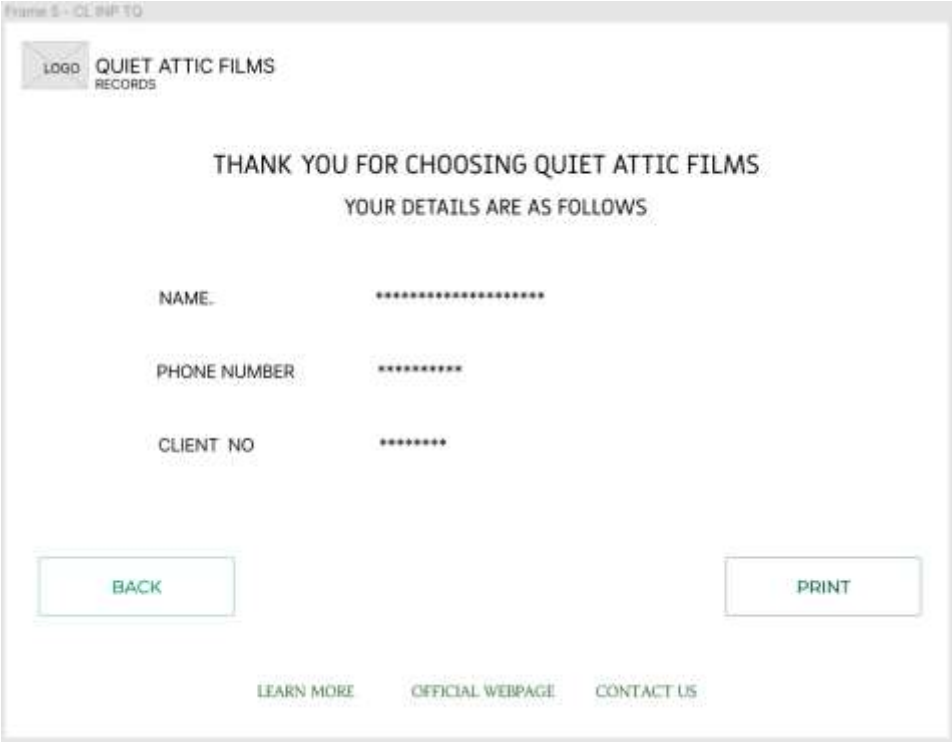

Figure 9-- Insert clients for quiet attic films confirmation output (Wireframe)

Update Client Data Interface

Figure 10-Update clients for quiet attic films (Wireframe)

The user can update client information here. The output page after updating is shown below.

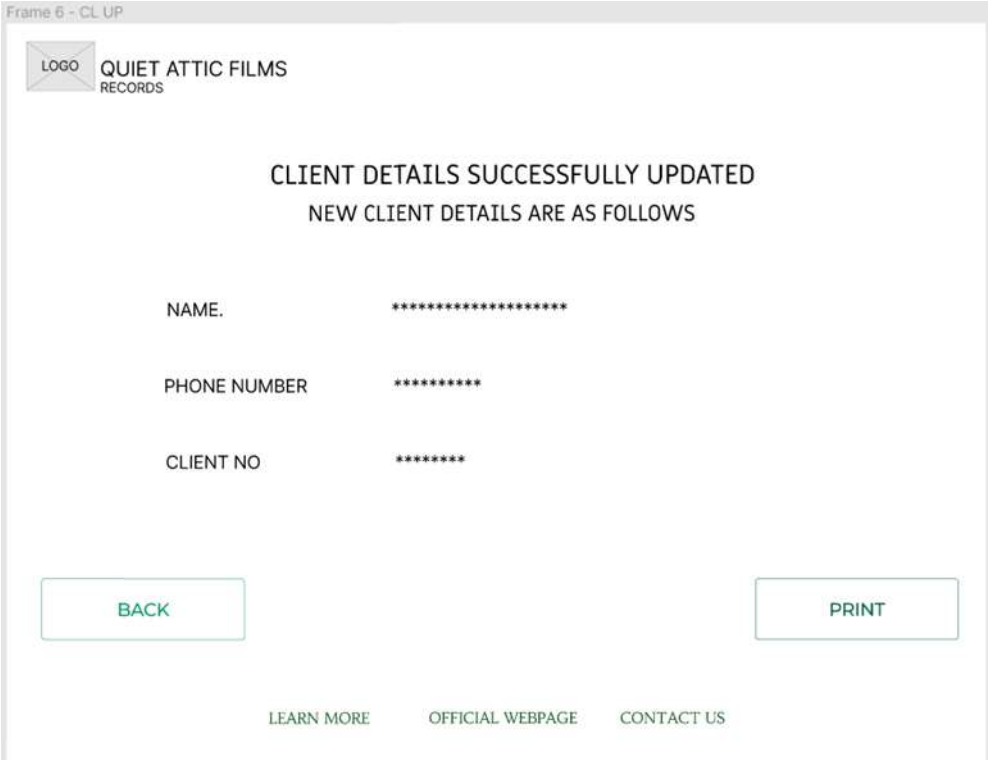

Figure 11-Update clients for quiet attic films confirmation output (Wireframe)

Delete Client Data Interface

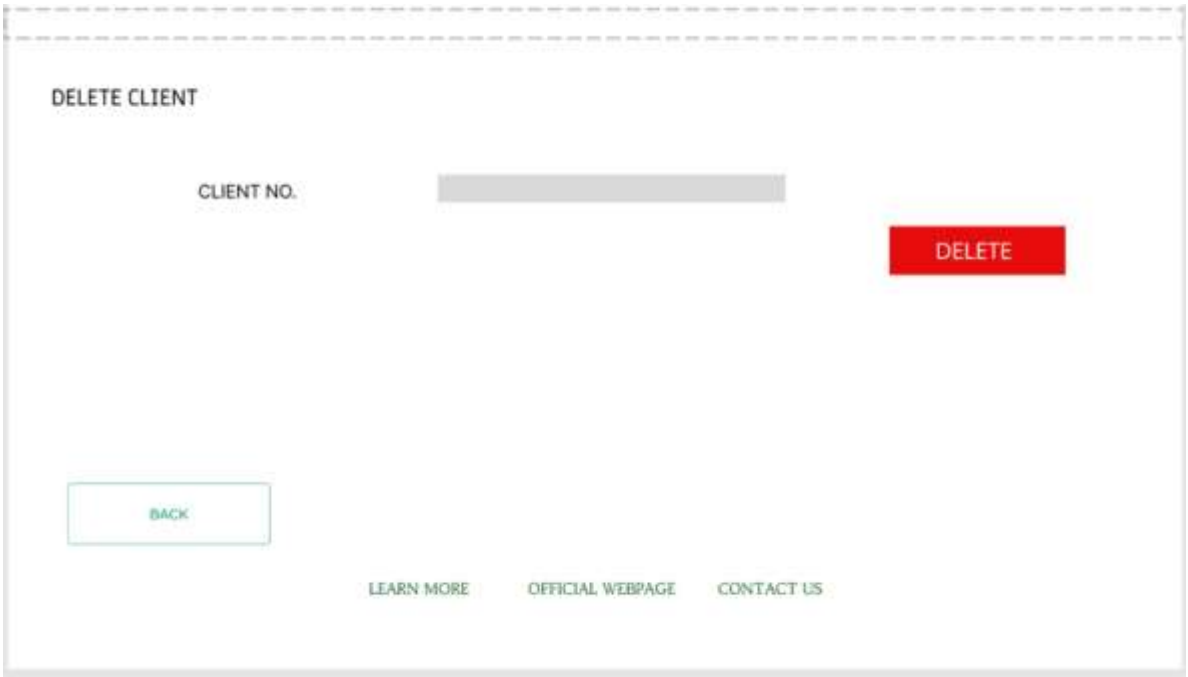

Figure 12-Delete clients for quiet attic films (Wireframe)

The user can delete client information here.

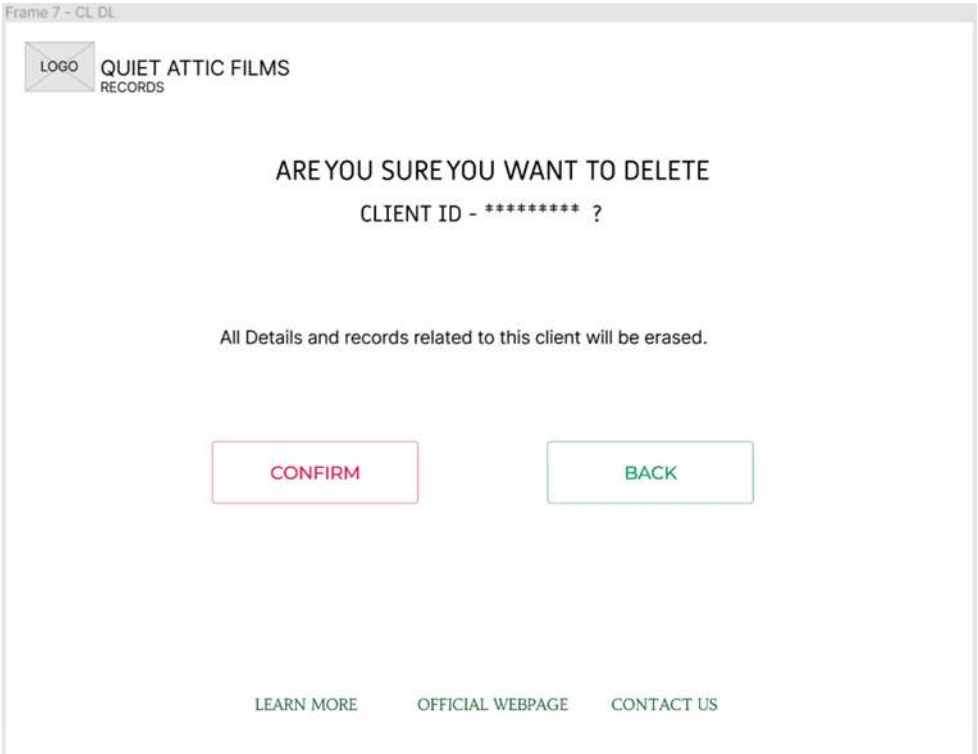

Figure 13-Delete clients for quiet attic films confirmation (Wireframe)

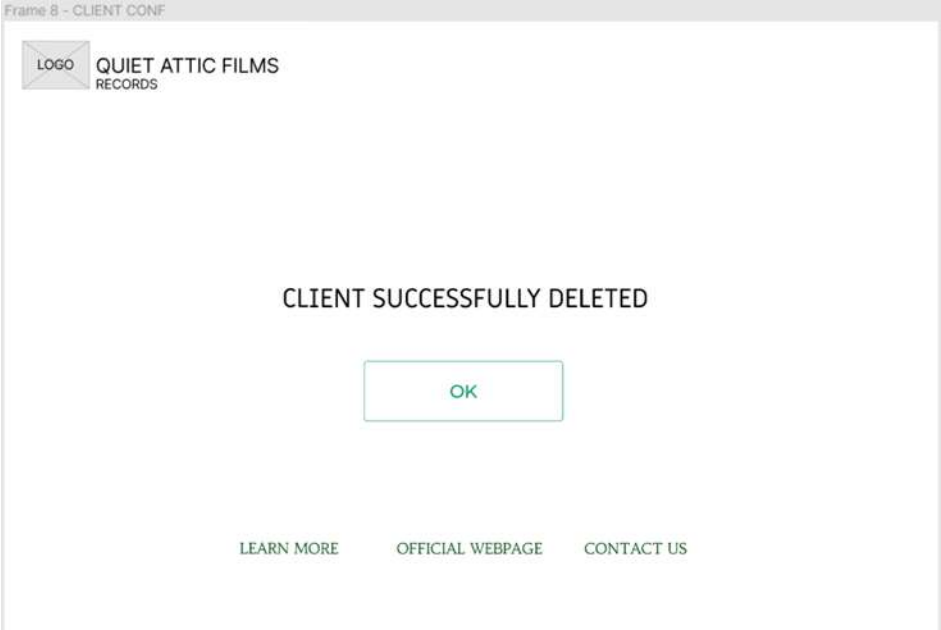

Figure 14-Delete clients for quiet attic films success (Wireframe)

Manage Productions Interface (Wireframe map)

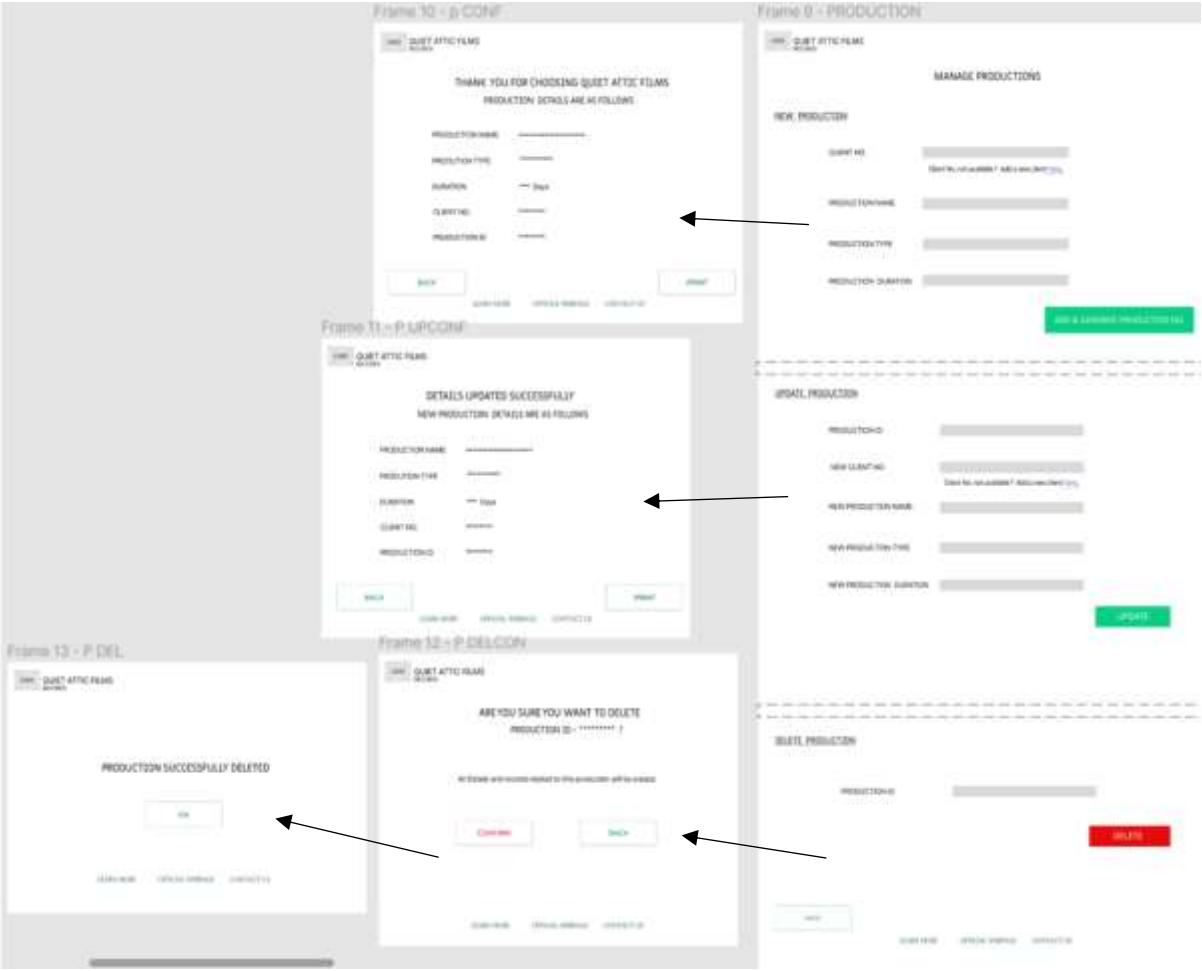

Figure 15-Manage productions for quiet attic films (Wireframe)

Shown above in figure 15 is the map of pages in the interface on managing the production data.

Insert Production Data Interface

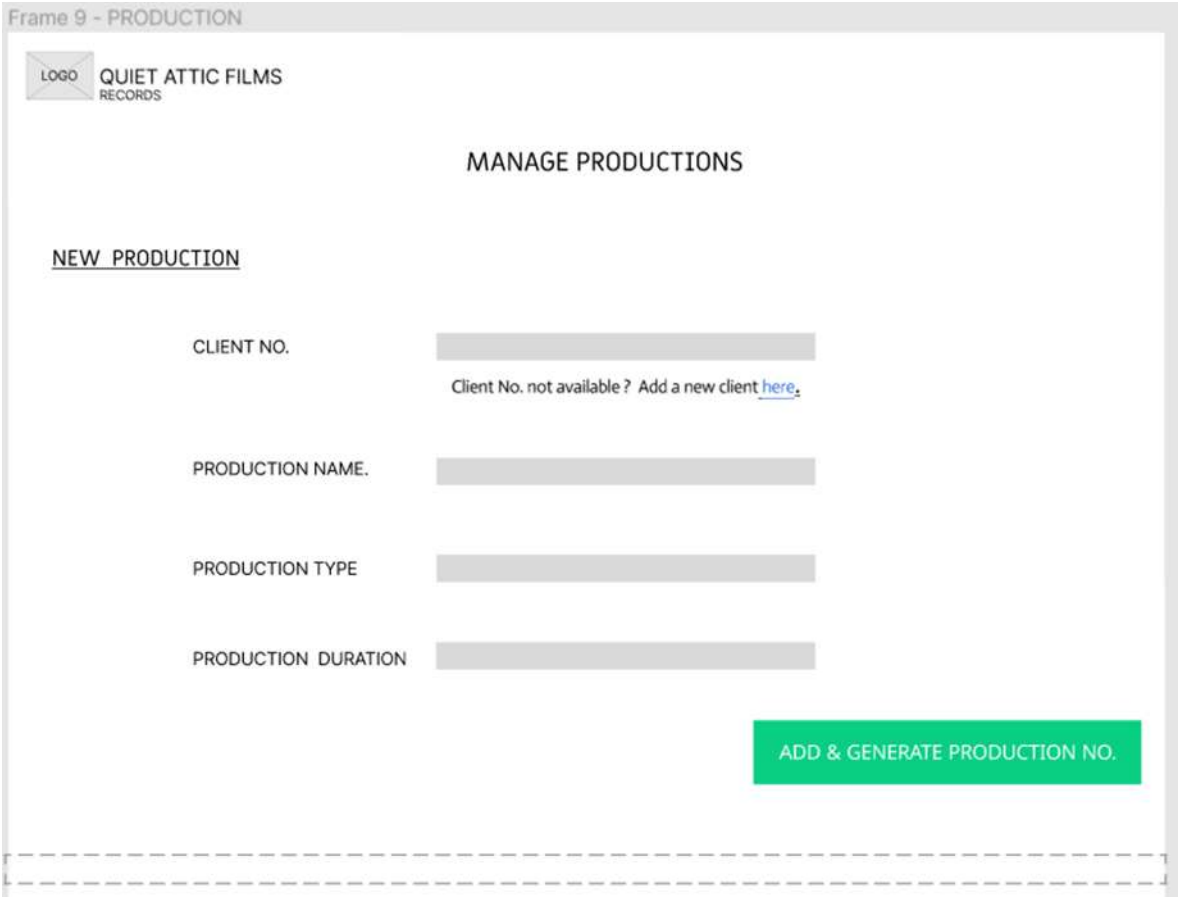

Figure 16- Insert Production for quiet attic films (Wireframe)

The User will insert the Client No. (If not available, can be created using the following link), Production name, Type, and duration and get a generated Production ID. The output page providing the user about the information of the client after inserting data is shown below.

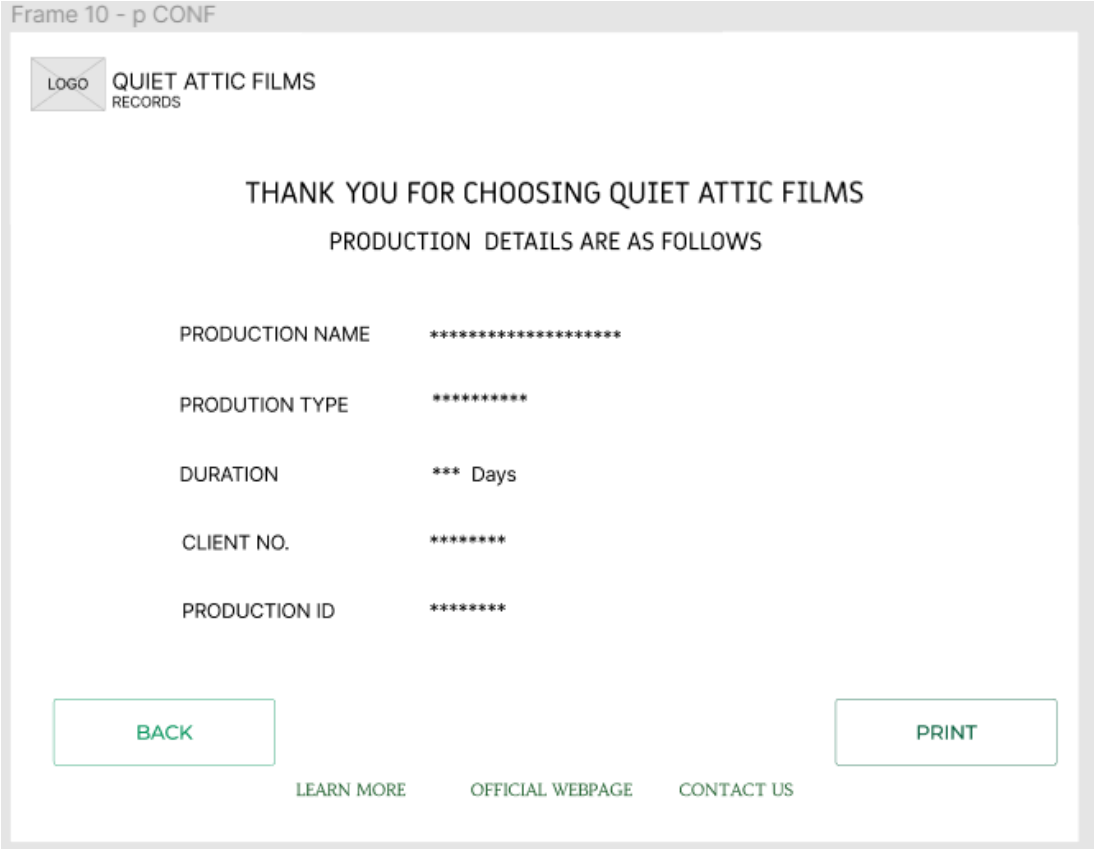

Figure 17- Insert Production for quiet attic films confirmation output (Wireframe)

Update Production Data Interface

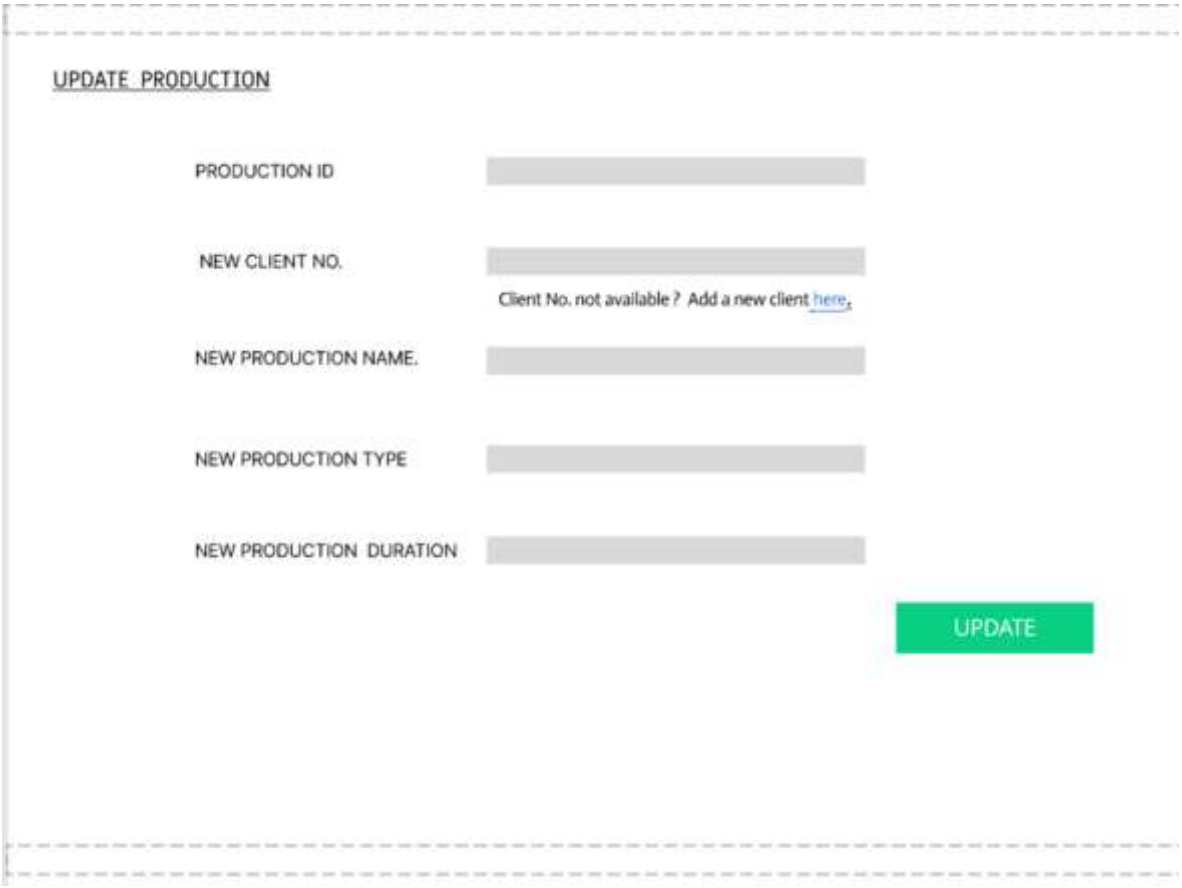

Figure 18 - Update Production for quiet attic films (Wireframe)

The user can update production information here. The output page after updating is shown below.

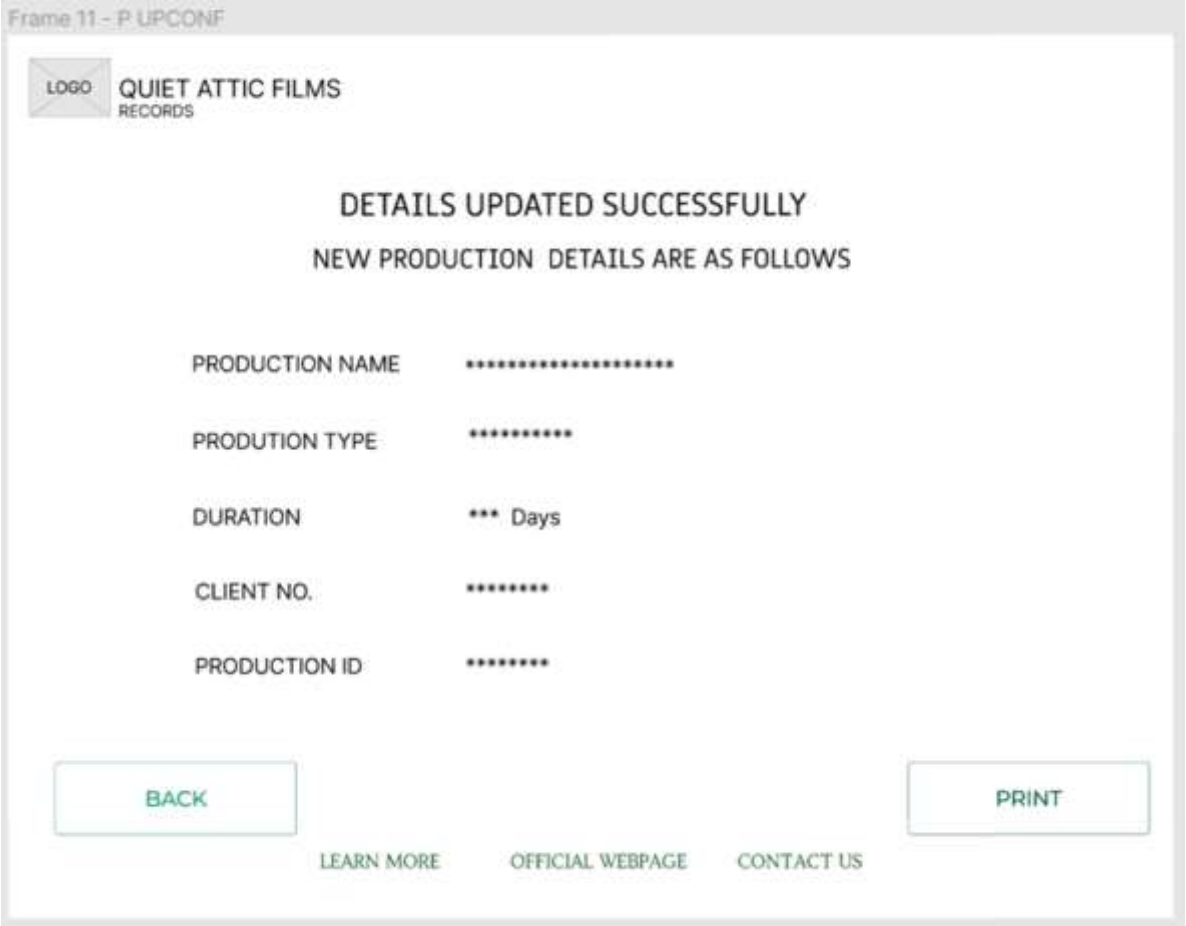

Figure 19- Update Production for quiet attic films confirmation output (Wireframe)

Delete Production Data Interface

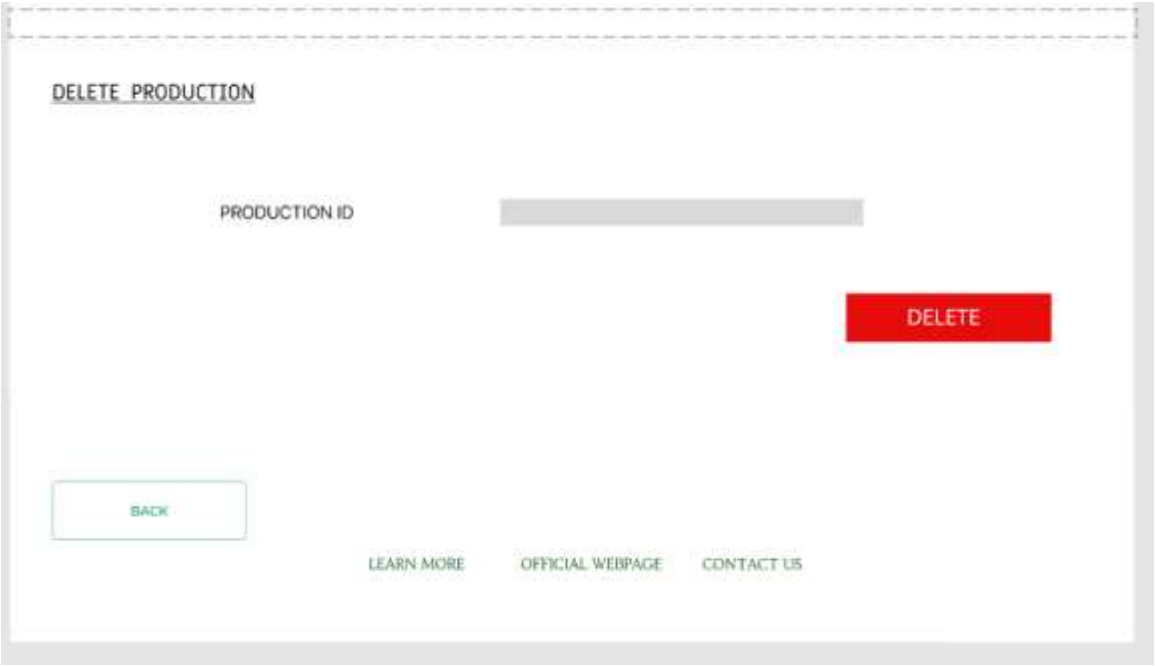

Figure 20- Delete Production for quiet attic films (Wireframe)

The user can delete production information here.

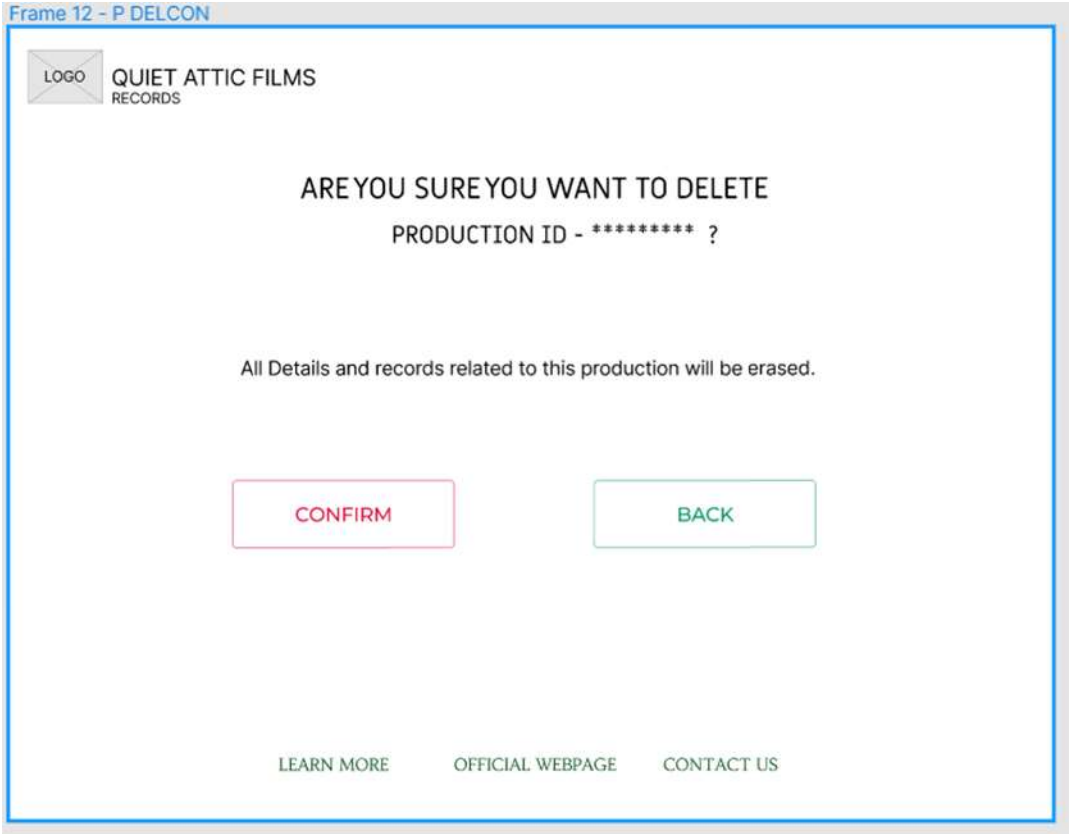

Figure 21-Delete Production for quiet attic films confirmation (Wireframe)

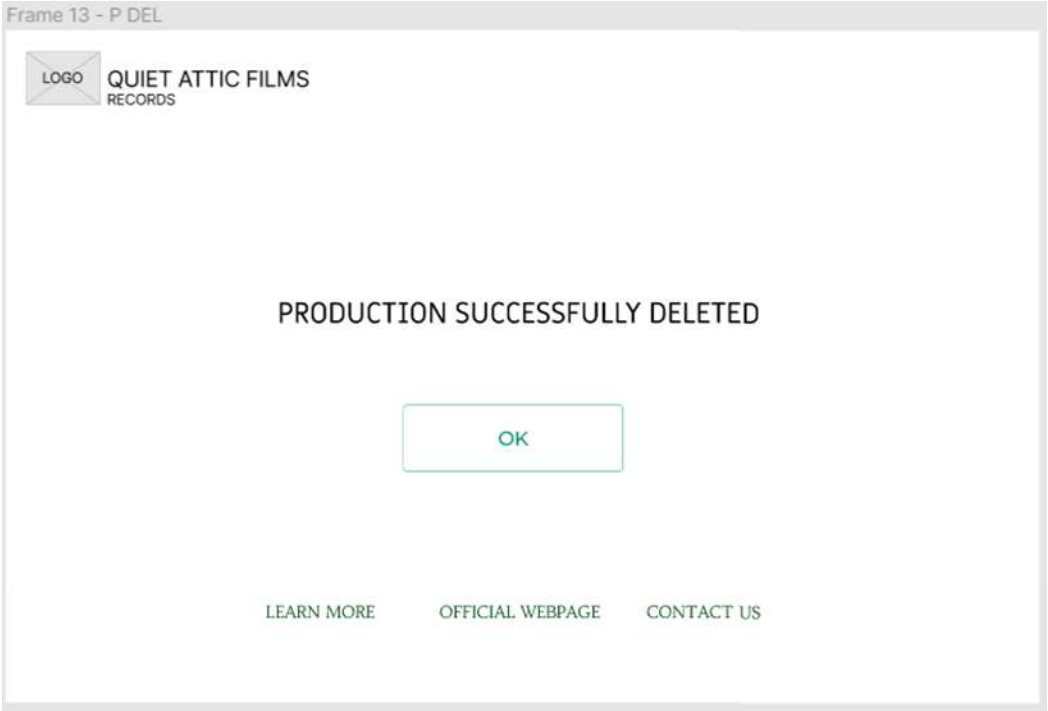

Figure 22- Delete Production for quiet attic films success (Wireframe)

The user will receive a confirmation page before deleting the production data as shown in figure 22 and a successfully deleted page after deleting.

Manage Staff Interface (Wireframe map)

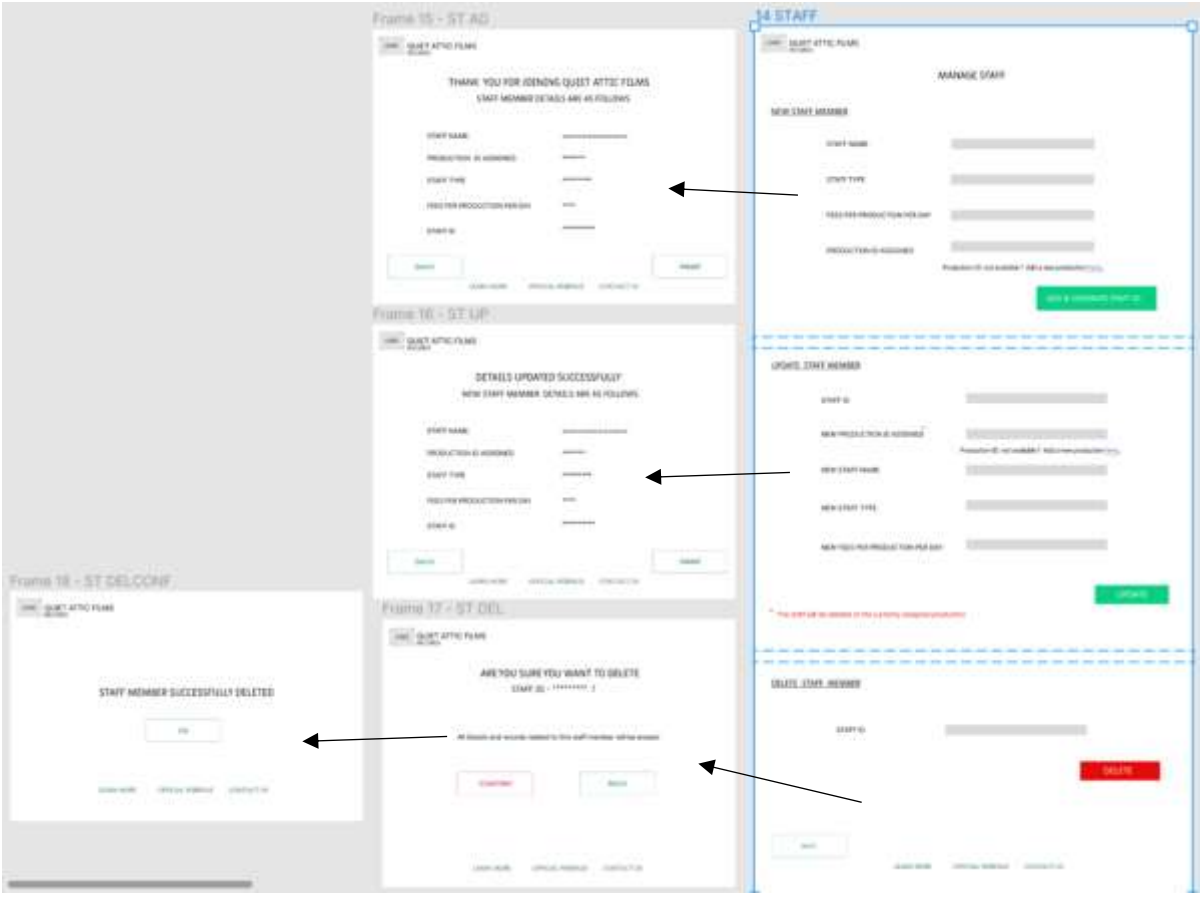

Figure 23 - Manage productions for quiet attic films (Wireframe)

Insert Staff Data Interface

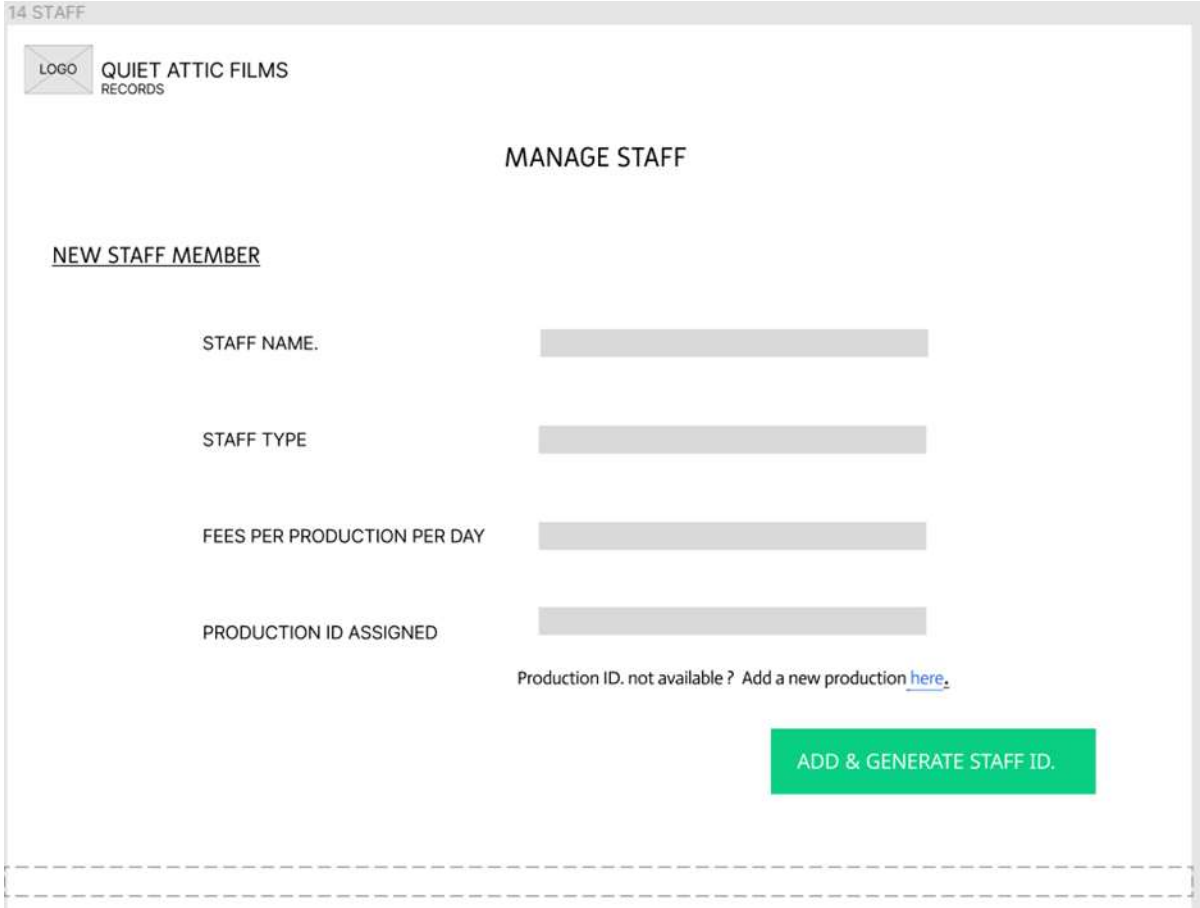

Figure 24- Insert staff for quiet attic films (Wireframe)

The User will insert the Staff name, Type, Fees per day and Production ID (If not available, can be created using the following link) and get a generated Staff ID.

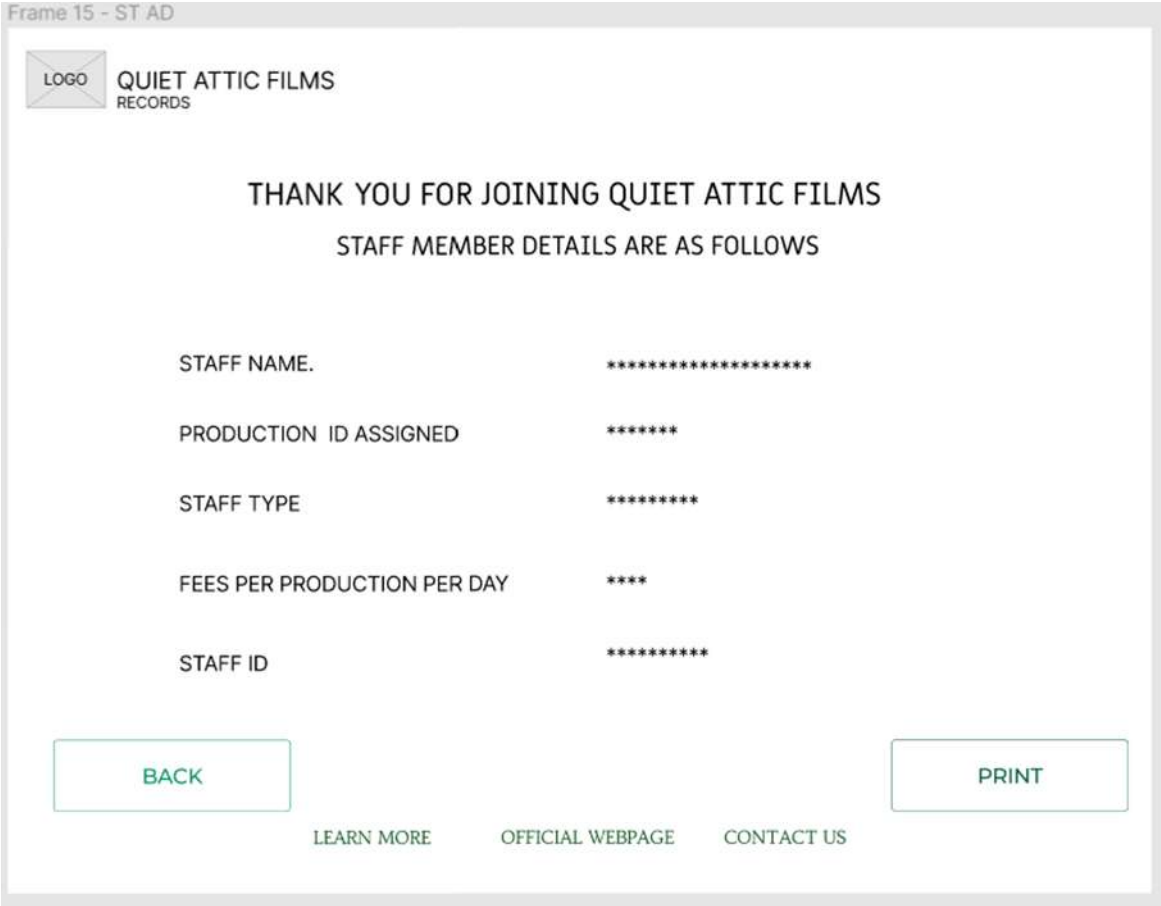

Figure 25- Insert staff for quiet attic films confirmation output (Wireframe)

Update Staff Data Interface

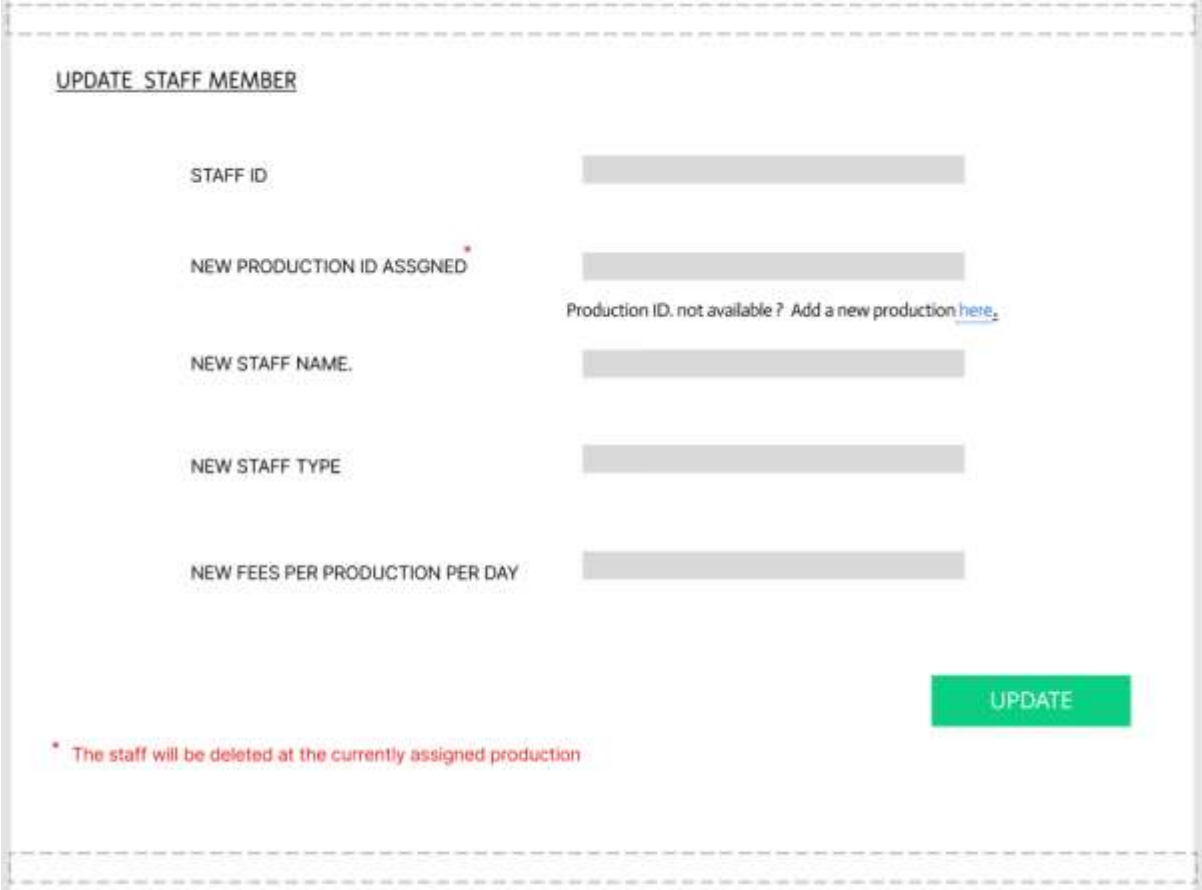

Figure 26- Update staff for quiet attic films (Wireframe)

The user can update staff information here. The output page after updating is shown below.

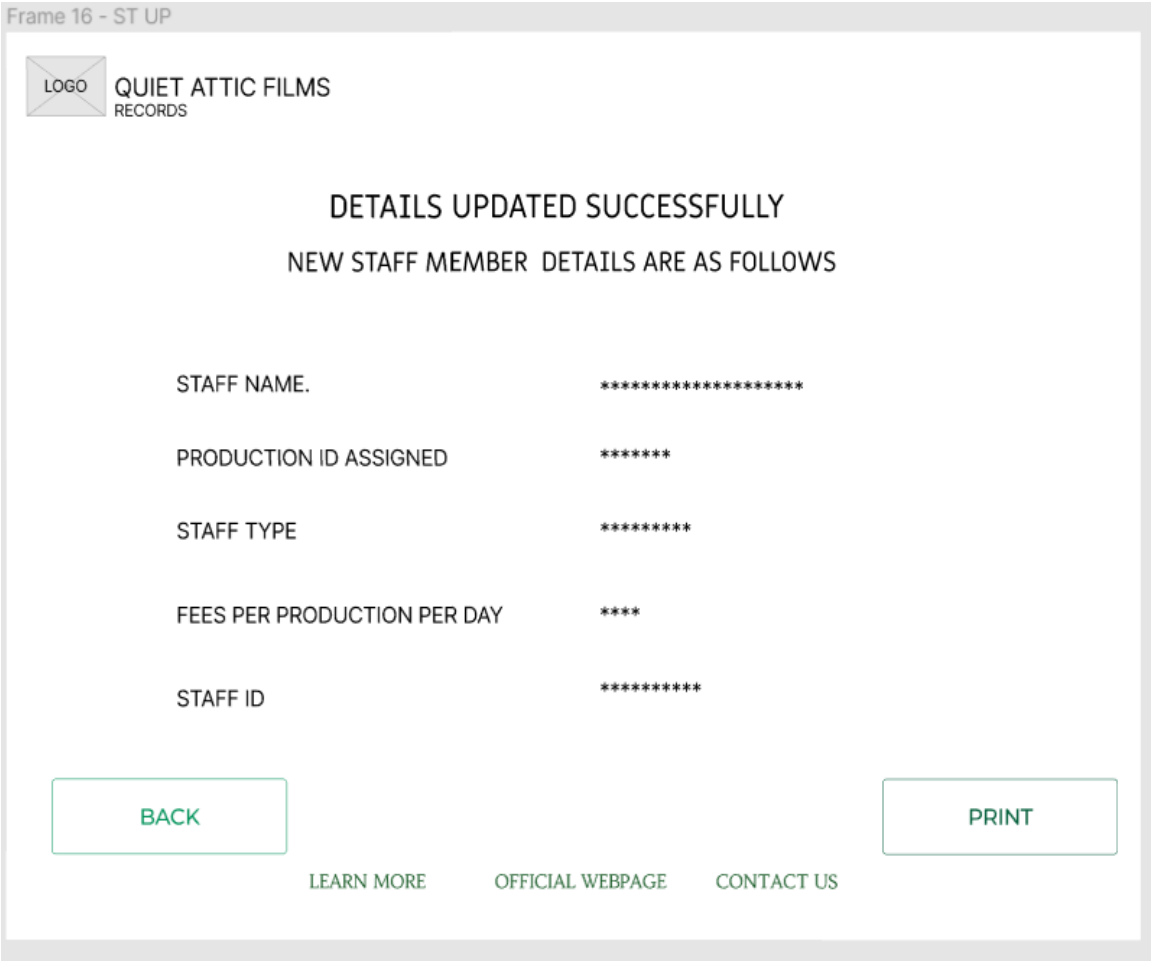

Figure 27 - Update staff for quiet attic films confirmation output (Wireframe)

Delete Staff Data Interface

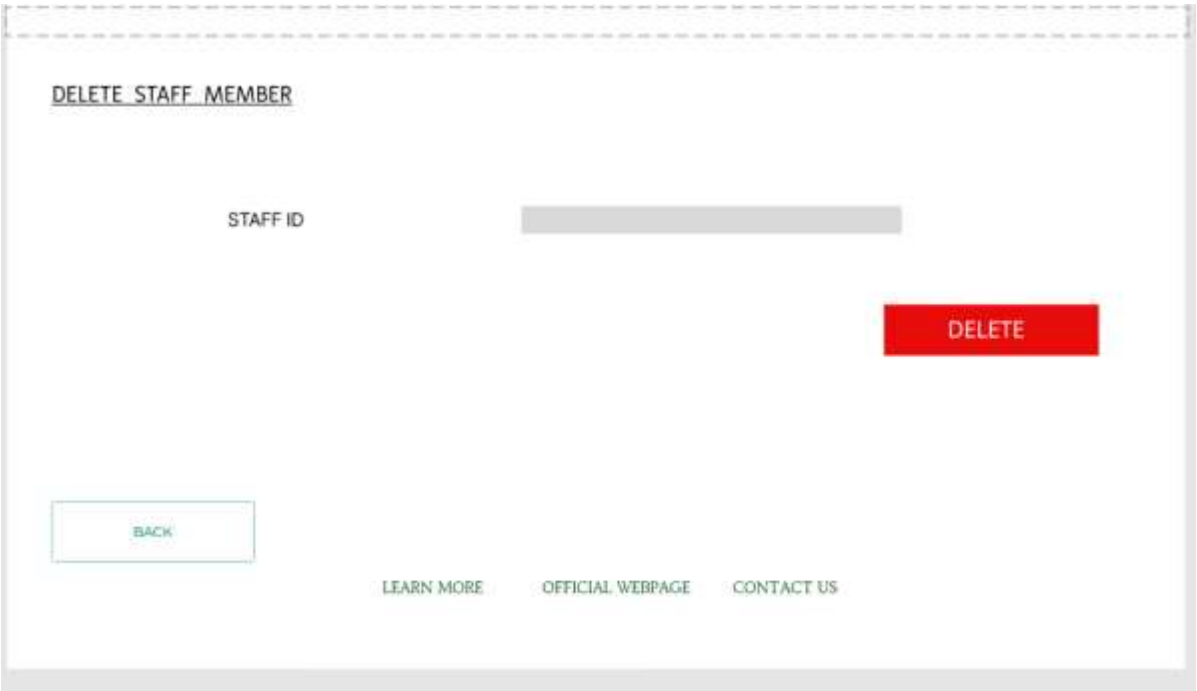

Figure 28- Delete staff for quiet attic films (Wireframe)

The user can delete staff information here.

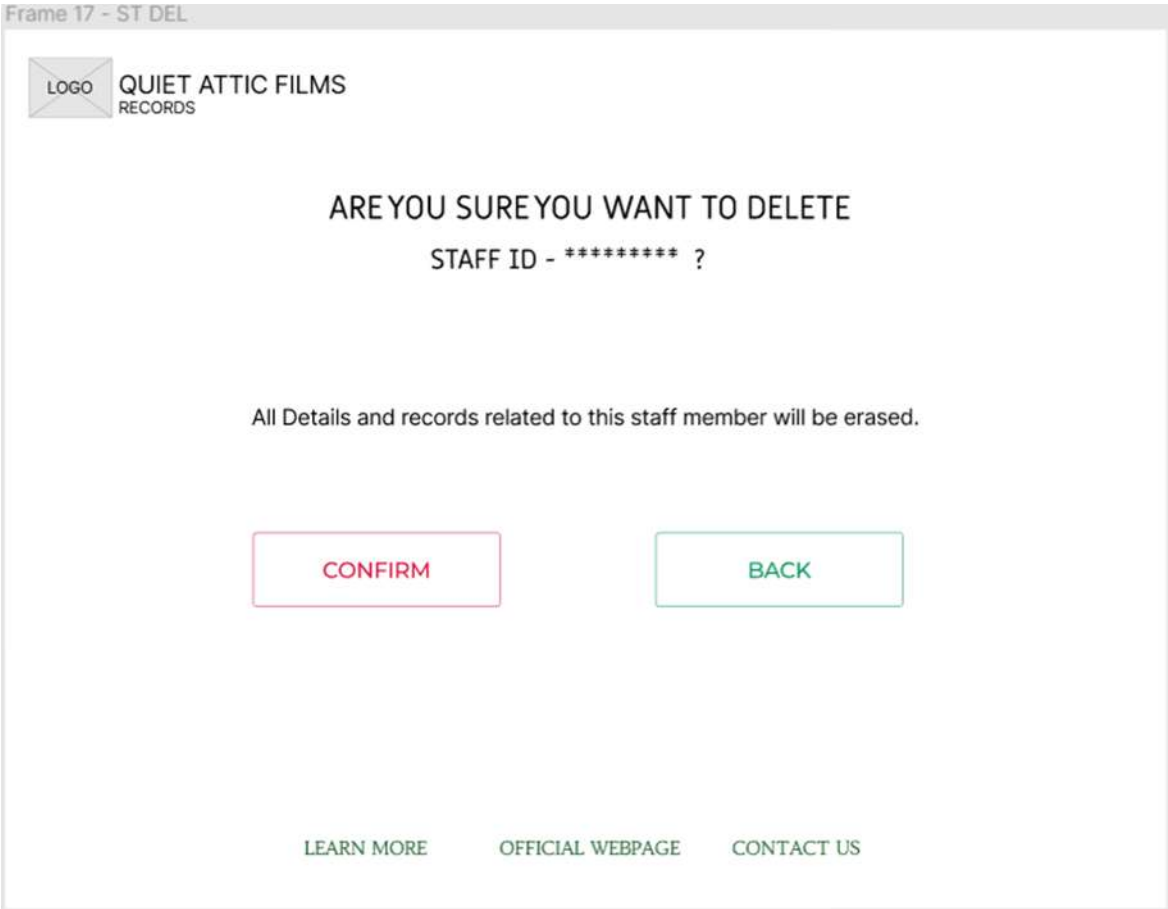

Figure 29- Delete clients for quiet attic films confirmation (Wireframe)

The user will receive a confirmation page before deleting the staff data as shown in figure 29 and a successfully deleted page after deleting.

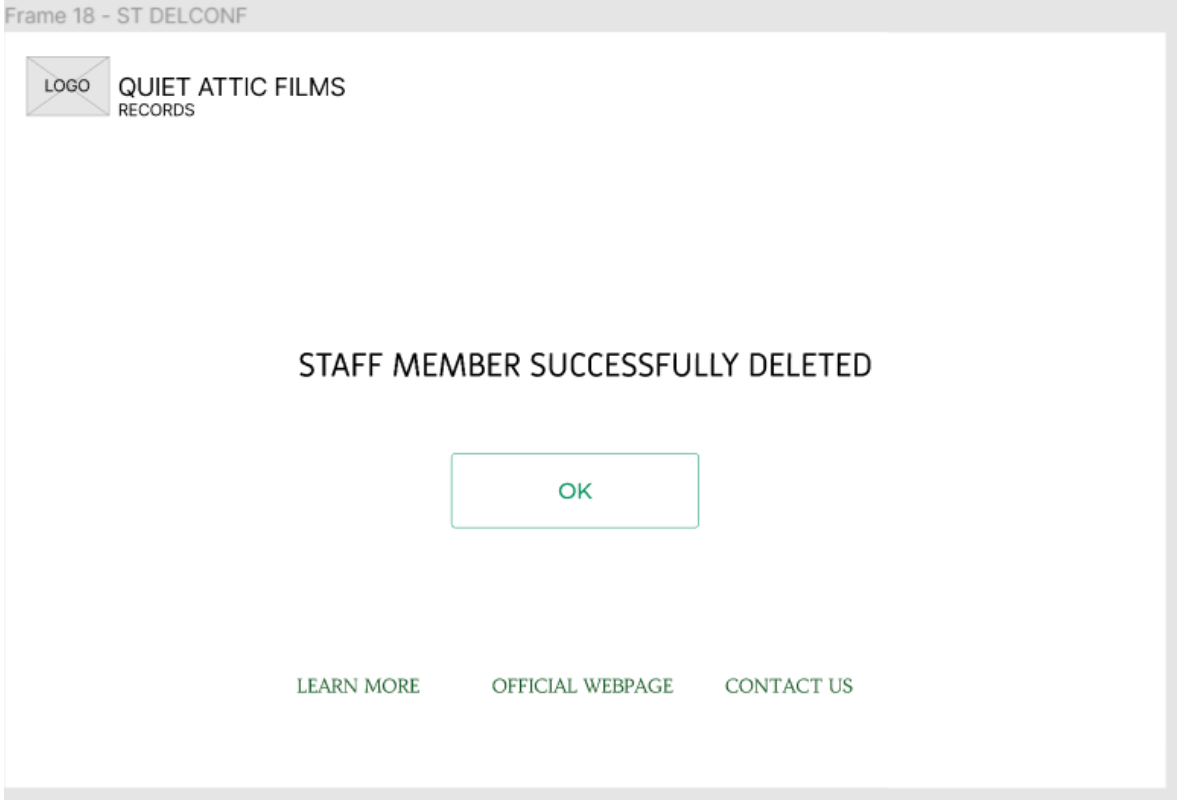

Figure 30- Delete clients for quiet attic films success (Wireframe)

‘

Manage location Interface (Wireframe map)

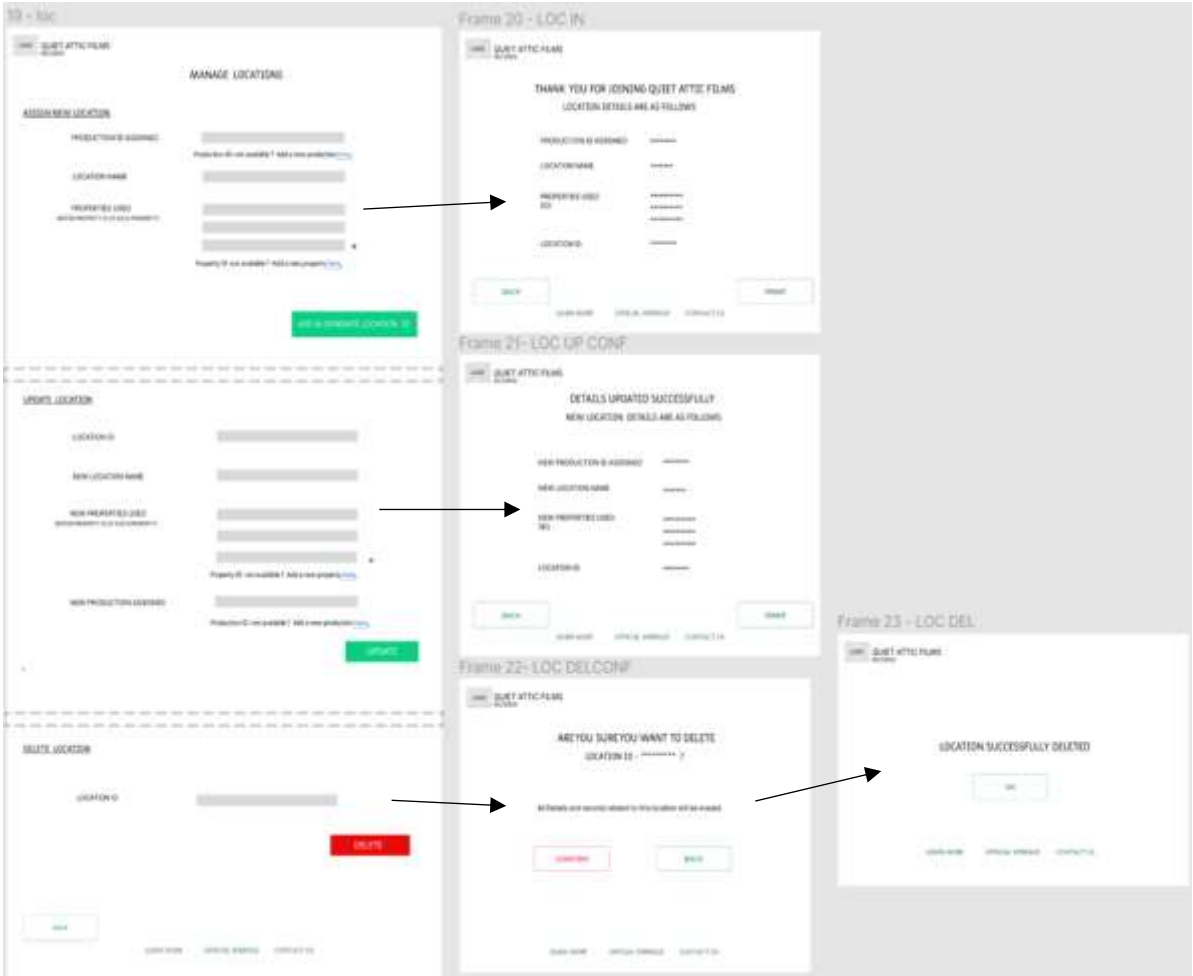

Figure 31- Manage location for quiet attic films (Wireframe)

Shown above in figure 31 is the map of pages in the interface on managing the client data.

Insert location Data Interface

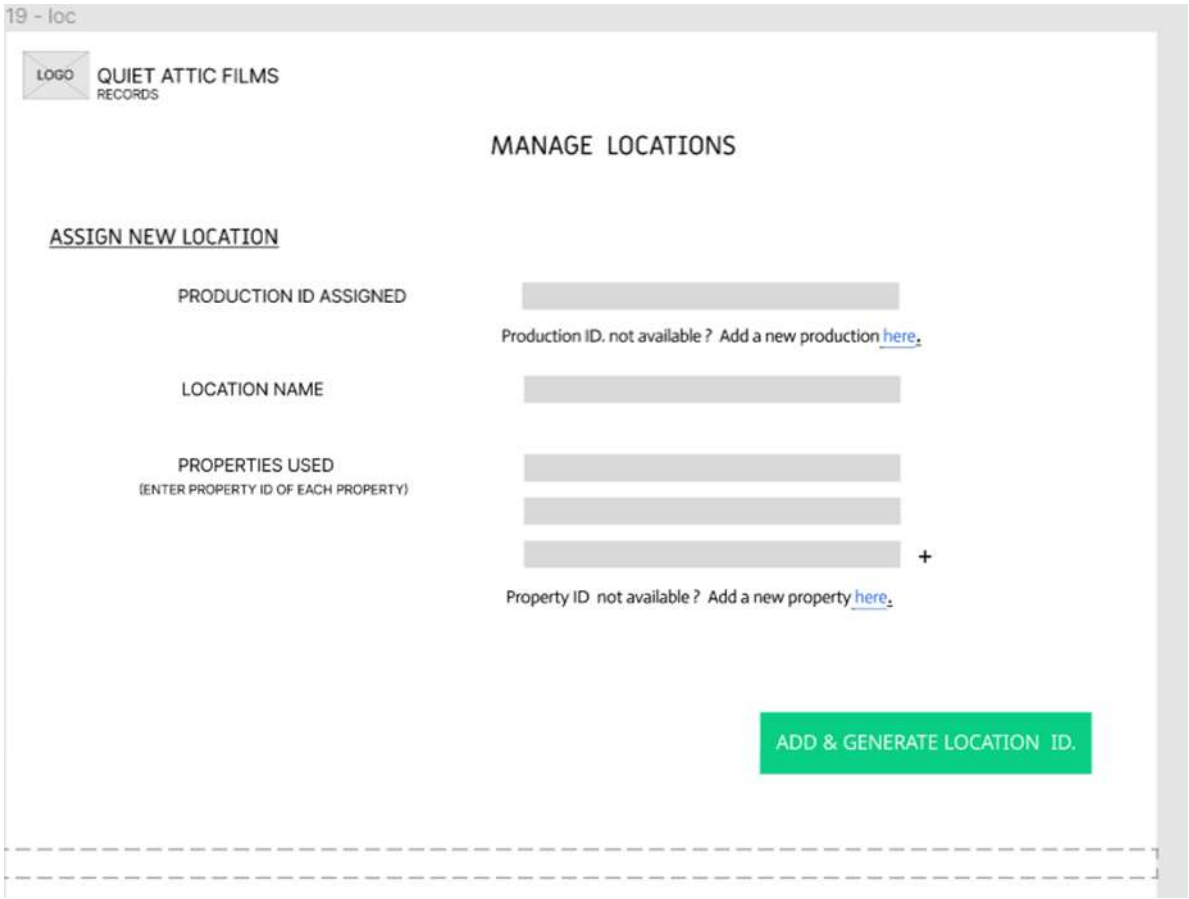

Figure 32-Insert location for quiet attic films (Wireframe)

Here the user can assign a new location by entering the production ID, Location name and Properties used and get a generated location ID.

Figure 33- Insert staff for quiet attic films confirmation output (Wireframe)

Update location Data Interface

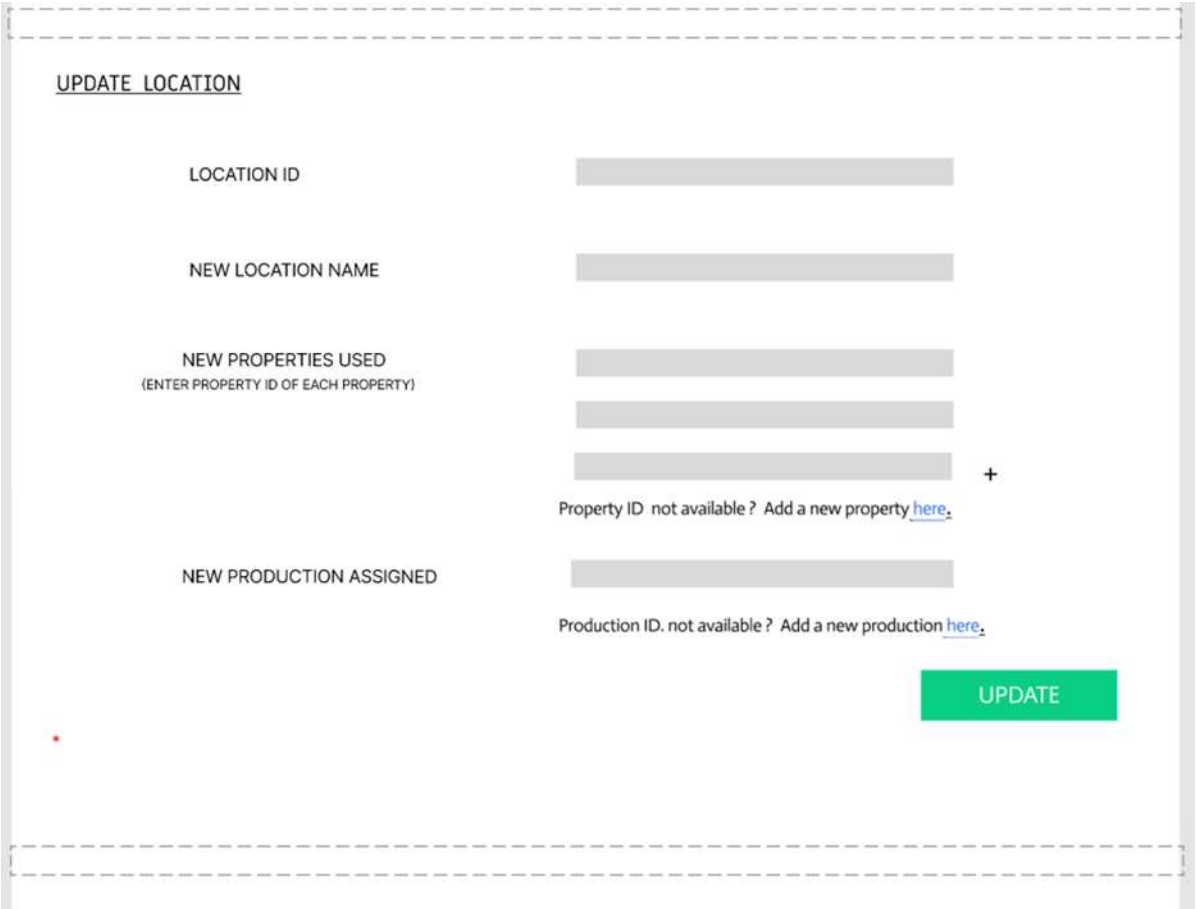

Figure 34- Update location for quiet attic films (Wireframe)

The user can update the location data here. The output page after updating is shown below.

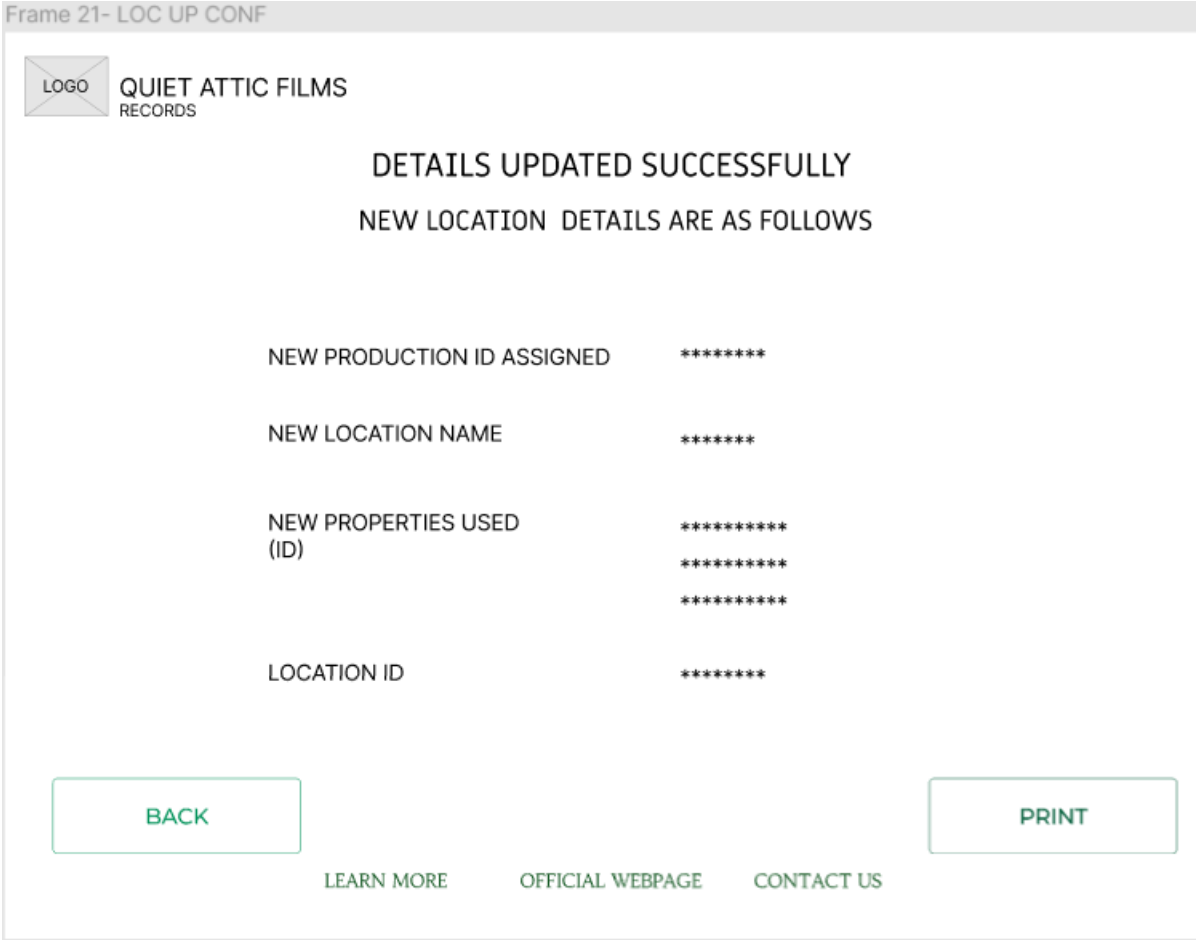

Figure 35- Update location for quiet attic films confirmation output (Wireframe)

Delete location Data Interface

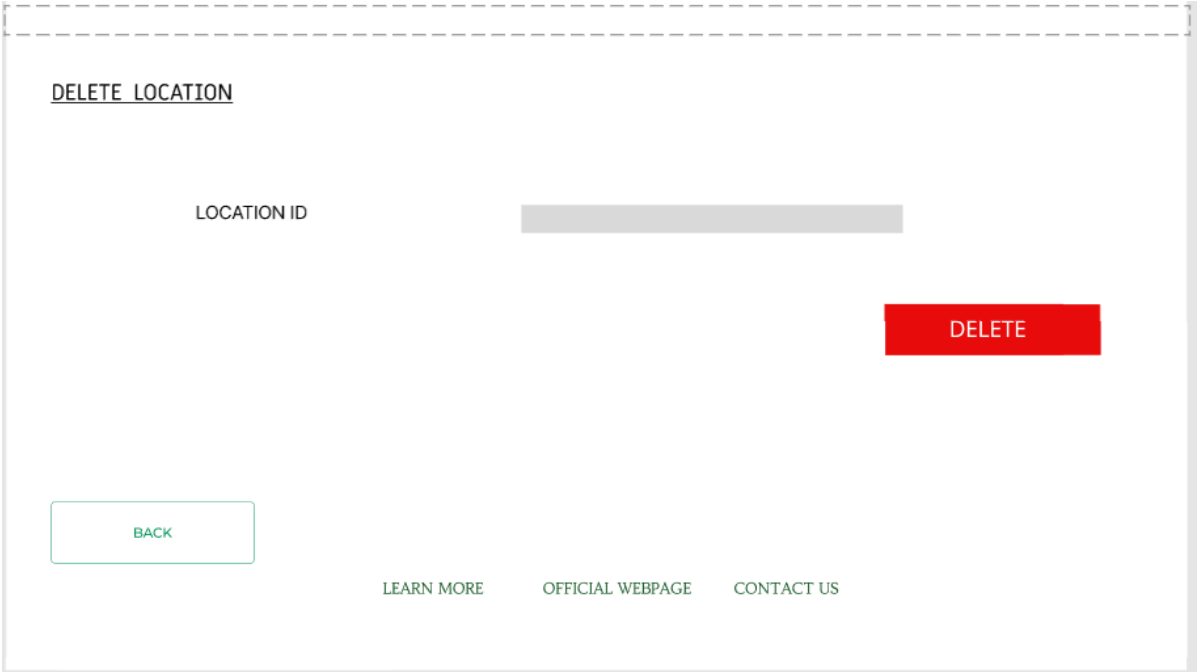

Figure 36- Delete location for quiet attic films (Wireframe)

Here the user can delete location data.

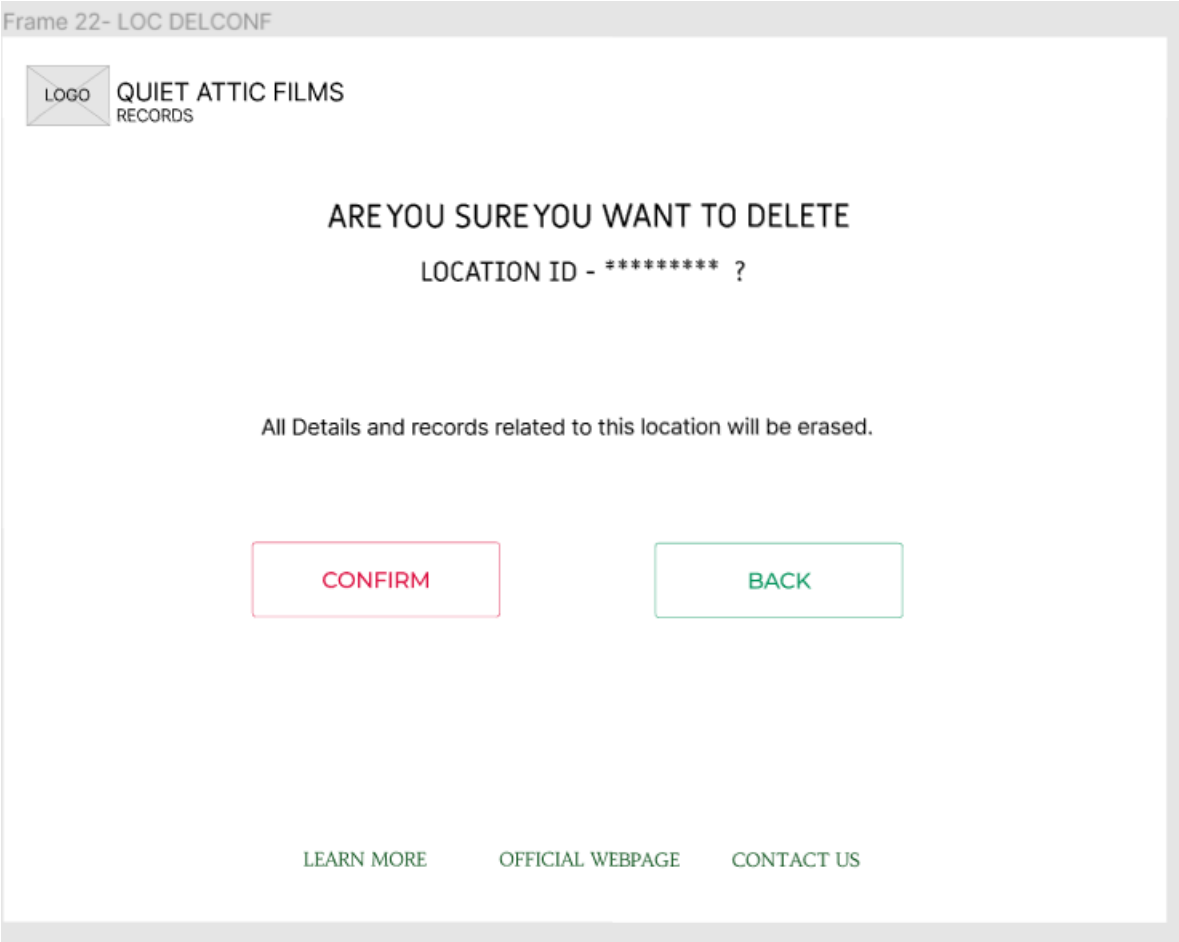

Figure 37- Delete location for quiet attic films confirmation (Wireframe)

The user will receive a confirmation page before deleting the location data as shown in figure 37 and a successfully deleted page after deleting.

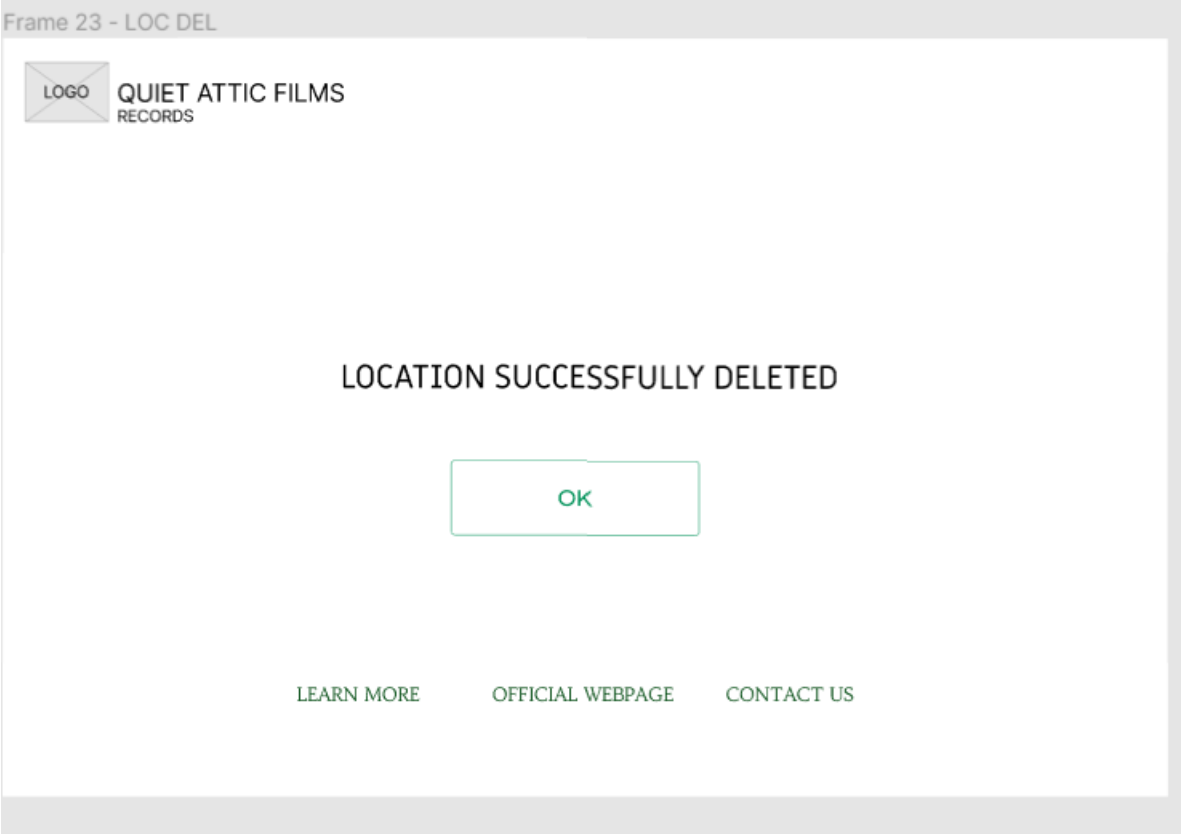

Figure 38 - Delete location for quiet attic films success (Wireframe)

Manage Property Interface

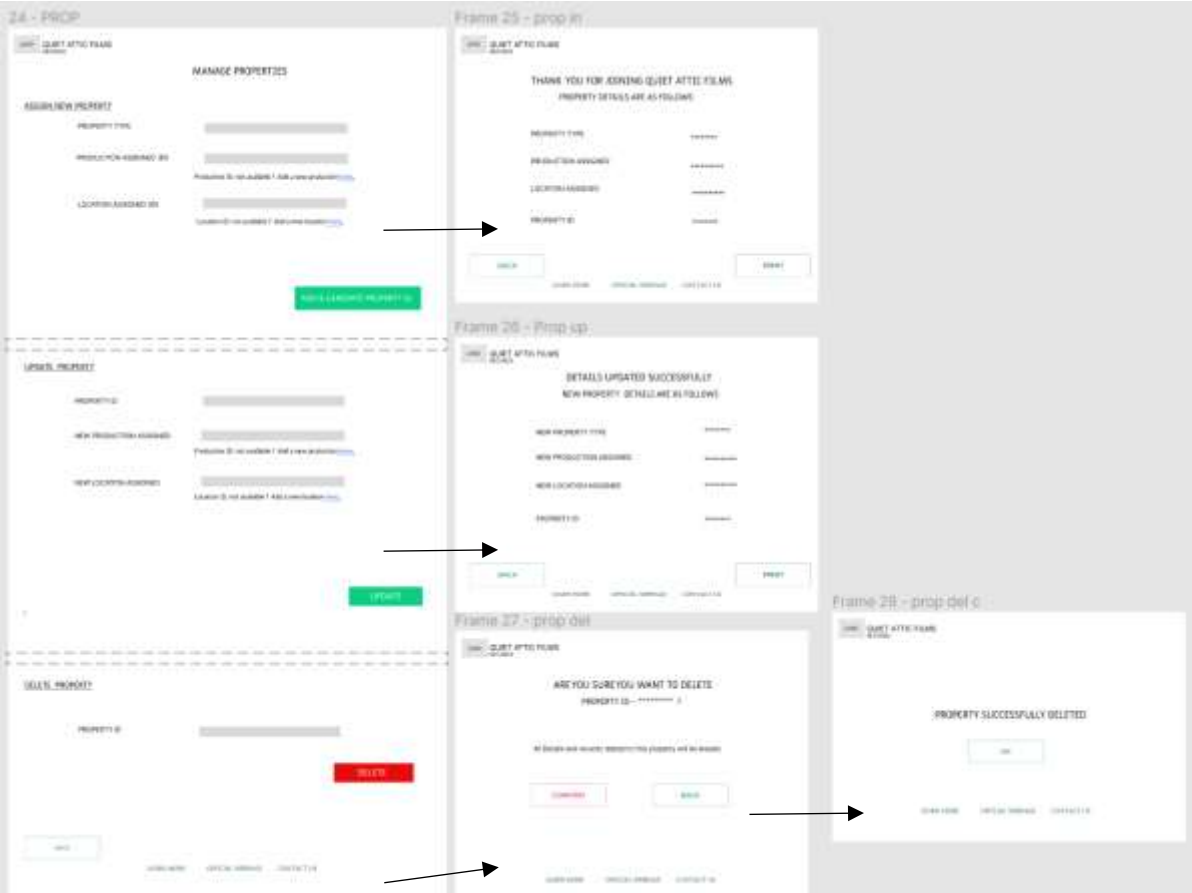

Figure 39-Manage properties for quiet attic films (Wireframe)

Insert Property Data Interface

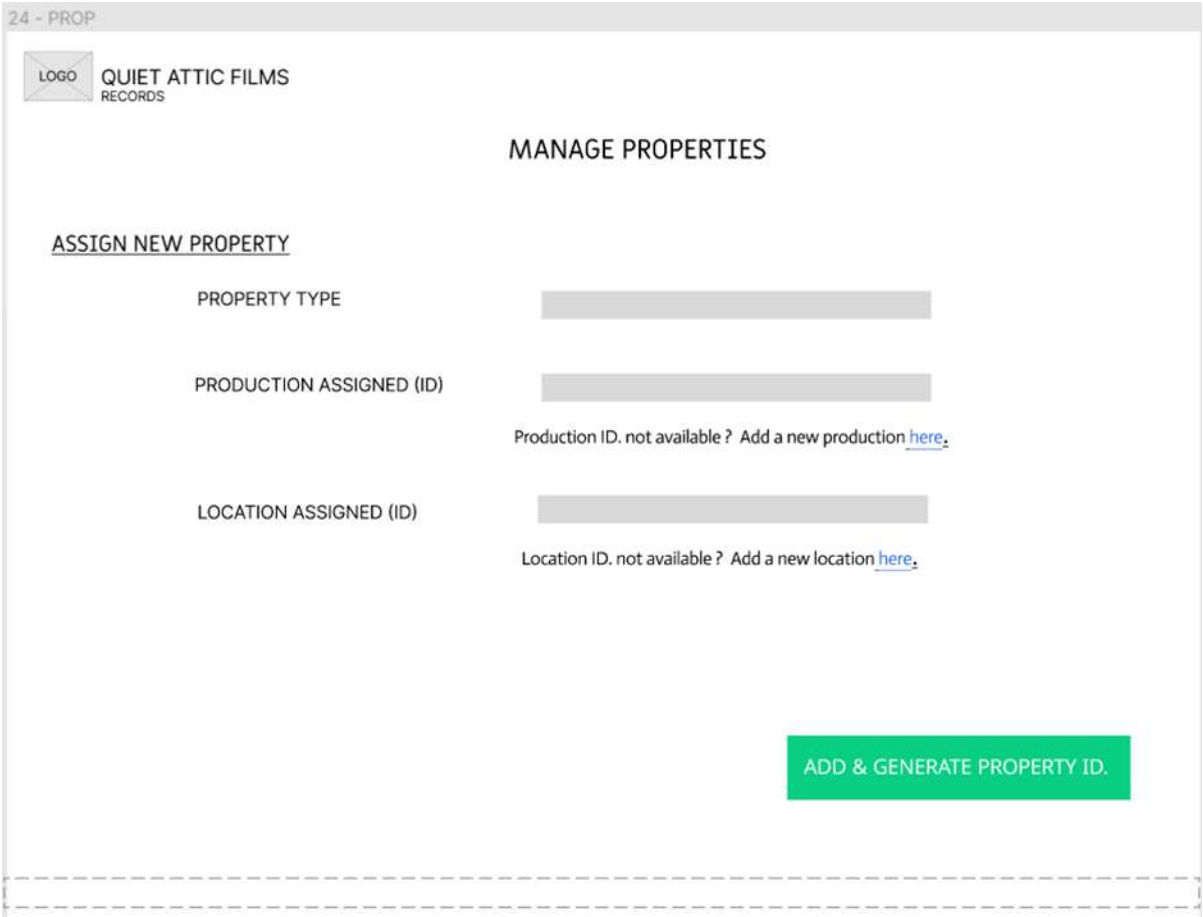

Figure 40-Insert properties for quiet attic films (Wireframe)

Here the user can insert property data by adding the property type, Production assigned and Location assigned and get a generated Property ID. Below is the output to be recieved after successfully inserting the property data

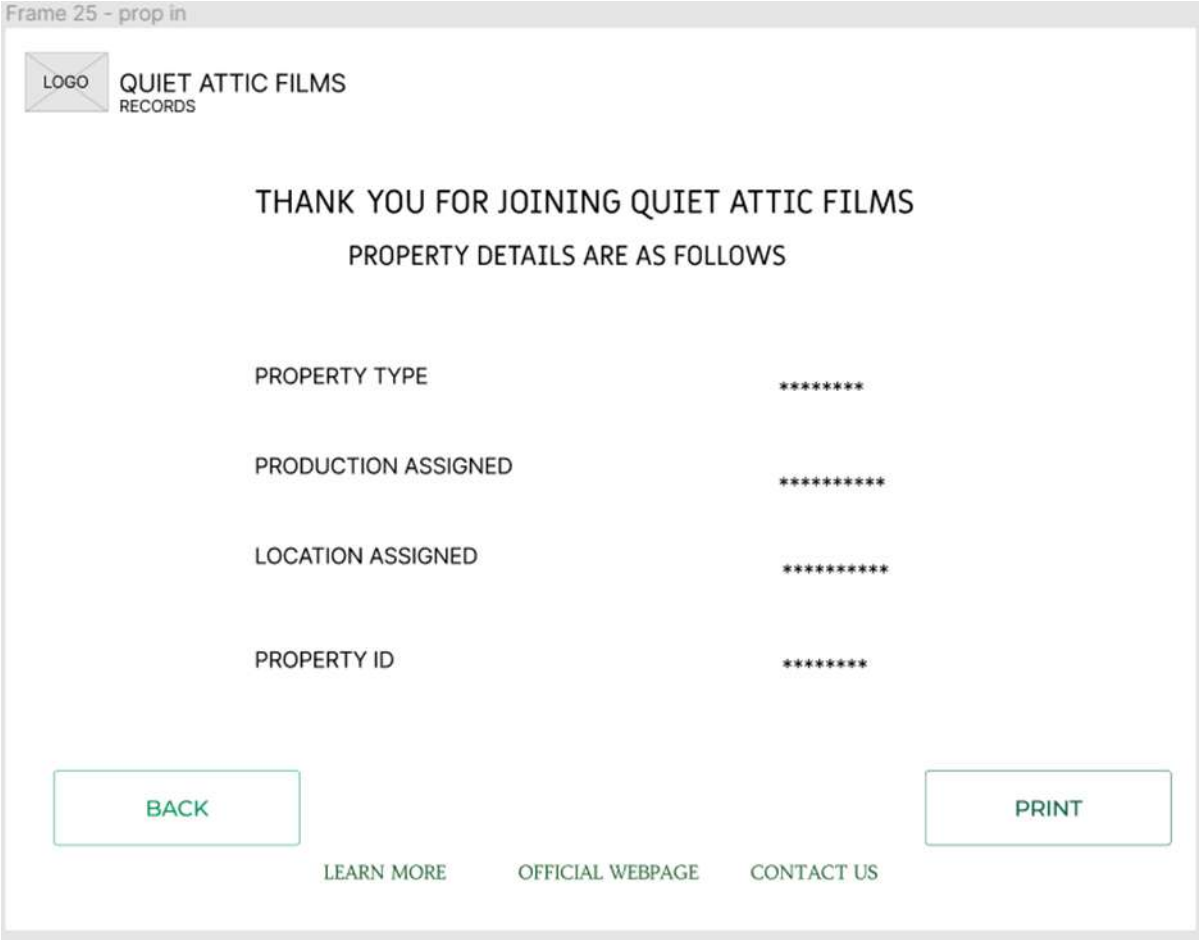

Figure 41- Insert property for quiet attic films confirmation output (Wireframe)

Update Property Data Interface

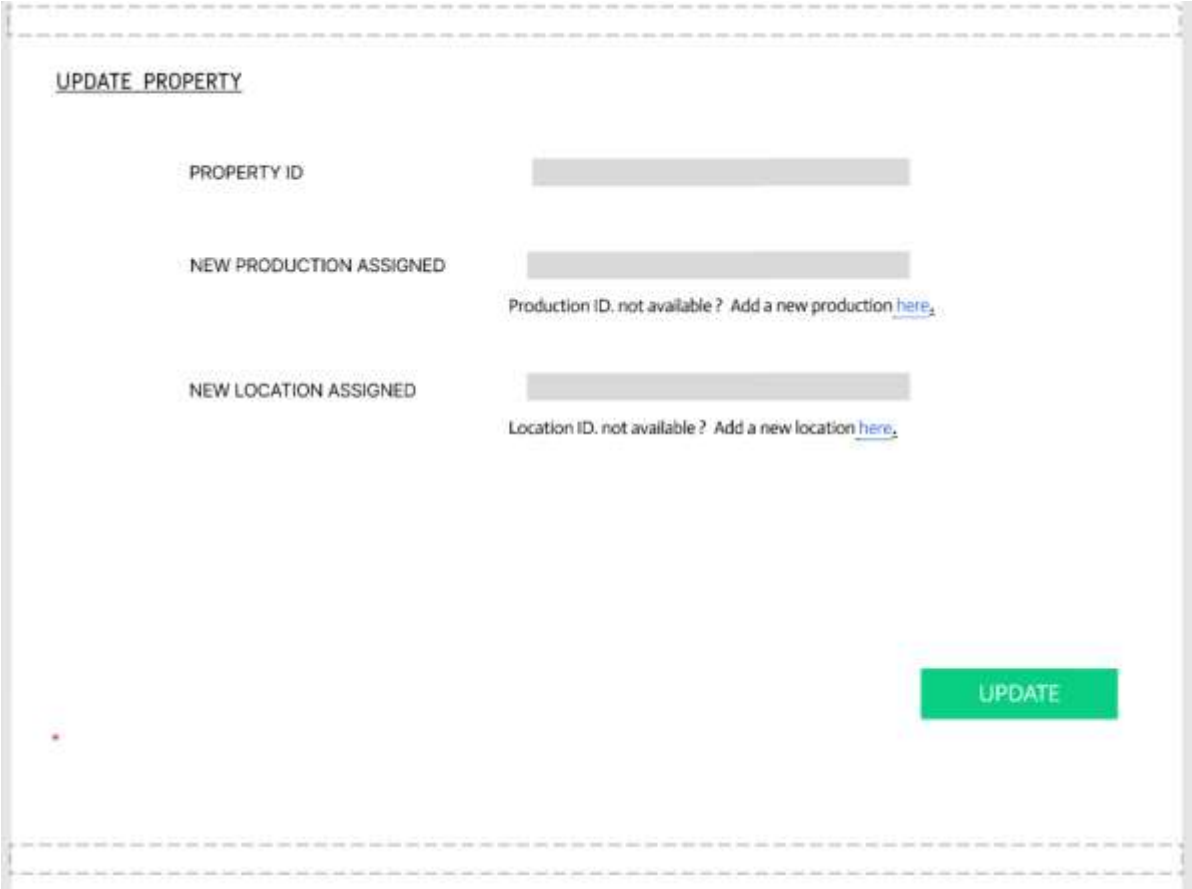

Figure 42-Update property for quiet attic films (Wireframe)

Here the user can update property data.

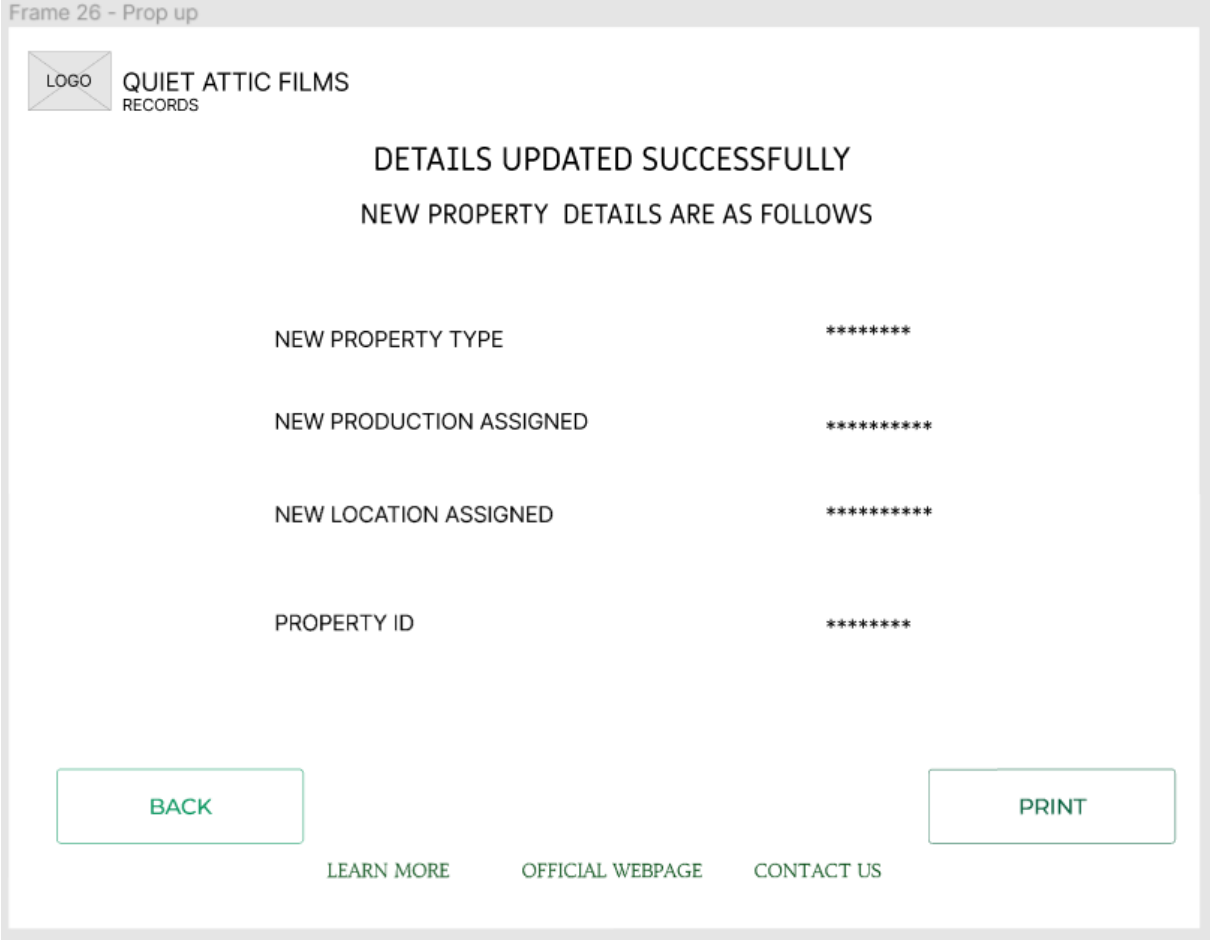

Figure 43- Update property for quiet attic films confirmation output (Wireframe)

Delete Property Data Interface

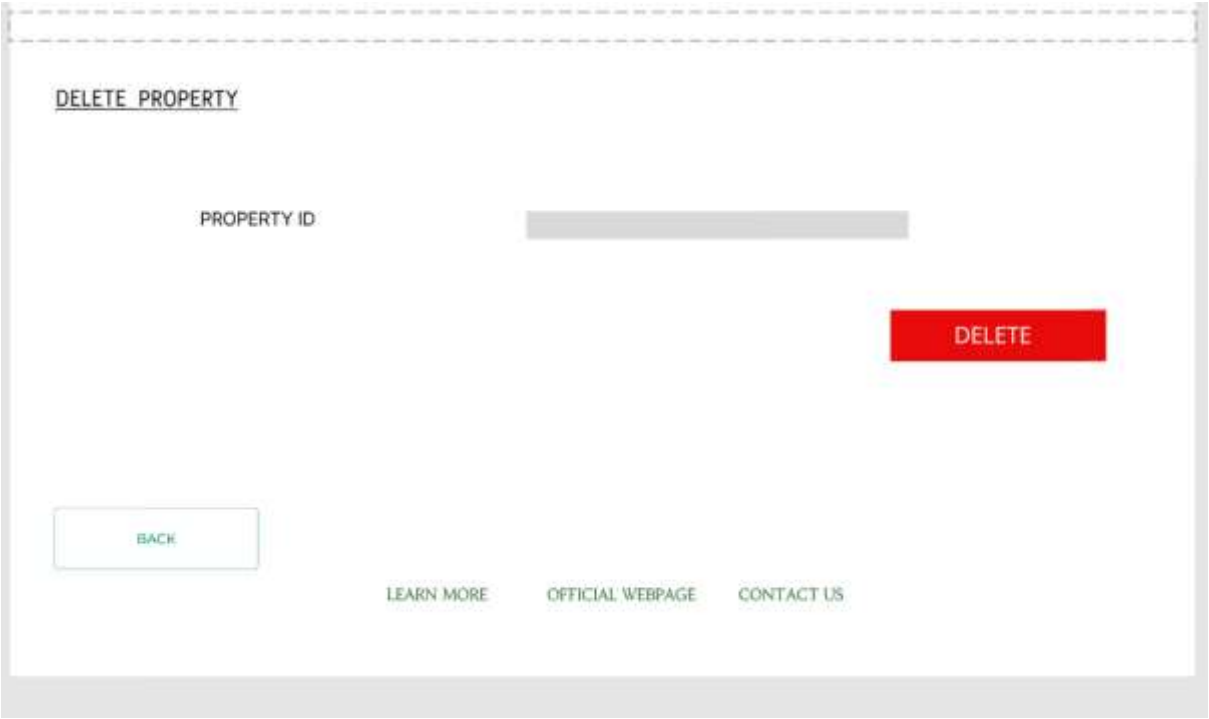

Figure 44-- Delete property for quiet attic films (Wireframe)

Here the user can delete property data.

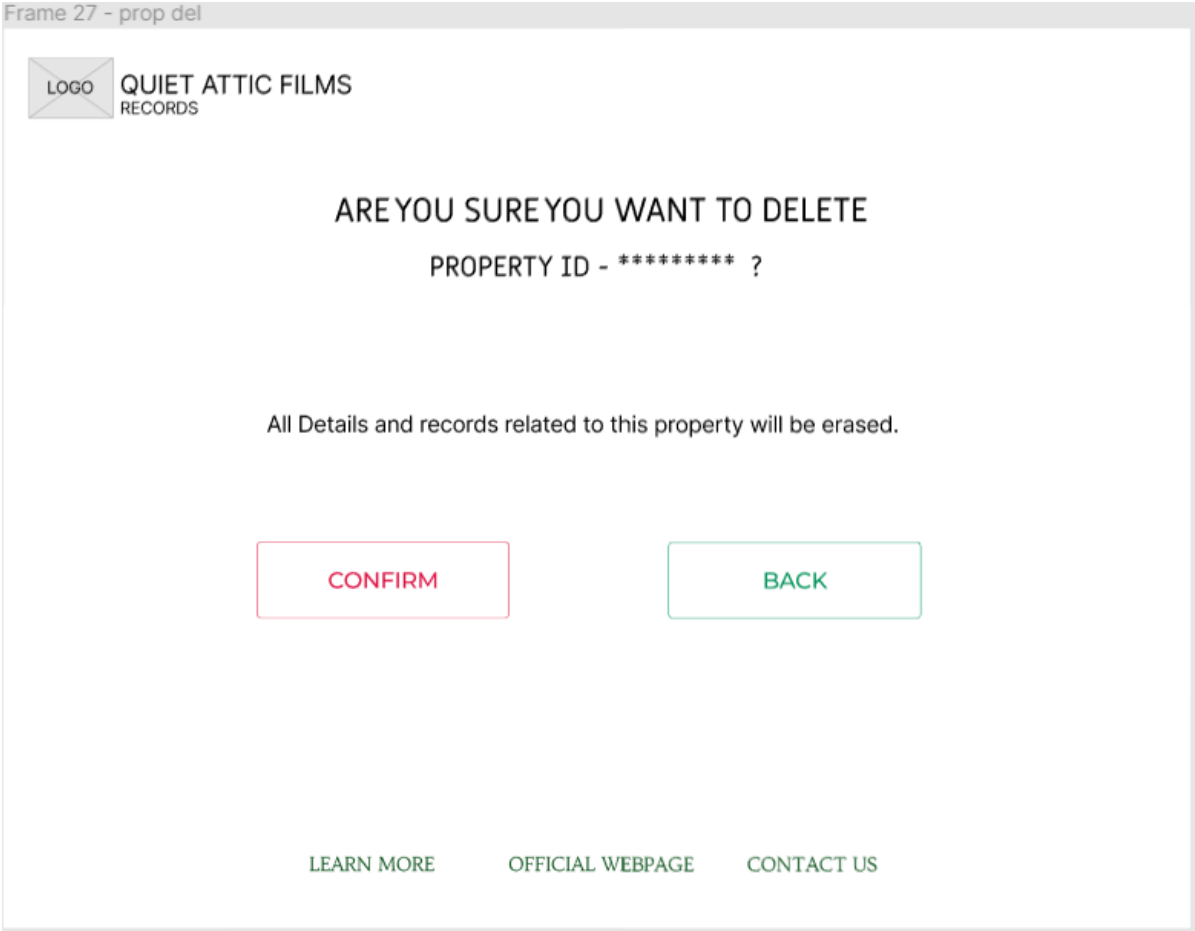

Figure 45—Delete property for quiet attic films confirmation (Wireframe)

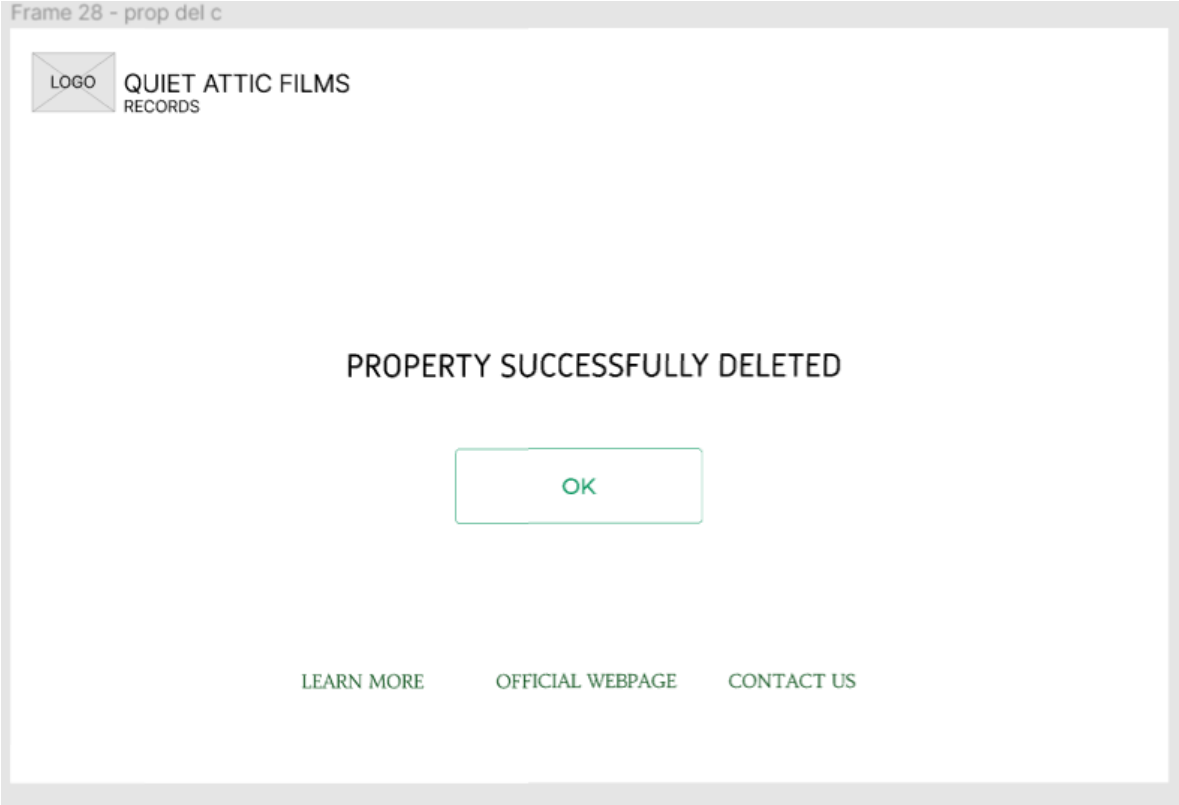

Figure 46-Delete location for quiet attic films success (Wireframe)

Explaining about the above comprehensive design

The above comprehensive design is specially designed for Quiet Attic films. The above design includes the interfaces that the users will experience while they are using the database. The user interfaces will be able to allow the users to insert, update, view and delete the data. The normalization for the design is done with the logical database design. The view page can be used to validate the data and necessary output pages will be shown when the the data are successfully managed in the database.

Evaluating the effectiveness of the design in relation to the User and system requirements.

Once again, the user requirements of the scenario are

- Organize data on the concept of production.
- Determine what productions belong to client and which client hired a production.
- Determine which production will be done at a particular location.
- Determine which production use a recognized property at a time.
- Insert, Update and Delete data in the above entities.

Considering the above design, it is clear that the data will be organized in the concept of production, as it allows the users to insert, update and delete data relating to the productions of quiet attic films. The data will be clearly organized tabular formats within the database itself for each entity described above with no redundancies or conflicts due to normalizations.

The users of the database system will be able to view which client ordered a production by entering the production ID in the view data page of the interface. The data for this will be stored in the production table itself as the client No. By obtaining the client no. the users can find out the client details as well within the same system.

According to this comprehensive design, it is clear that the users will be able to determine which production will be done at a particular location. The user can simply enter the location ID in the view data page and observe which production is allocated in the particular location.

The users will also be able to determine which production use a recognized property at a time. The users can simply extract the data stored in the database by entering the desired property ID in the view data page. The user will be able to identify the property, see which productions use it and which location the property is allocated to at the time.

The comprehensive design is mainly based on inserting, updating and deleting data based on the entities of the scenario, The entities of the scenario are, Production, Staff, client, location and property. The user will be able to manage all these entities in one single set of user interface pages. Additionally, this interface and comprehensive design provides the planning for a fully functional system for quiet attic films,

The comprehensive design of quiet attic films is made based to the system requirements determined to the scenario. Once again, the system requirements determined are

Hardware requirements

-The relational database management system requires the following hardware requirements as it is to be run using Microsoft SQL server.

- A processor with minimum of processing speed 1.5GHz 64-bit, a processor over 2.0GHz with multi cores is recommended to run Microsoft SQL Server and the created relational database system.
- A minimum system RAM of 2GB and 4GB RAM is recommended depending to the workload of Quiet Attic Films.
- A hard disk drive space of over 15 GB (8 GB to install Microsoft SQL server + additional space over 7GB to store databases, backups and other data of Quiet Attic Films)
- An effective Network Connection in order to communicate with clients and access other resources online.

Software requirements

-These are the requirements which are required for the relational database management system of quiet attic films.

- Microsoft SQL Server; Which includes the software requirements below
    - Windows OS -Widows10 /8/ 8.1/ 7 (64-bit)
    - Microsoft .NET framework 4.8 or later.
- SQL Server Management Studio (SSMS) for database administrations.
- SQL Server Data Tools (SSDT) for database project developments.
- Microsoft Visual Studio for further maintenances.

Considering the system requirements, the above comprehensive design is made in compatible and run smoothly with any device containing the above hardware components as mentioned. Other than that, the system requires Microsoft SQL server as the above comprehensive design will be based on it.

The databases mentioned in the comprehensive design will be administrated using SSMS (SQL server Management Studio). And SSDT (SQL server Data Tools) will be used in order to implement database project developments.

The program interfaces will be implemented and the SQL database will be connected to the program using Microsoft Visual Studio. Visual studio will be a software suitable for maintaining the system as well. The interfaces can be deigned and coded as windows forms in

a suitable language and the SQL database can be connected to the system using a connection string.

Creating the database and objects for Quiet Attic Films

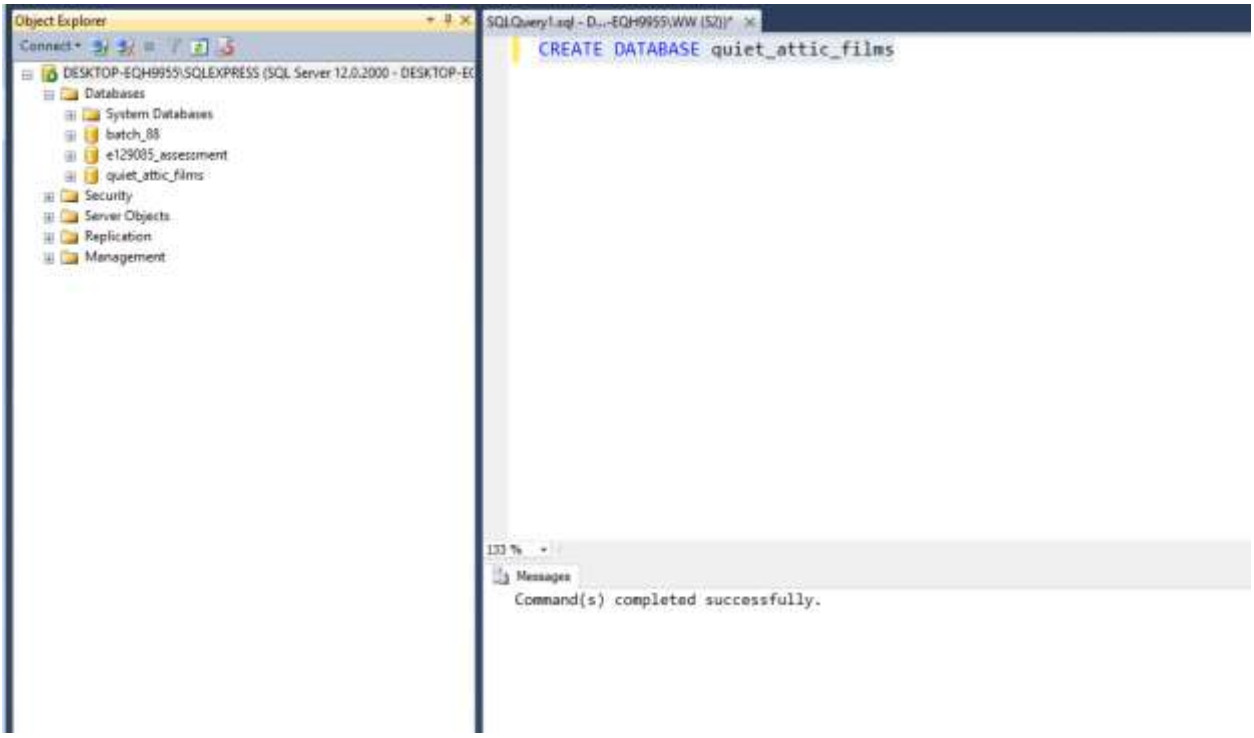

Figure 47- Creating the Database for quiet attic films

The above query is executed in order to create the database for quiet attic films. In the left pane (Object explorer) it can be observed that “quiet_attic_films” database is created. The “CREATE” statement is used in order to create the database.

Creating the Client Table

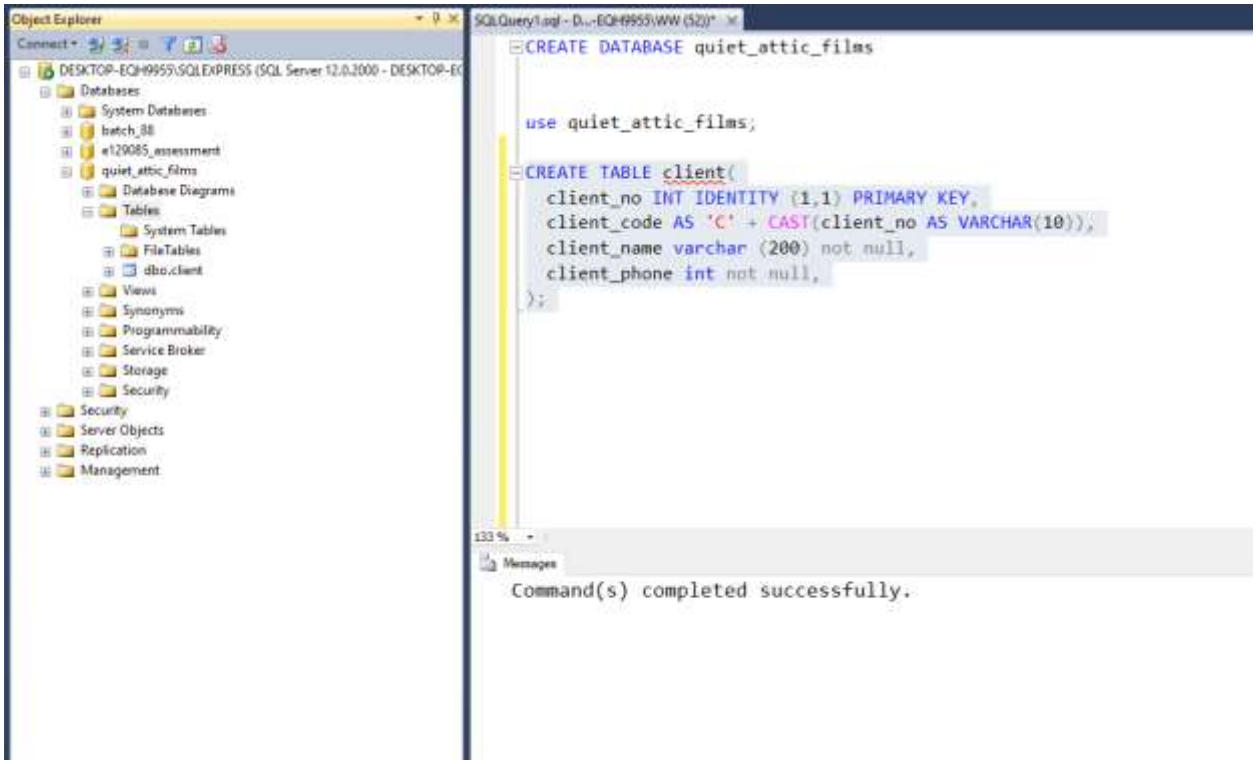

Figure 48-creating client table

The above query is used to create the Client table to quiet attic films. To create the table the “CREATE” statement is used.

The line “client_no INT IDENTITY (1,1) PRIMARY KEY” defines a column called "client_no" of type INT as the primary key to the table. The IDENTITY (1,1) means that the values for this column will be automatically generated and incremented by 1 beginning from 1.

The line “client_code AS 'C' + CAST (client_no AS VARCHAR (10))” creates a column named "client_code" in the table which will be a temporary column to perform calculations. The "AS" keyword is used to define an expression that calculates the value for this column. In this table, it will be the string 'C' with the value of "client_no". The "CAST" function coverts “C” to a string value. Therefore, the output values for this column will be calculated depending the values position in the "client_no" column.

After that, the below lines are simply the query to add the desired columns to the table using the column name, data type and constraint.

Creating the Production Table

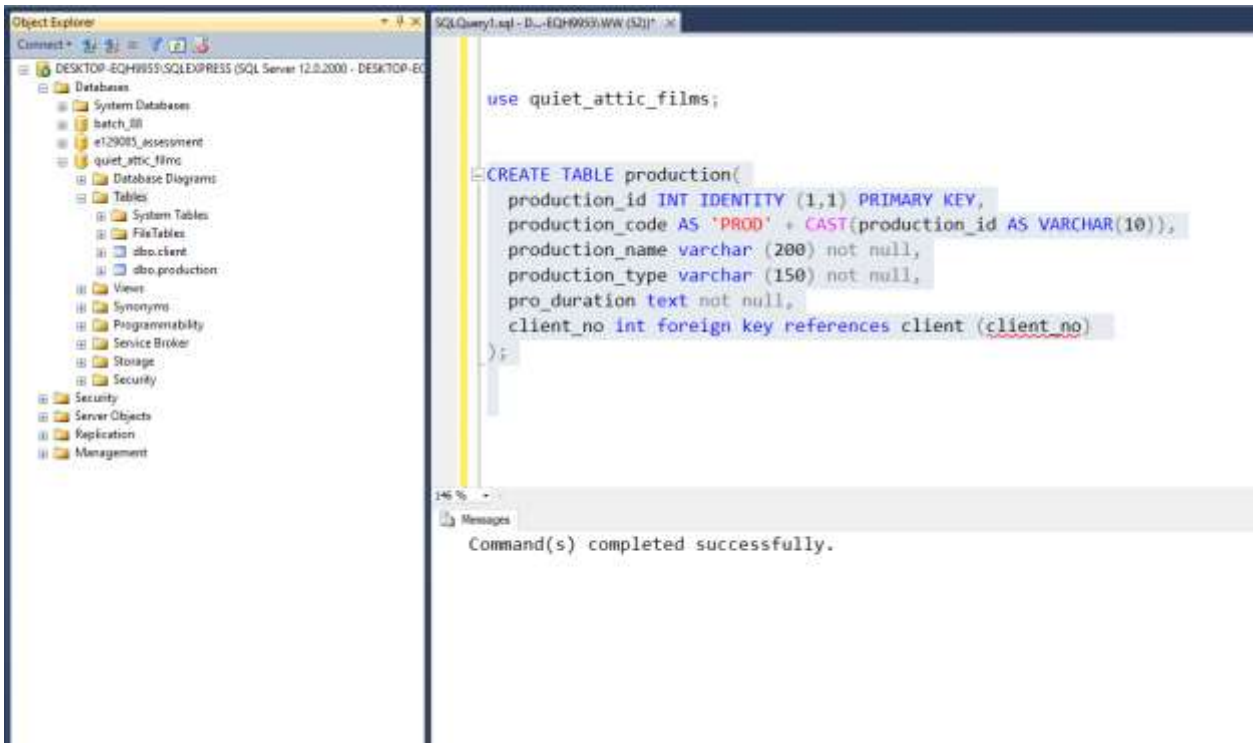

Figure 49- creating the Production table

The above query is used to create the production table to quiet attic films. As above, the “CREATE” statement is used. The same methodology as the client table is used here to generate the primary key.

After that, the below lines are simply the query to add the desired columns to the table using the column name, data type and constraint.

The line “client_no int foreign key references client (client_no)” is used to define a foreign key to the table. Here “client_no” of the client table will be a foreign key of the production table.

Creating the Staff Table

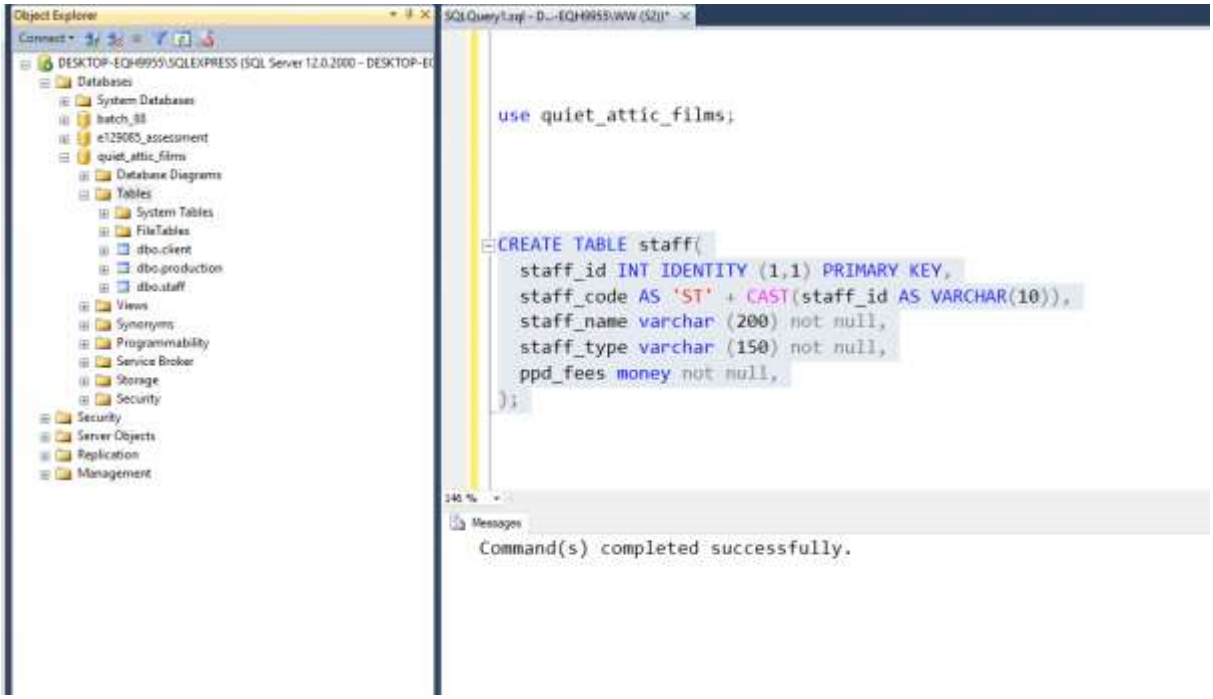

Figure 50-Creating Staff table

This query is used to create the staff table to quiet attic films. As above, the “CREATE” statement is used.
The same methodology as the client table is used here to generate the primary key.
After that, the below lines are simply the query to add the desired columns (staff_name, staff_type and ppd_fees) to the table using the column name, data type and constraint.

Creating the property Table

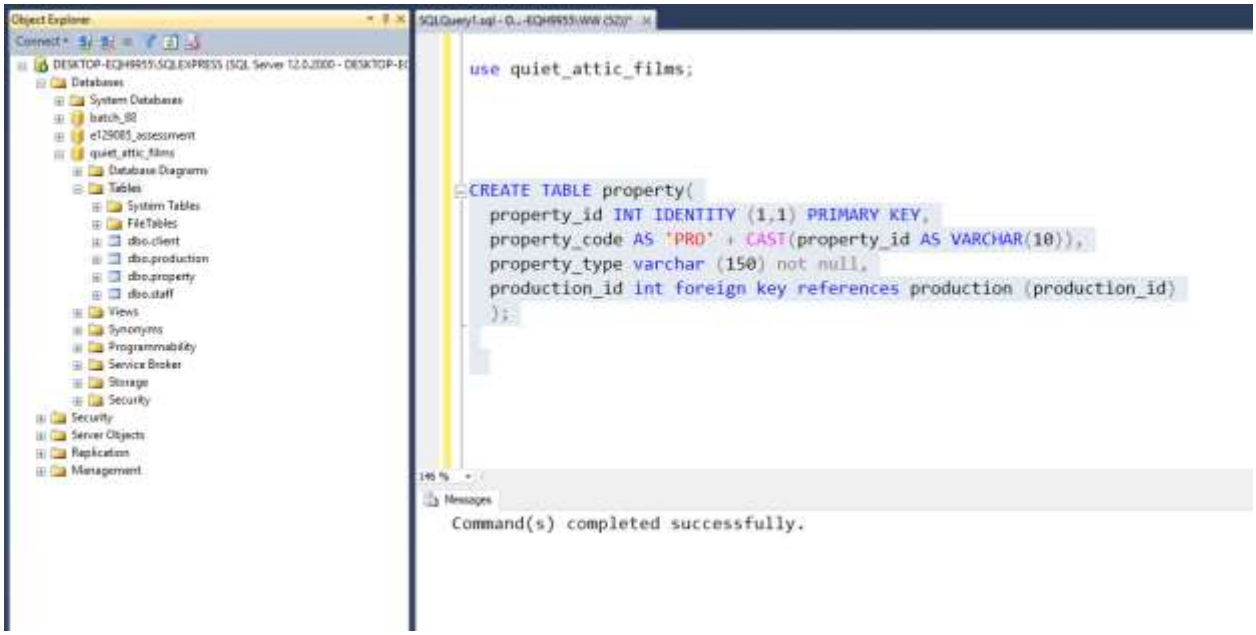

Figure 51-creating property table

The above query is used to create the property table for quiet attic films. The “CREATE” statement is used in order to create the table.
The same methodology as the client table is used here to generate the primary key of the property table.
After that, the “property_type” column is added to the table by entering the column name, data type and constraint.
Finally, “production_id” of the production ID is identifies as a foreign key of this table.

<u>NOTE</u>

The location_id is also a foreign key to this table. But in order to add that column, first the location table should be created. Then it will be added to the table using the ALTER statement.

Creating the location Table

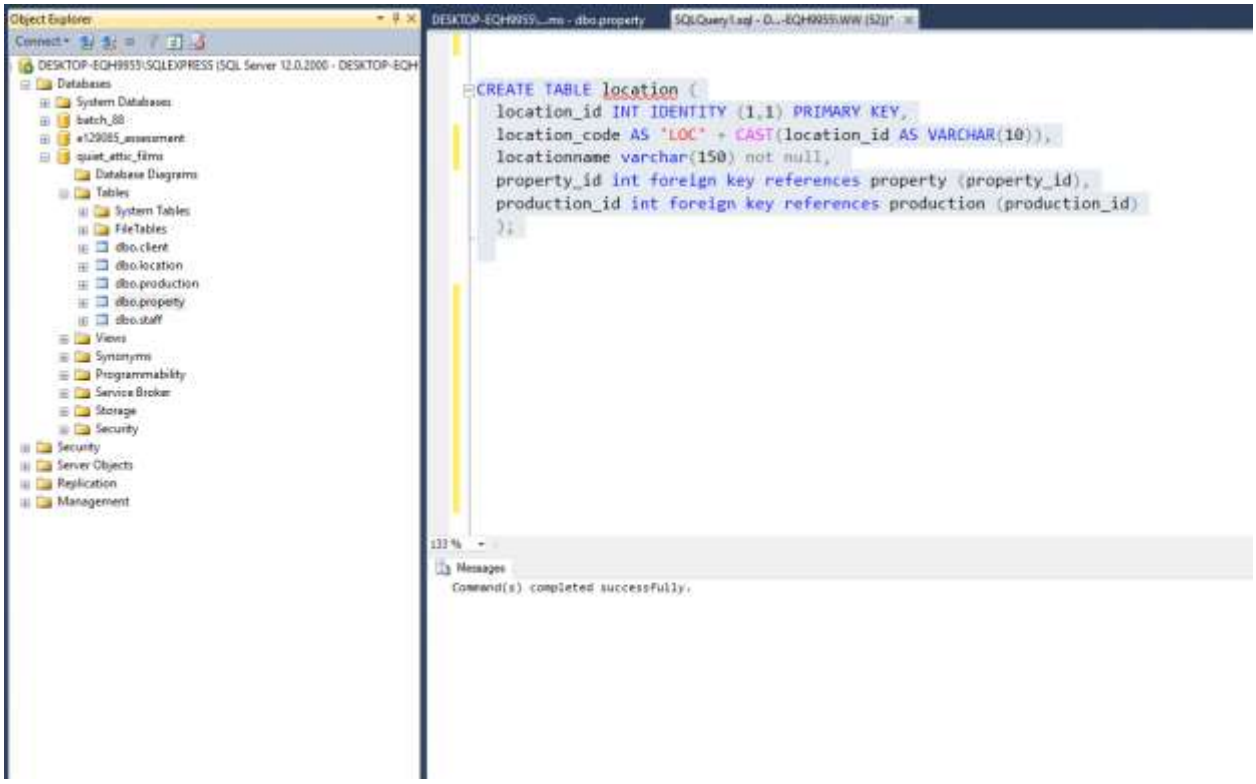

Figure 52-creating location table

The above query is used to create the location table for quiet attic films. The “CREATE” statement is used in order to create it.
The same methodology as the client table is used here to generate the primary key of the location table.
Then there are two foreign keys defined as “property_id” of the property table and “production_id” of the production table.

Adding the “location_id” foreign key to property tablke

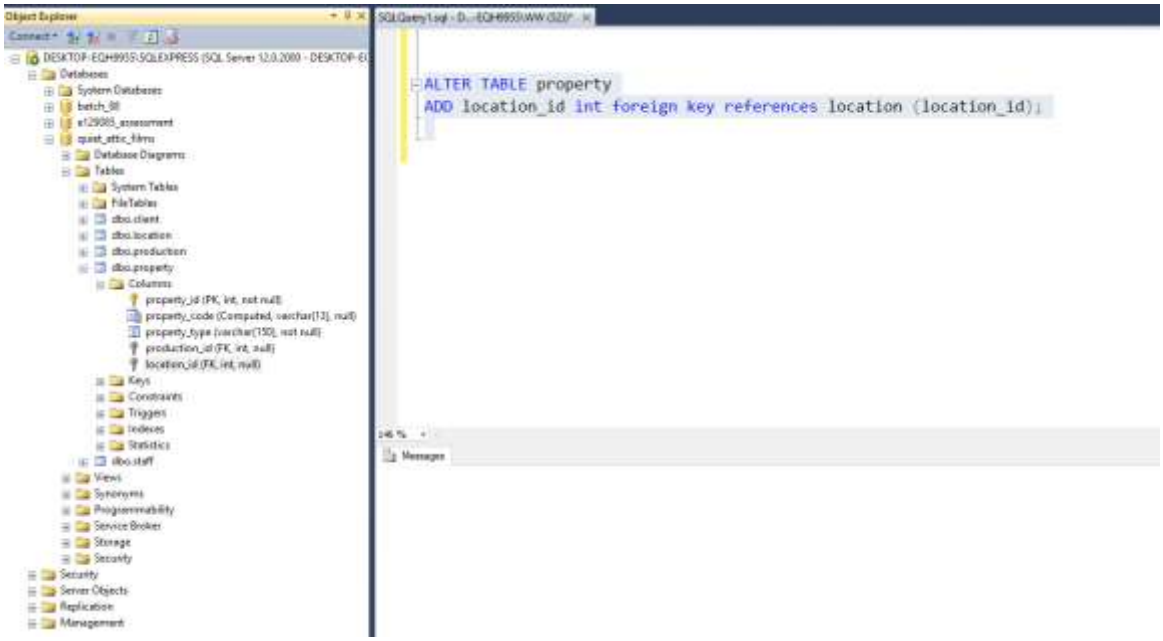

Figure 53- Adding the location foreign key to property table

This query is where “location_id” will added to the property table as the location table was not created while the property table is being created. The “ALTER.. ADD’ statement is used here.

Inserting the paper records which are currently kept by Quiet Attic Films and examples to the relational database system.

Below are samples of the paper records currently kept by the company.

Initial Production Sheet at outset of a production

Production ID	Client	Locations	Production Type	Staff	Properties	Number of Days
2	Epom Motors	1. Greenwich Park, London 2. Windsor Castle Grounds	Advertisement	2 x Camera Crew 1 x Runner 1 x Producer 3 x Actors	Featured Vehicle for both locations. Wrecked Car for location 1.	5
6	Ministry of Agriculture, Fisheries and Food	1. River Cam, Cambridge 2. Quiet Attic Studio	Information Film	2 x Camera Crew 1 x Runner 1 x Producer 1 x Voice Actor	None	1
7	Ministry of Agriculture, Fisheries and Food	1. St James's Park, London	Training Film	2 x Camera Crew 1 x Runner 1 x Producer 1 x Voice Actor	Tractor	2

Figure 54-The examples provided in the assignment brief of Quiet Attic films

Inserting Clients (Paper records)

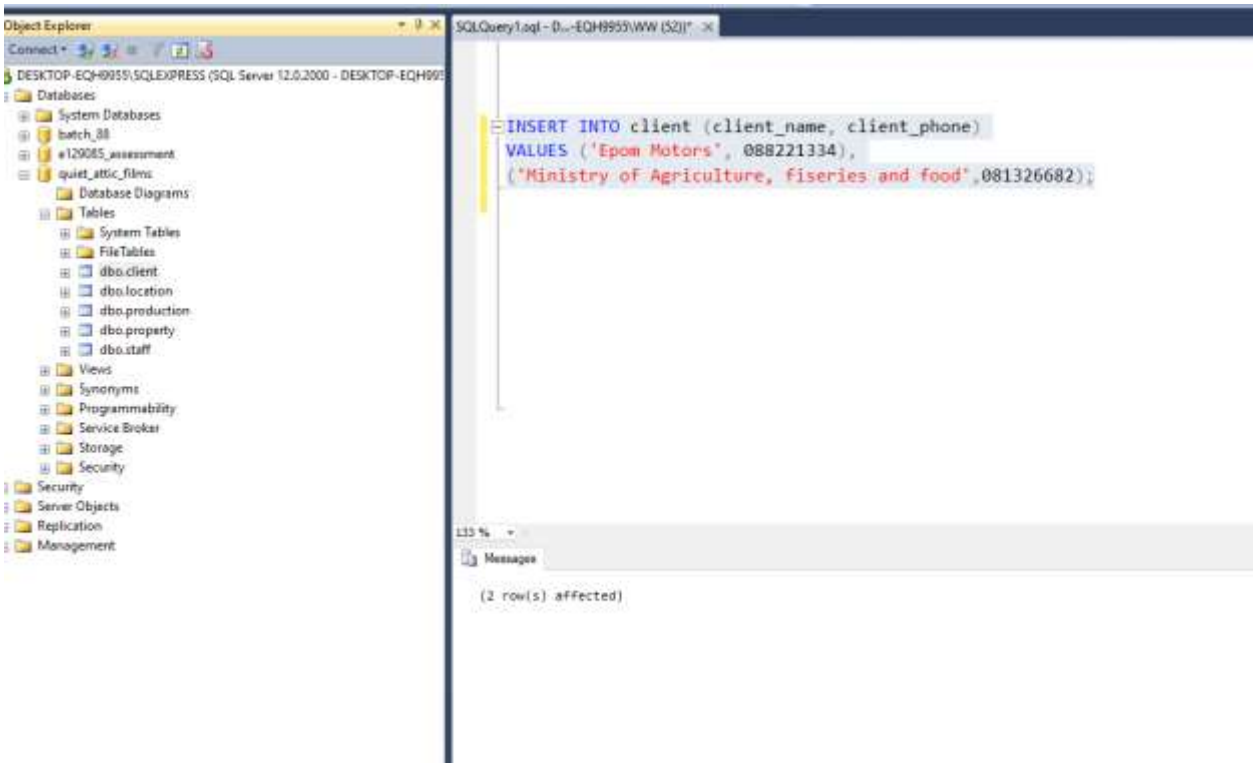

Figure 55 - Inserting clients to the relational database (paper records)

Inserting Clients (examples)

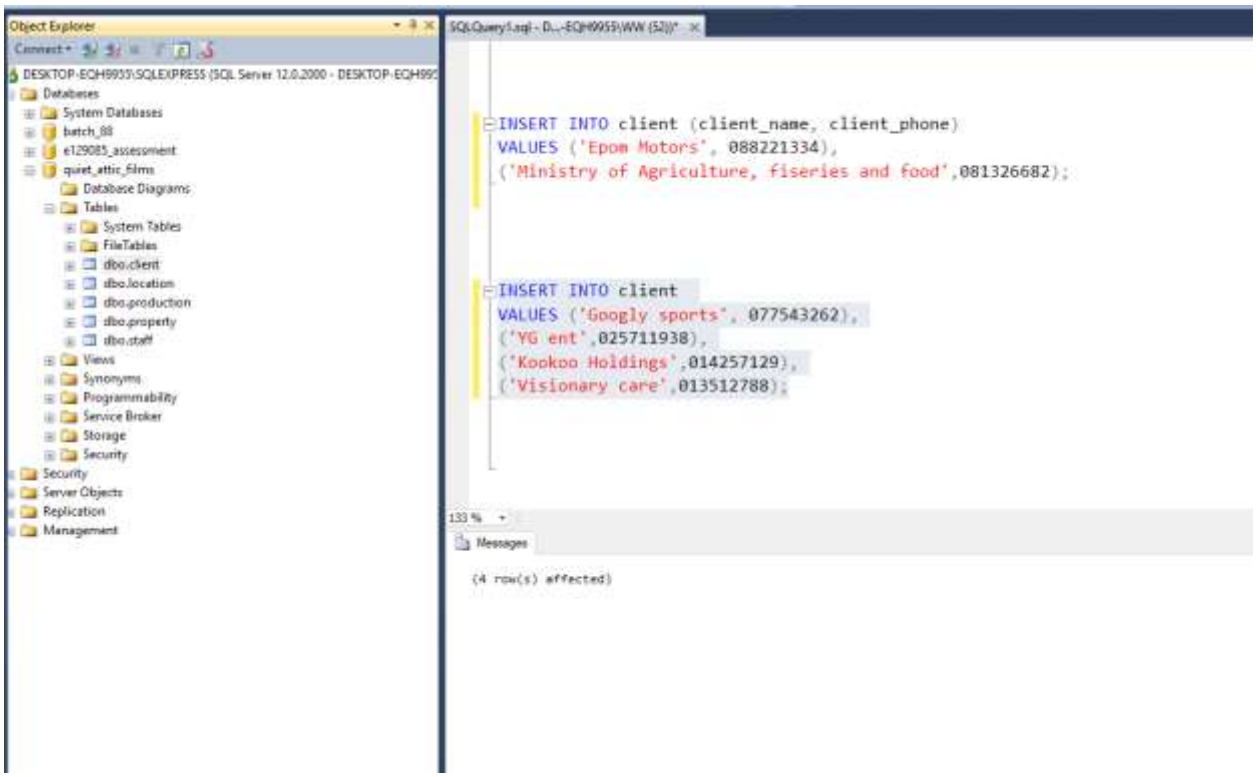

Figure 56 - Inserting clients to the relational database(example)

Inserting Productions (paper records + examples)

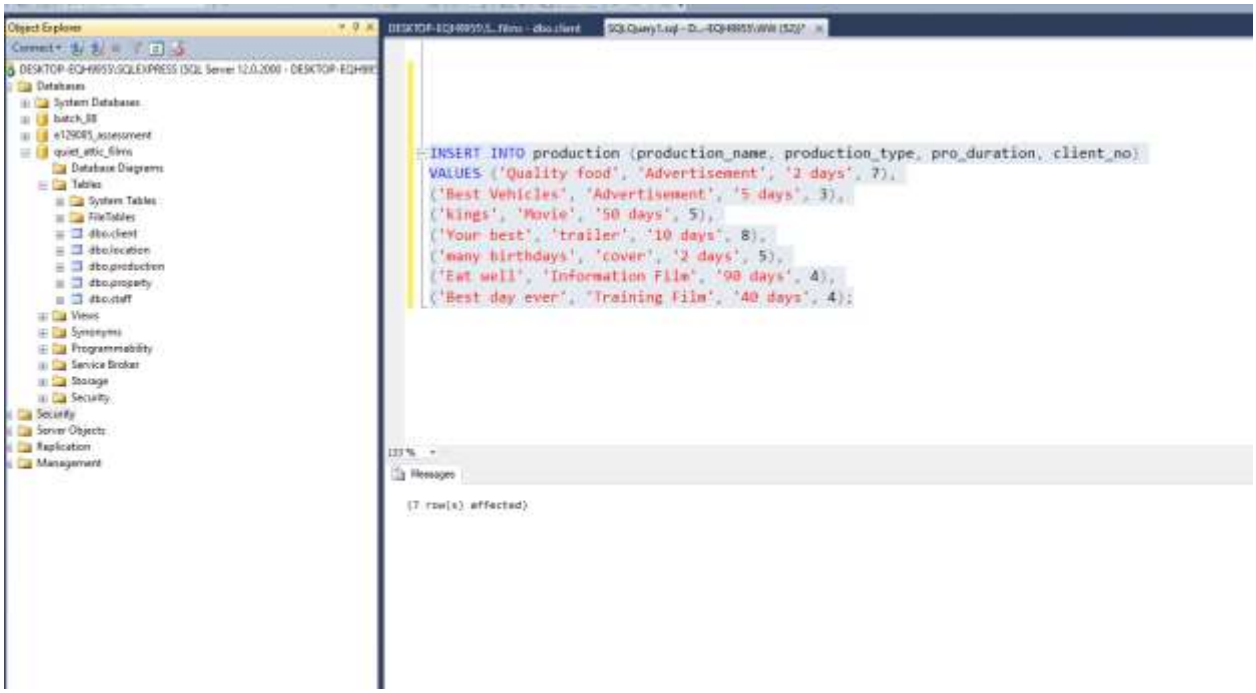

Figure 57 -Inserting Productions (paper records + examples)

Inserting Staff (paper records + examples) (UNNORMALIZED)

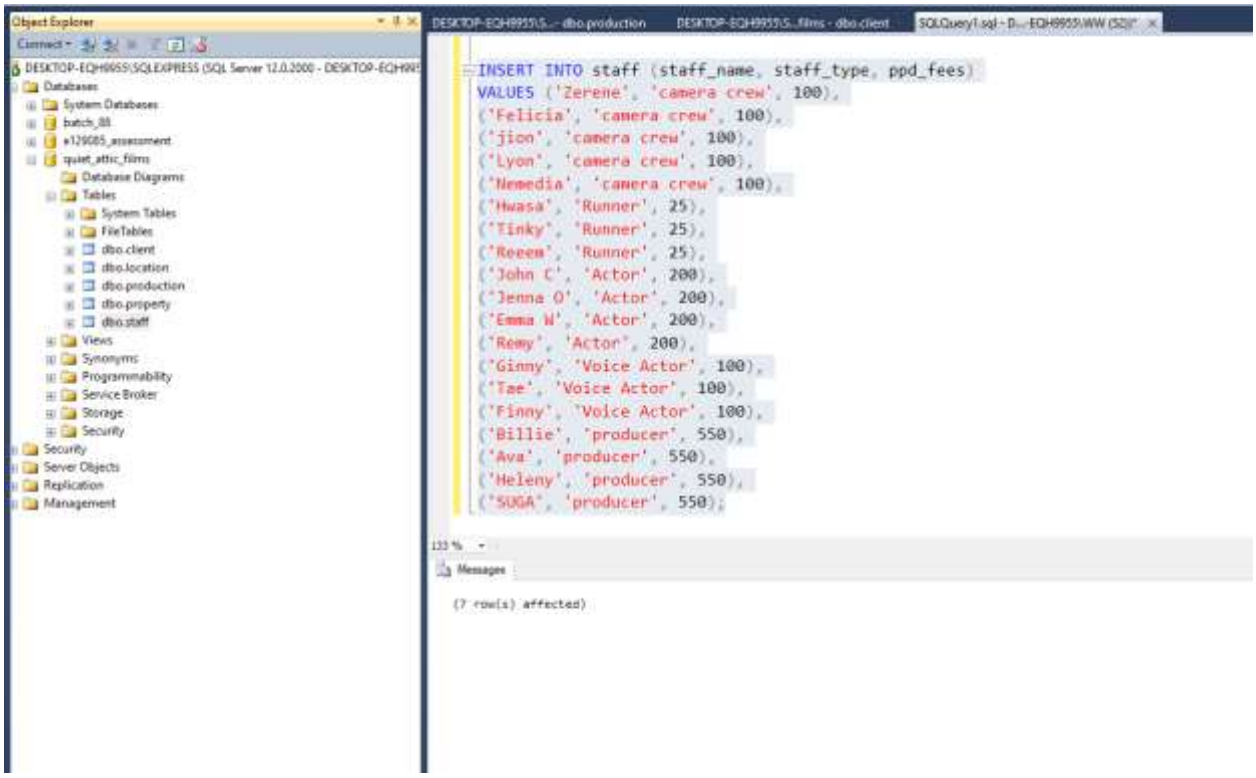

Figure 58 - Inserting Staff (paper records + examples)

Inserting properties (paper records)

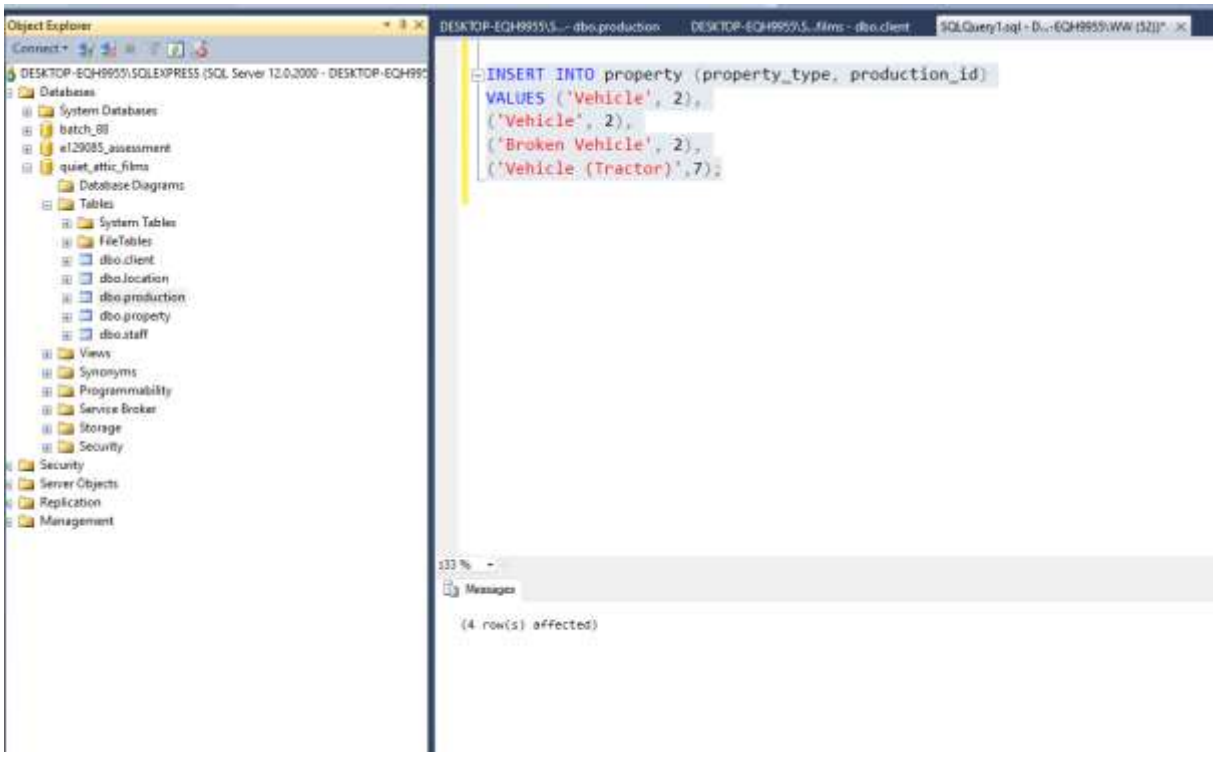

Figure 59 - inserting properties

Inserting locations (paper records)

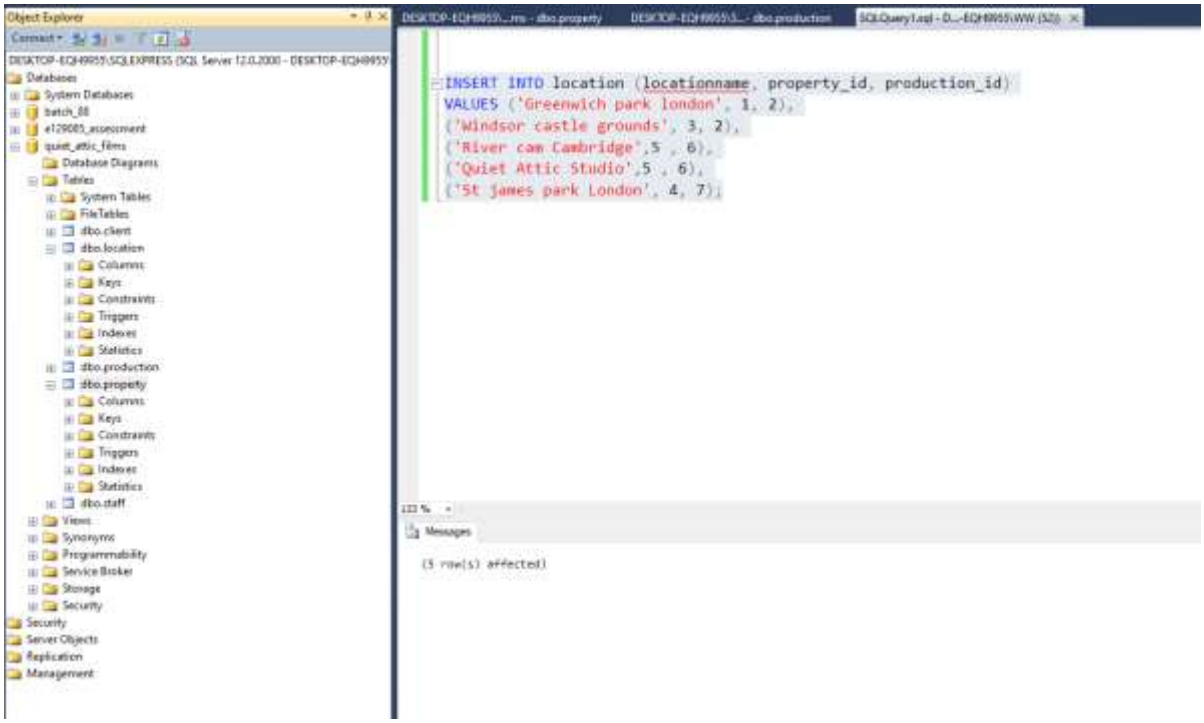

Figure 60 - inserting Locations

Examples on using the relational database considering the desired user requirements

Once again, the desired user requirements are,

- Organize data on the concept of production.
- Determine what productions belong to client and which client hired a production.
- Determine which location/locations the production will be done and which production will be done at a particular production.
- Determine which properties are used at productions.
- Insert, Update and Delete data in the above entities.

Using the Select Statement to view each entity data as an overview

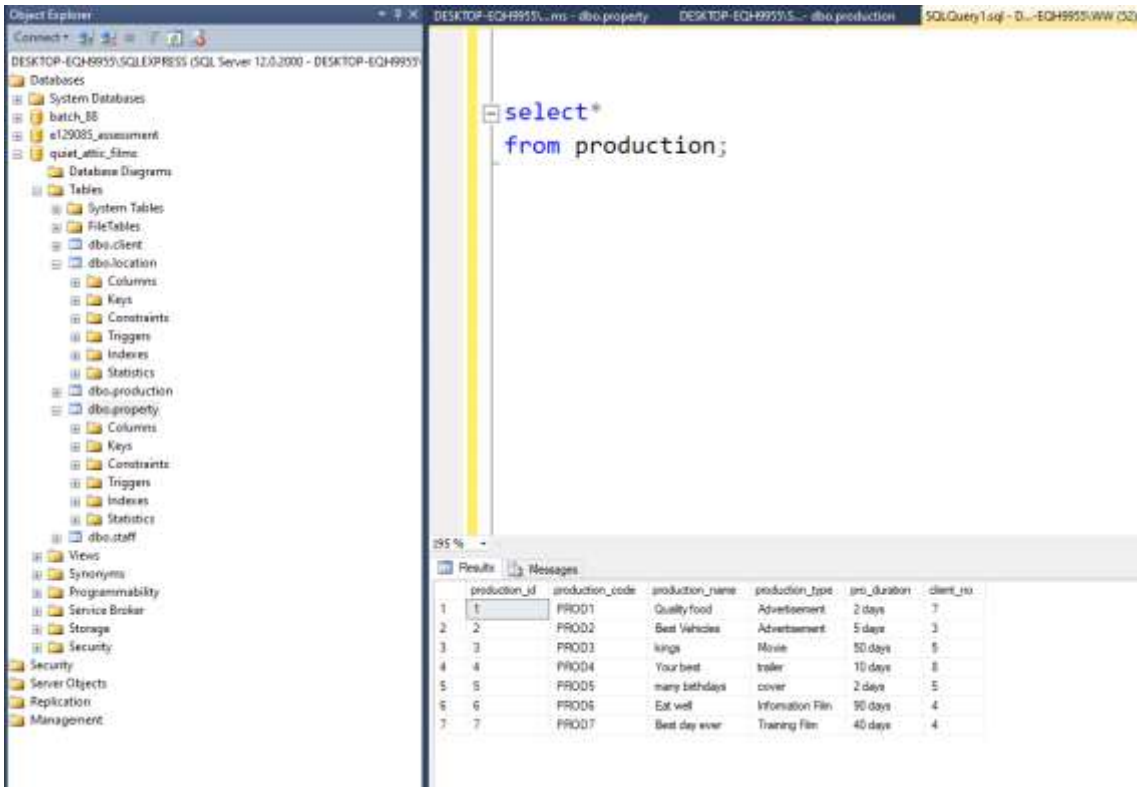

Figure 61 - select statement

As seen above the data are cleanly organized.

Locating or finding data using the "Where" statement. Organized data is depicted.

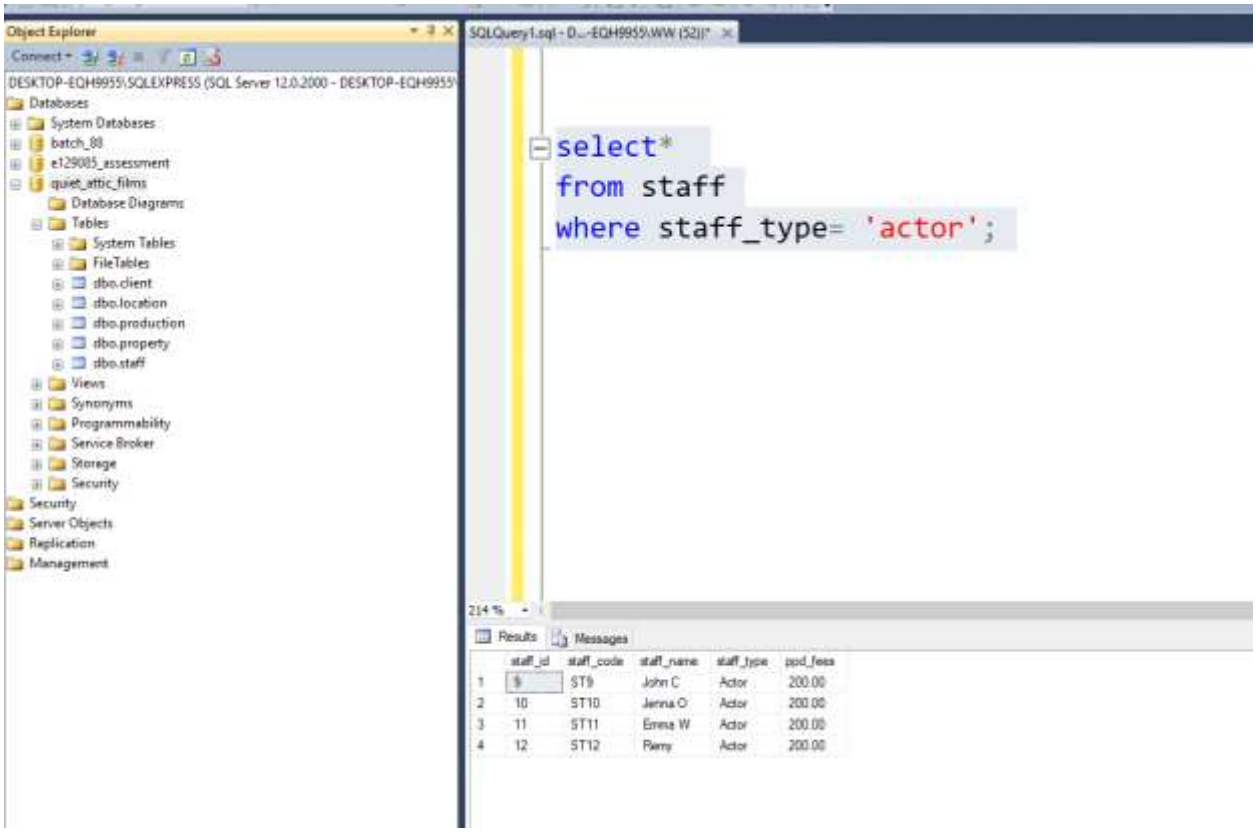

Figure 62 - Where statement

Determine what productions belong to client and which client hired a production. (Using the JOIN ON functions)

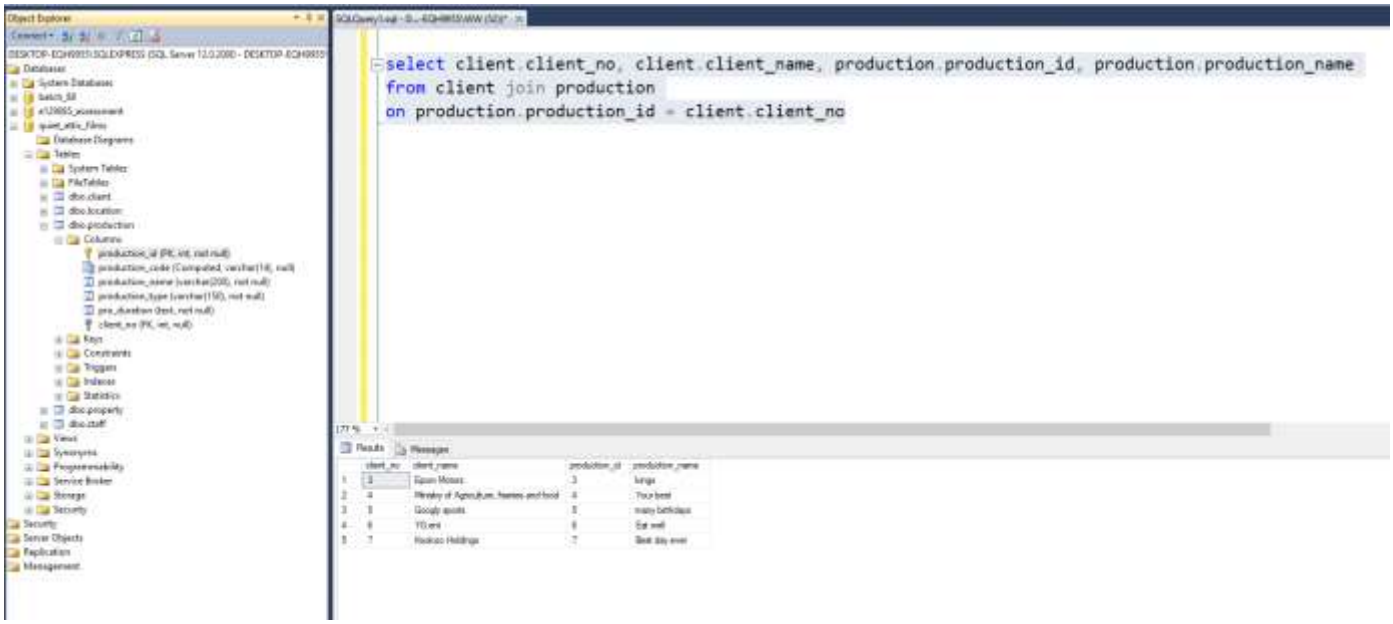

Figure 63- Determine what productions belong to client and which client hired a production.

Determine which location/locations the production will be done and which production will be done at a particular production.

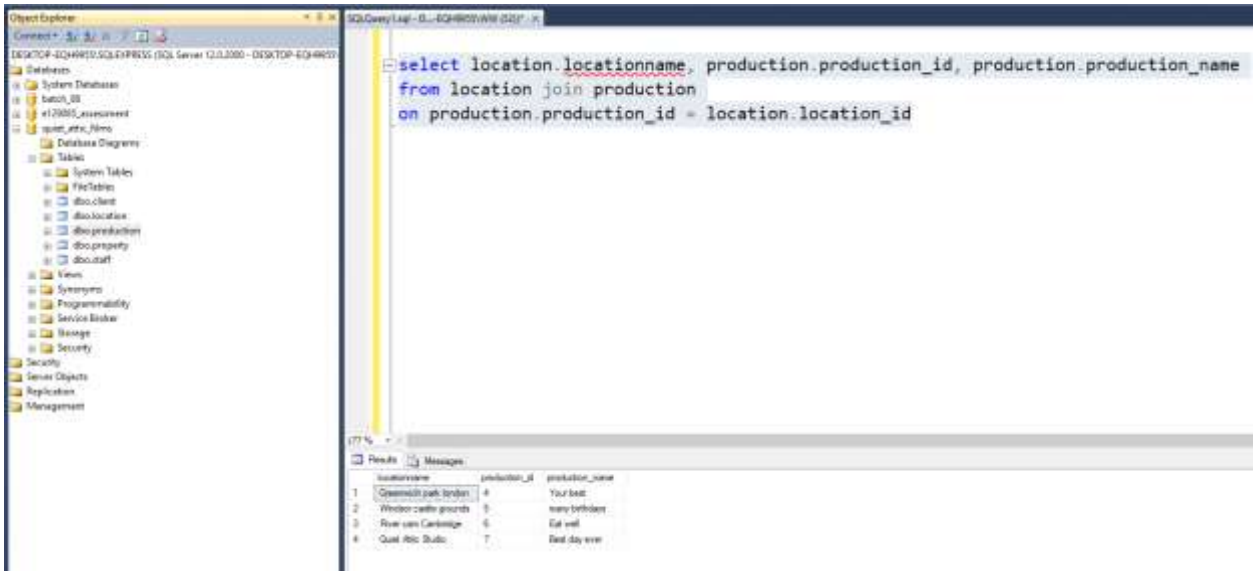

Figure 64 - Determine which location/locations the production will be done and which production will be done at a particular production.

Determine which properties are used at productions

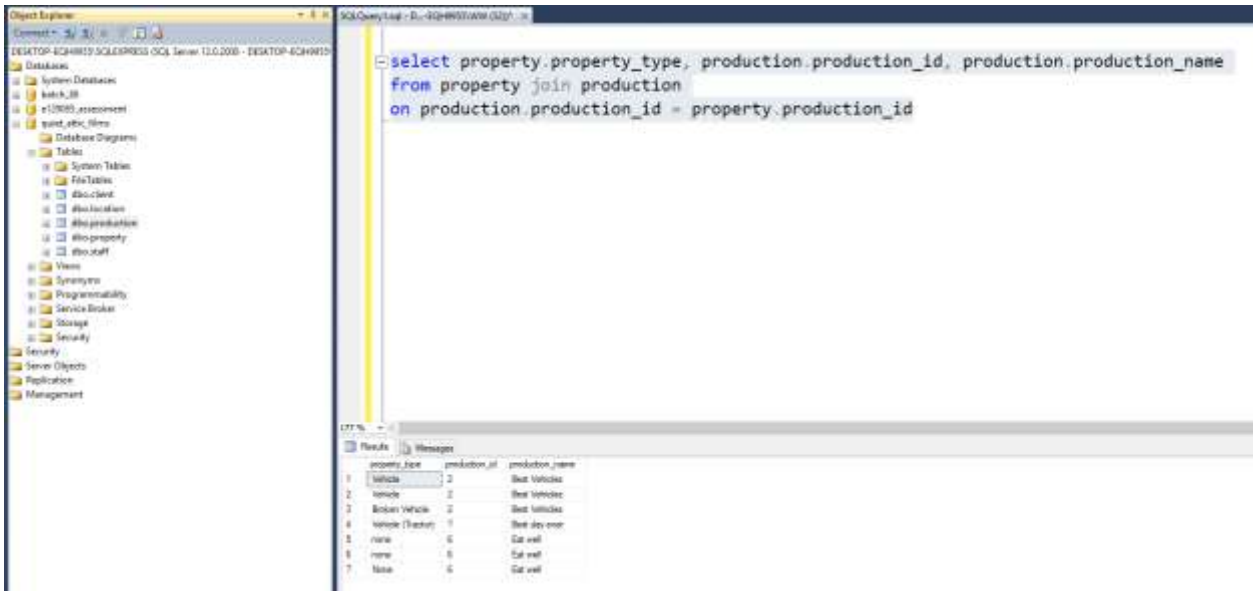

Figure 65 - Determine which properties are used at productions

Using the UPDATE statement to update data

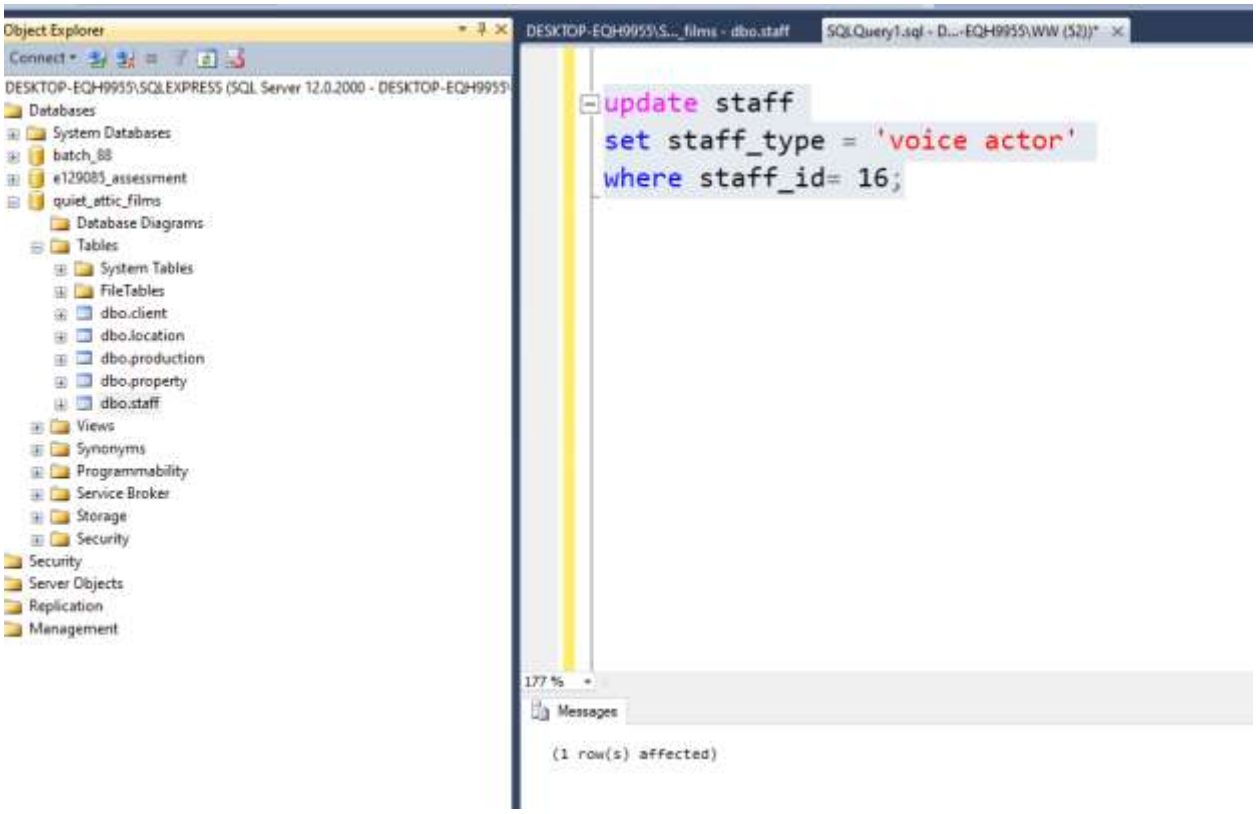

Figure 66 - using the Update Statement

Using the BETWEEN statement to get ppd fees accordingly to a desired limit

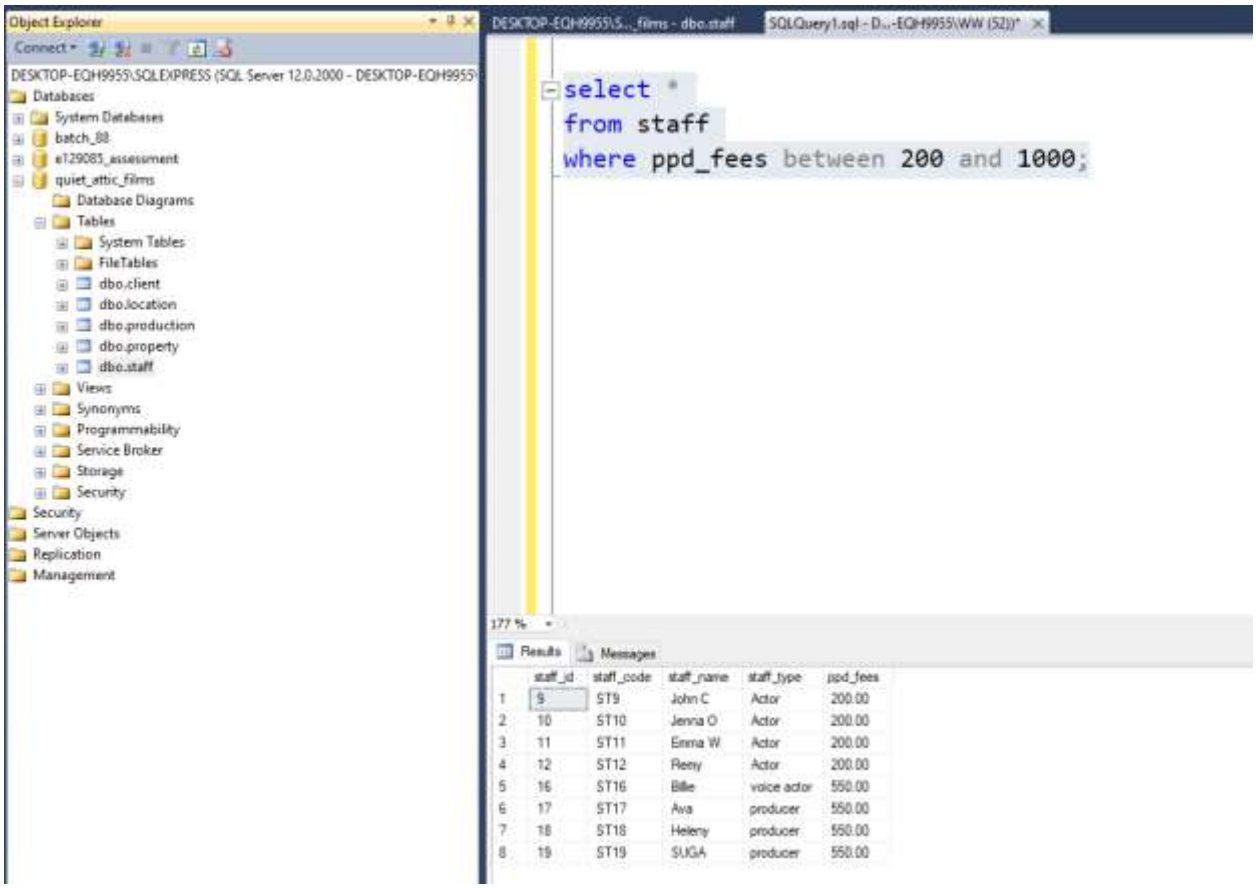

Figure 67 - Using the between Statement

Using the IN statement to retrieve the desired category of attributes.

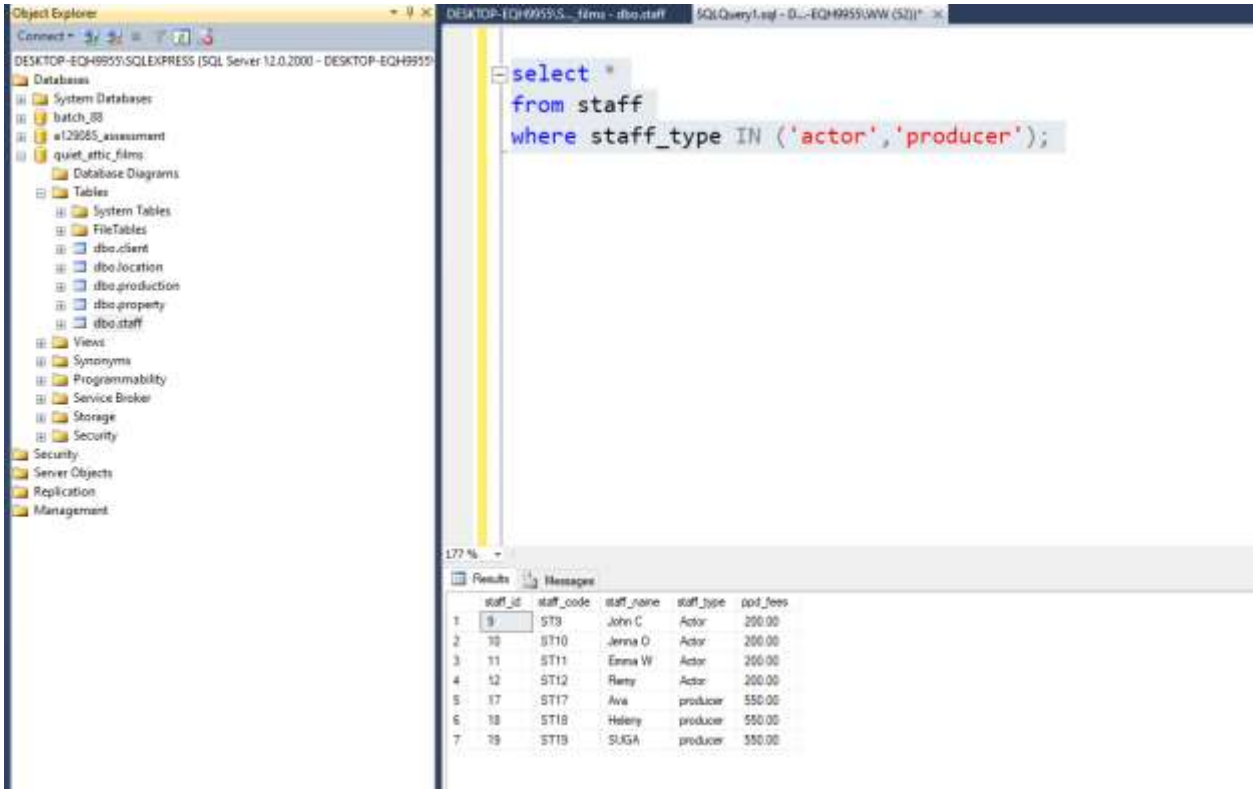

Figure 68 - Using the IN statement

Using the Group by statement to group a certain result set.

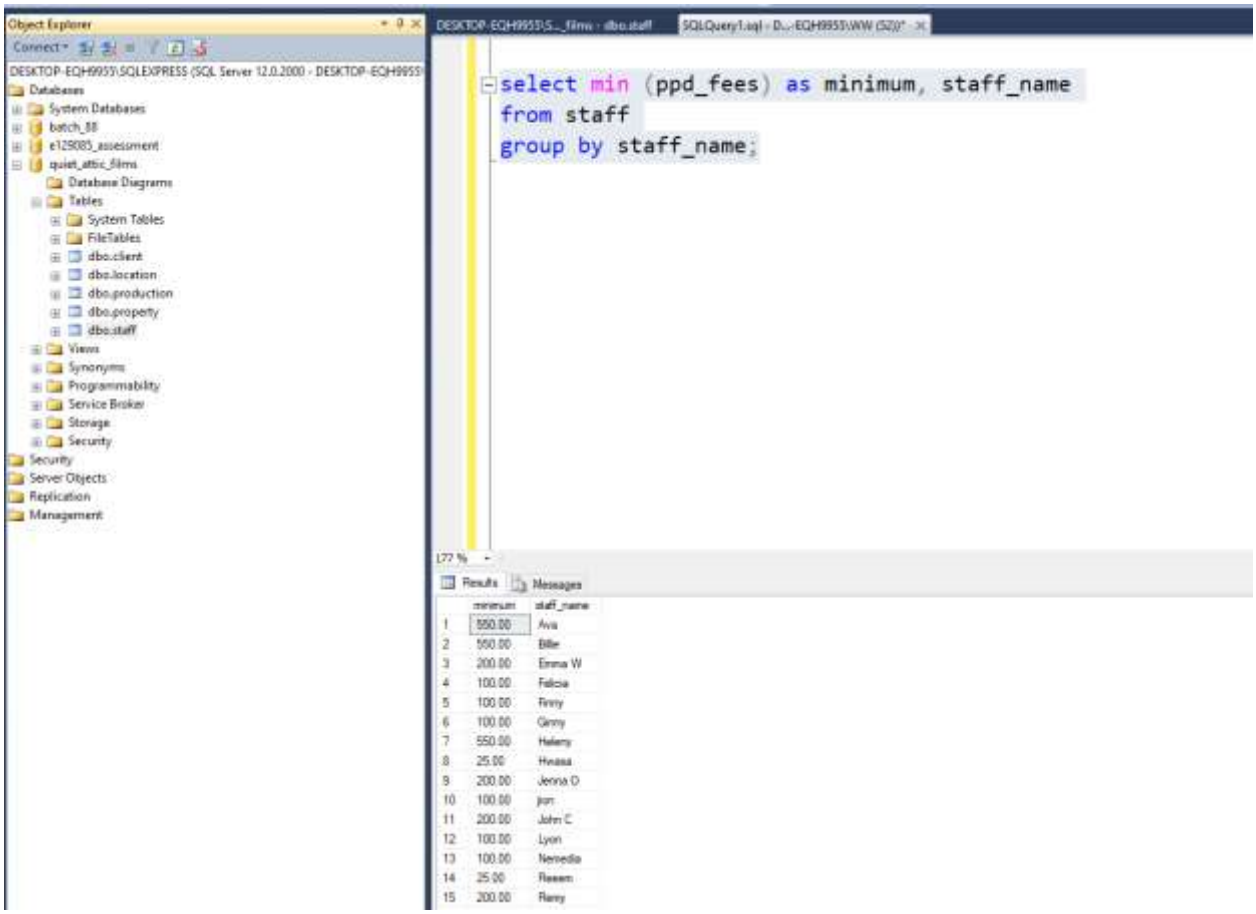

Figure 69 - Using the group by statement

Using the Having statement

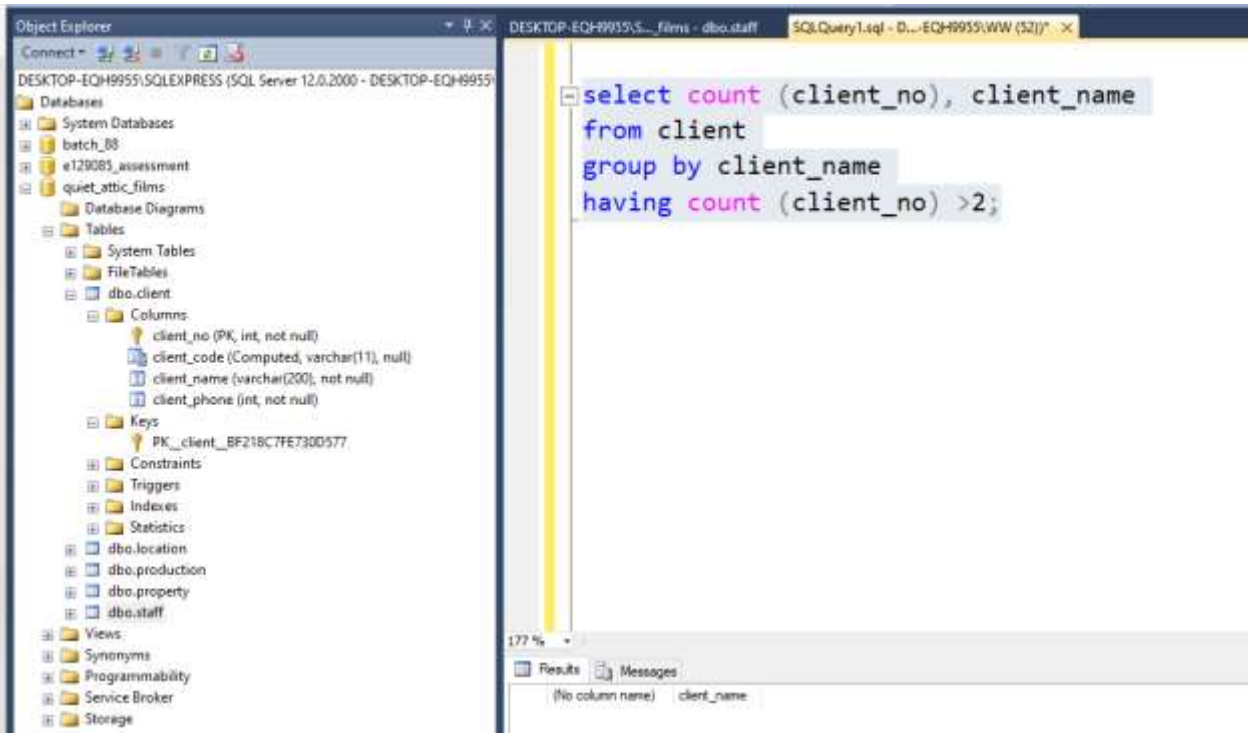

Figure 70 - Having statement

Designing a fully functional System for the relational database of Quiet Attic Films.

In activity 1, a wireframe model has been created to create an interface for the database system of Quiet attic films. Below is the designed interface for it using Microsoft Visual Studio 2013.

1. **Startup Page**

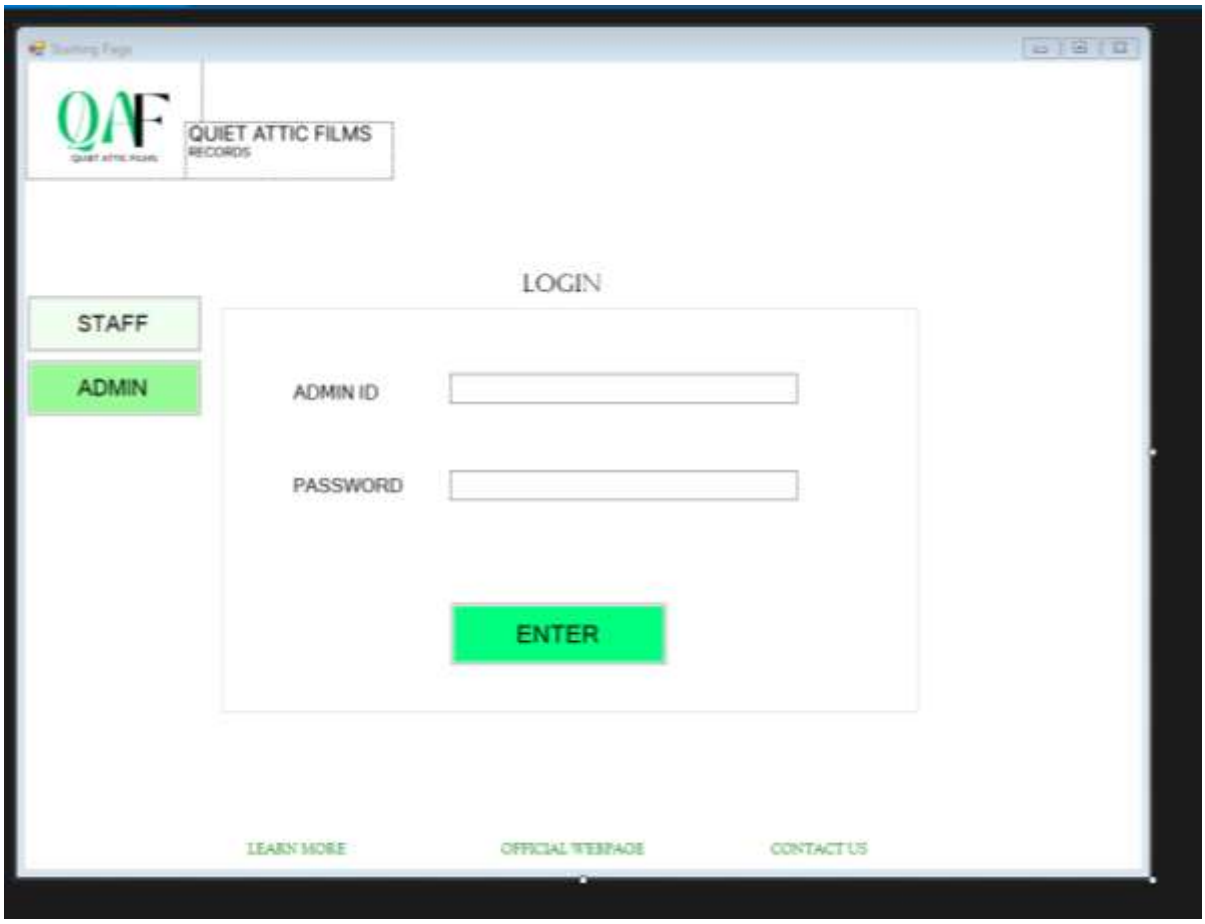

Figure 71 - Interface startup page

This is the startup page for quiet attic films interface. The user will be able to login here.

2. **Selection page**

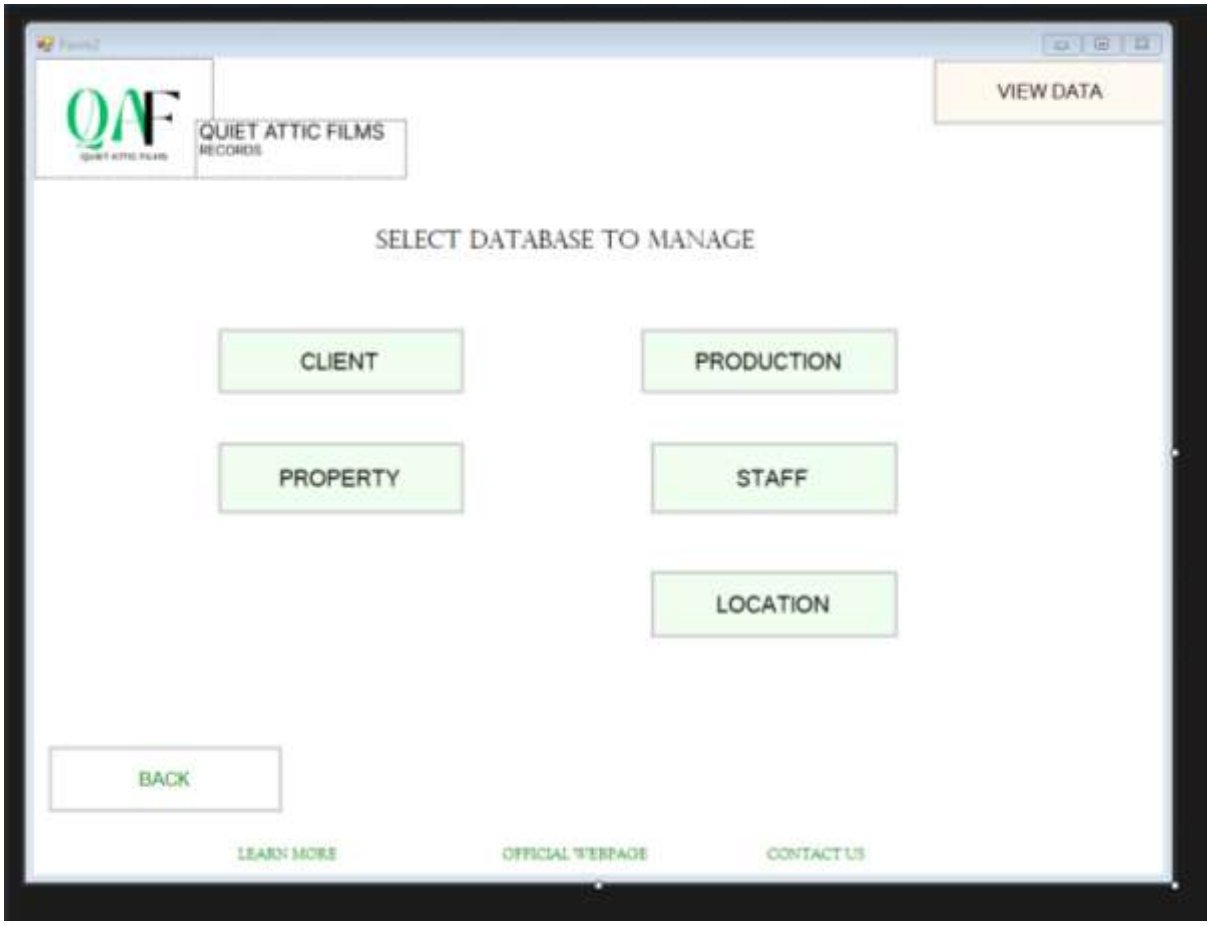

Figure 72 - interface Selection page

This page is where the user can select which entity to manage.

3. **View data page**

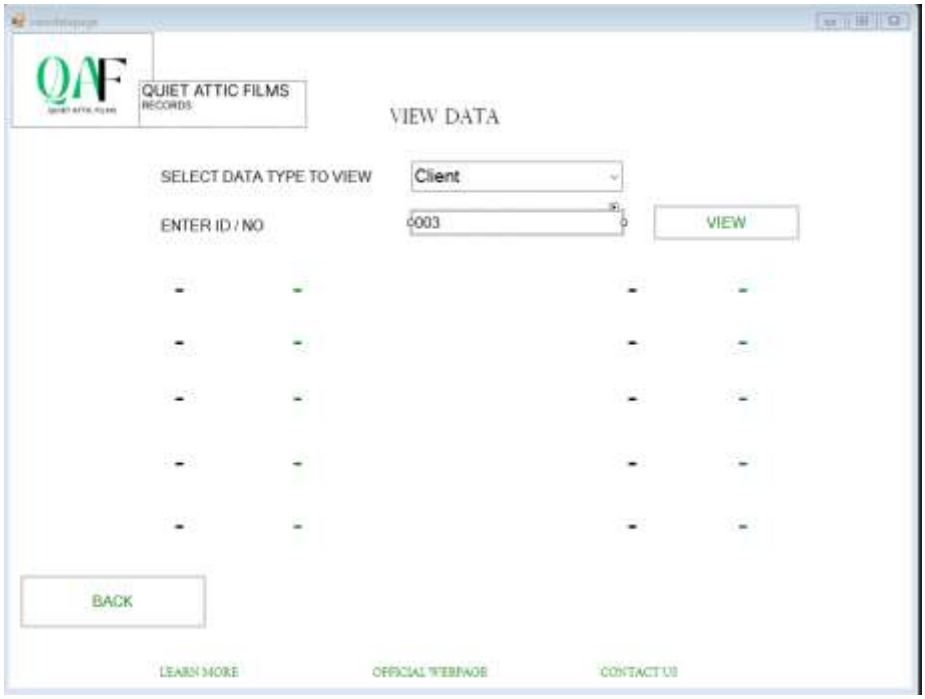

Figure 73 - View page interface

This page will be displayed if “View page” button is clicked from the selection page. The user can select the desired entity and view desired data here.

4. Manage Clients page

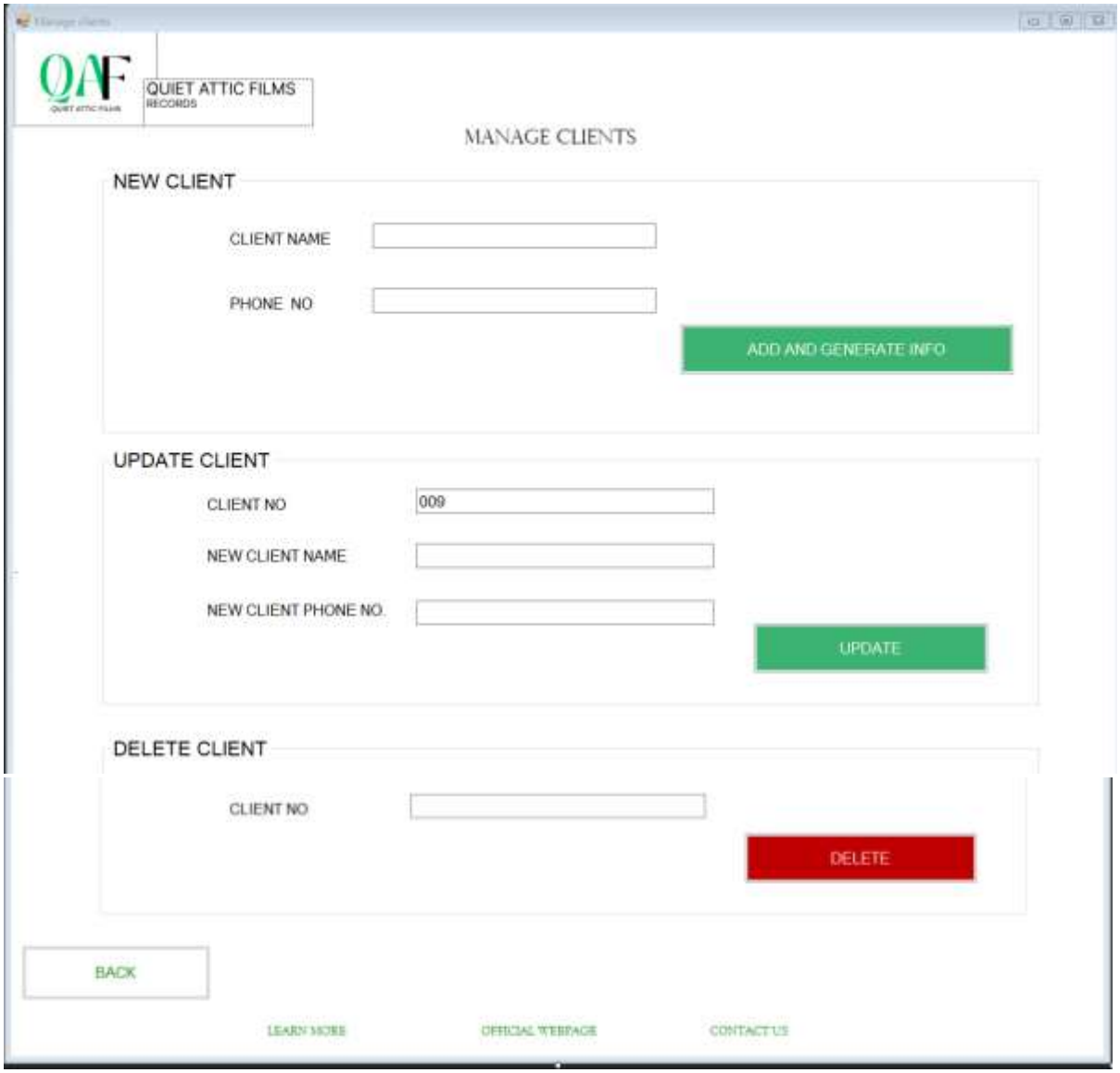

Figure 74 - Manage Clients page

Here the user can manage client data in the database.

4.1 Client Successfully Added Page

Figure 75 - client successfully added page

4.2 Client successfully updated page

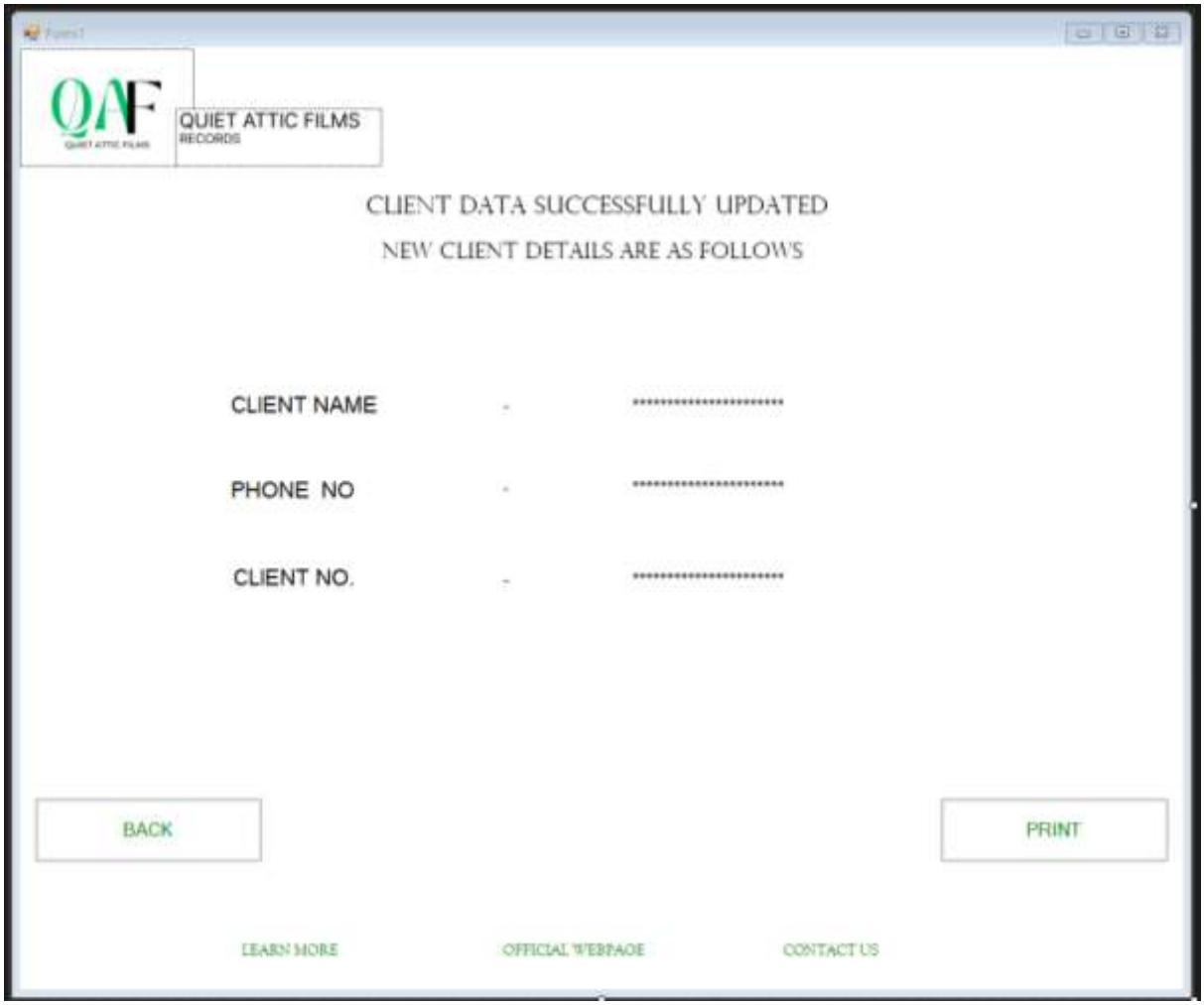

Figure 76 - client successfully updated page

4.3 Delete client Confirmation page

Figure 77 - delete client confirmation page

4.4 client successfully deleted page.

Figure 78 - client successfully deleted page

\

5. **Manage production page**

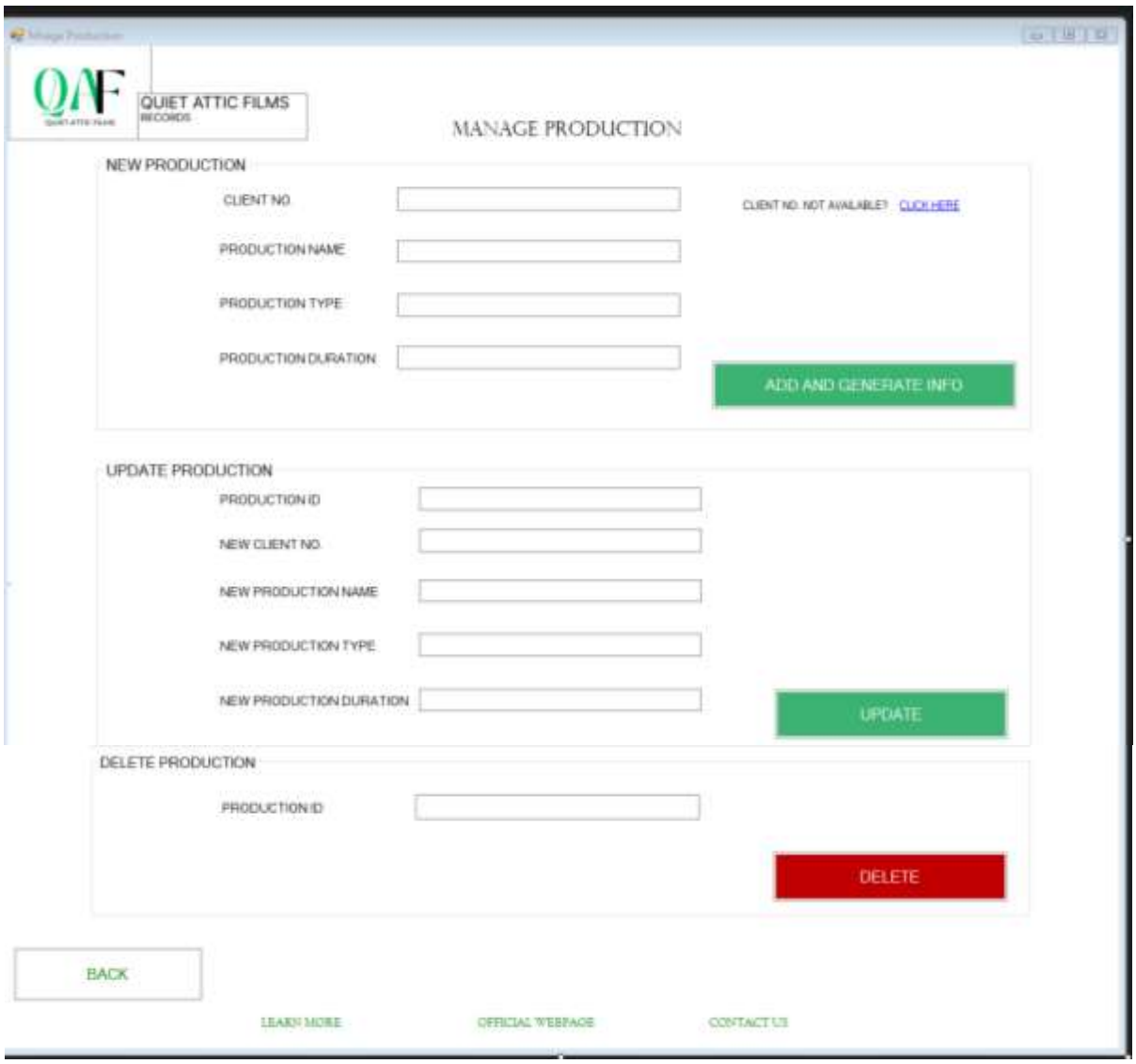

Figure 79 - Manage productions page

Here the user can manage production data in the database.

5.1 Production Successfully Added Page

Figure 80 - Production Successfully Added Page

5.2 Production Successfully Updated Page

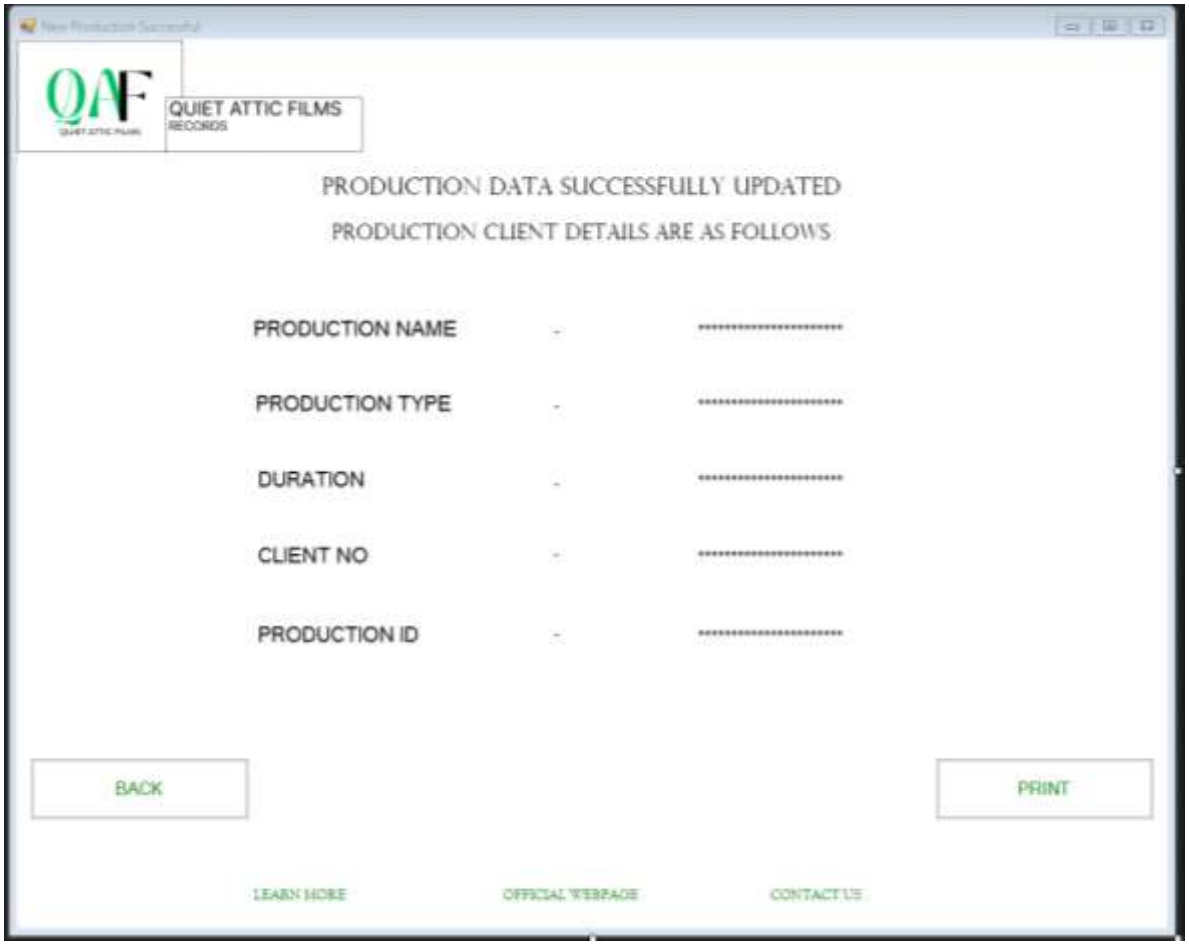

Figure 81- Production Successfully updated Page

5.3 Delete Production Confirmation page

Figure 82 - Delete Production Confirmation page

5.4 production successfully deleted page

Figure 83 - production successfully deleted page

6. **Manage Staff Page**

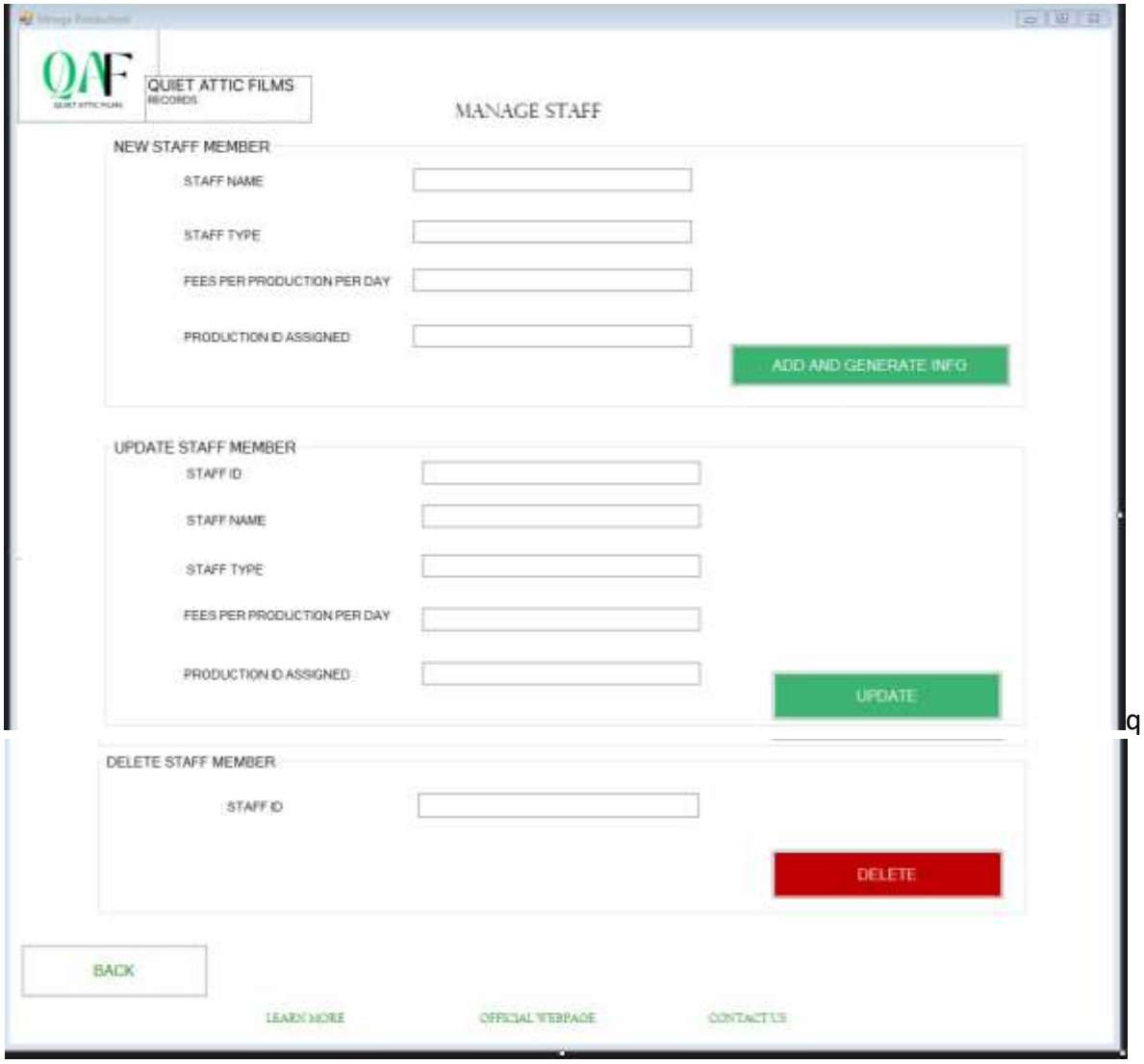

Figure 84- Manage Staff Page

Here the user can manage staff data in the database.

6.1 Staff Successfully Added Page

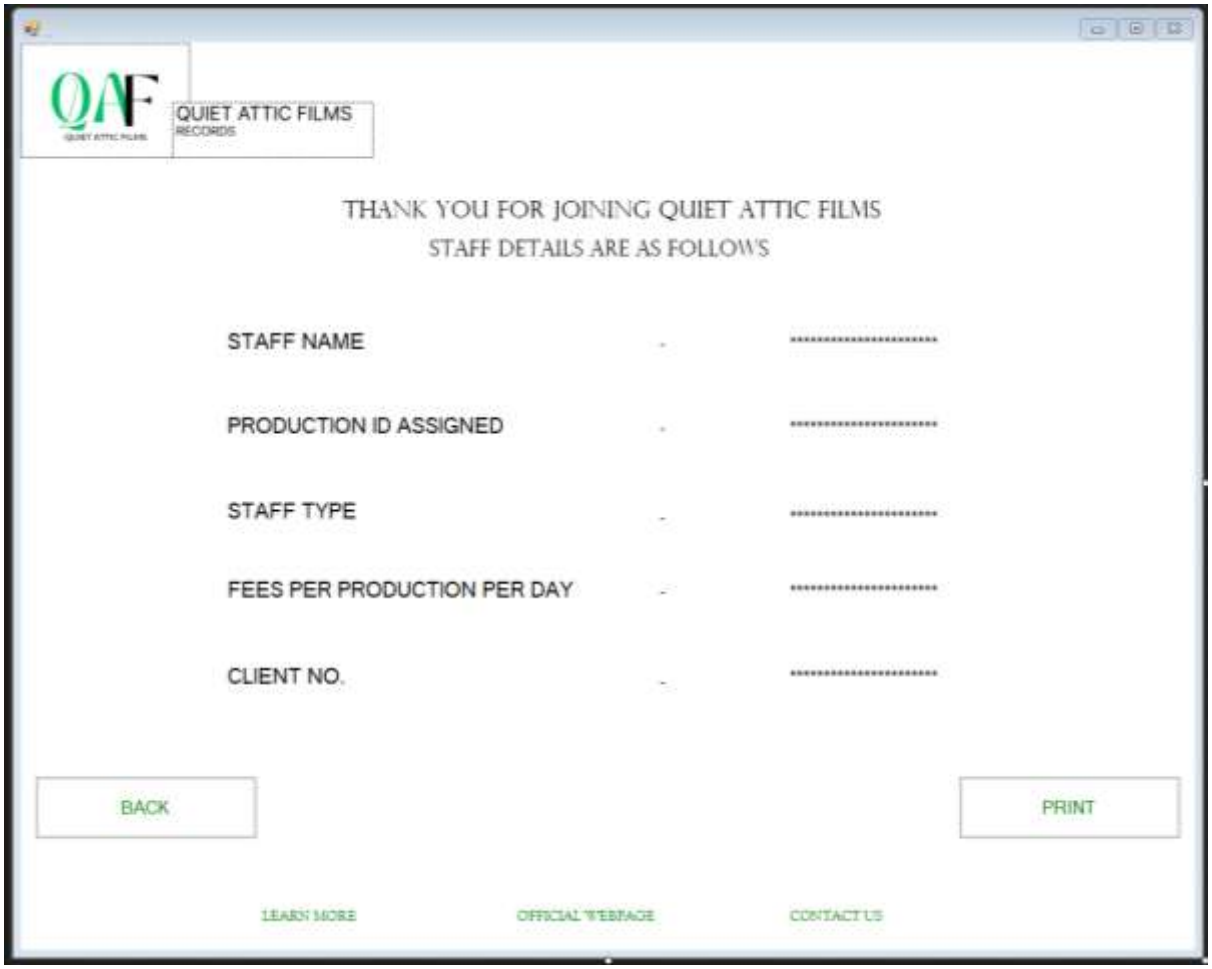

Figure 85 - Staff Successfully Added Page

6.2 Staff Successfully Updated Page

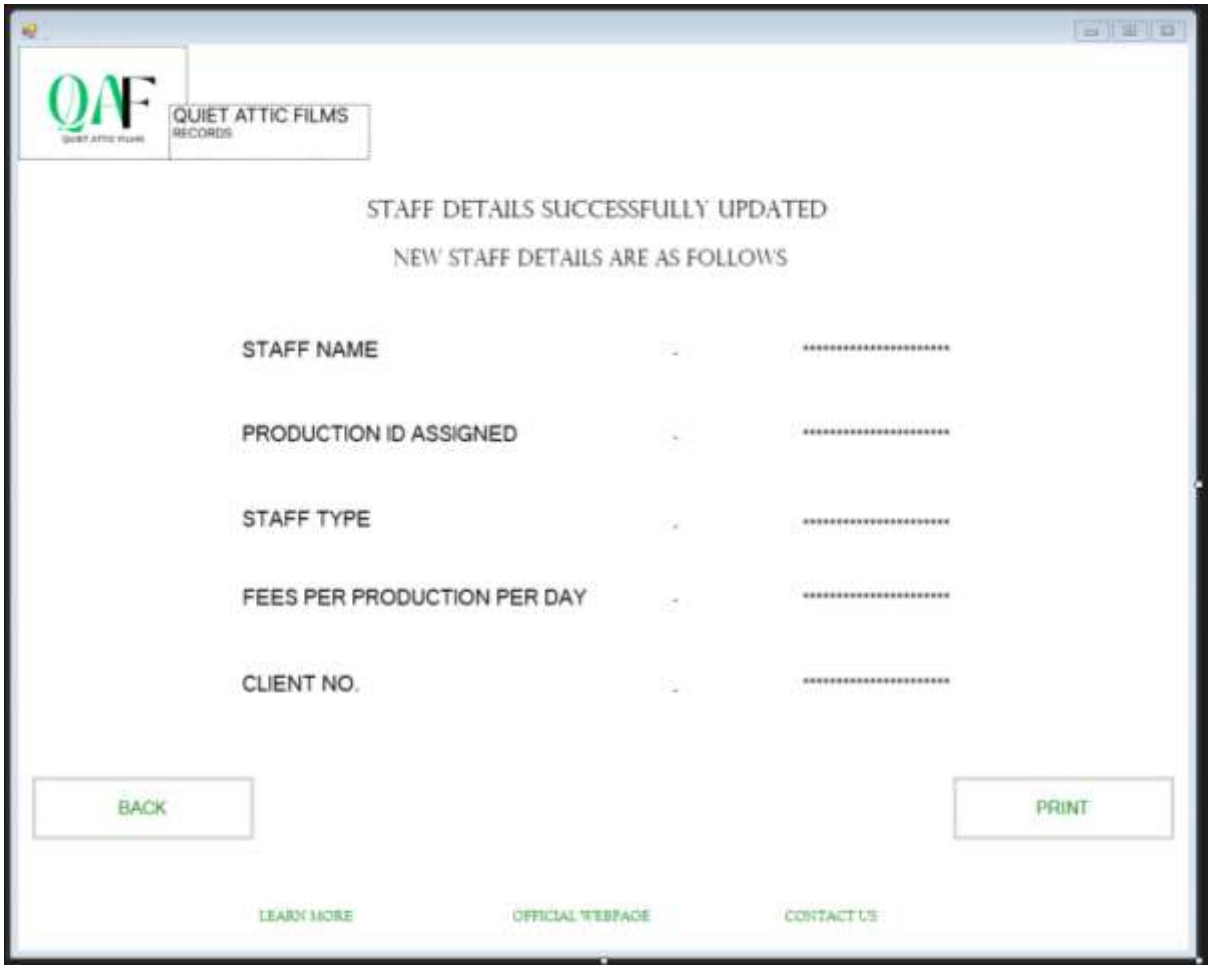

Figure 86 - Staff Successfully Updated Page

6.3 Delete Staff Confirmation page

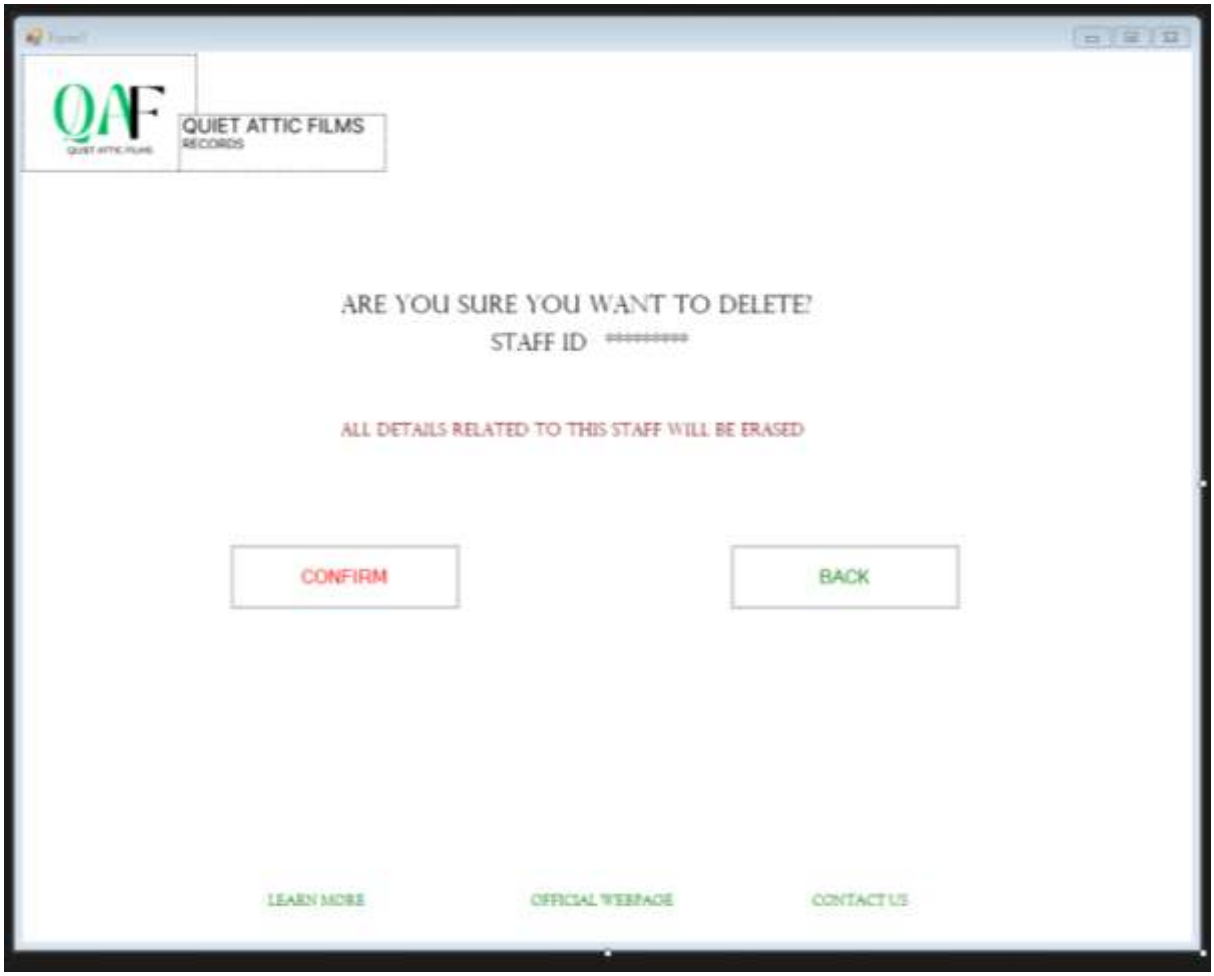

Figure 87 - Delete Staff Confirmation page

6.4 Staff successfully deleted page

Figure 88 - Staff successfully deleted page

7. Manage Location Page

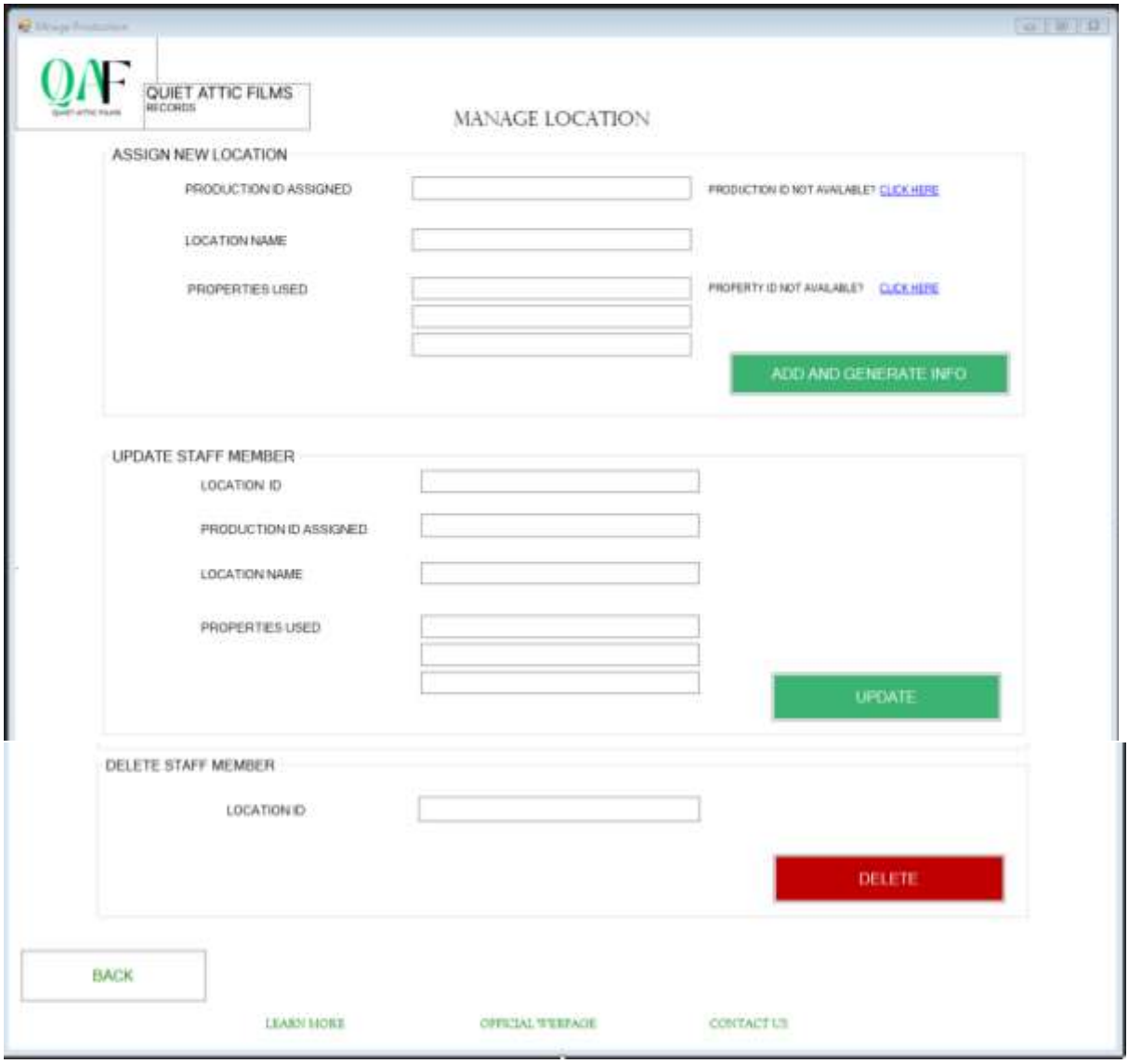

Figure 89 - Manage Location Page

Here the user can manage location data in the database.

7.1 Location Successfully Added Page

Figure 90 - Location Successfully Added Page

7.2 Location Successfully Updated Page

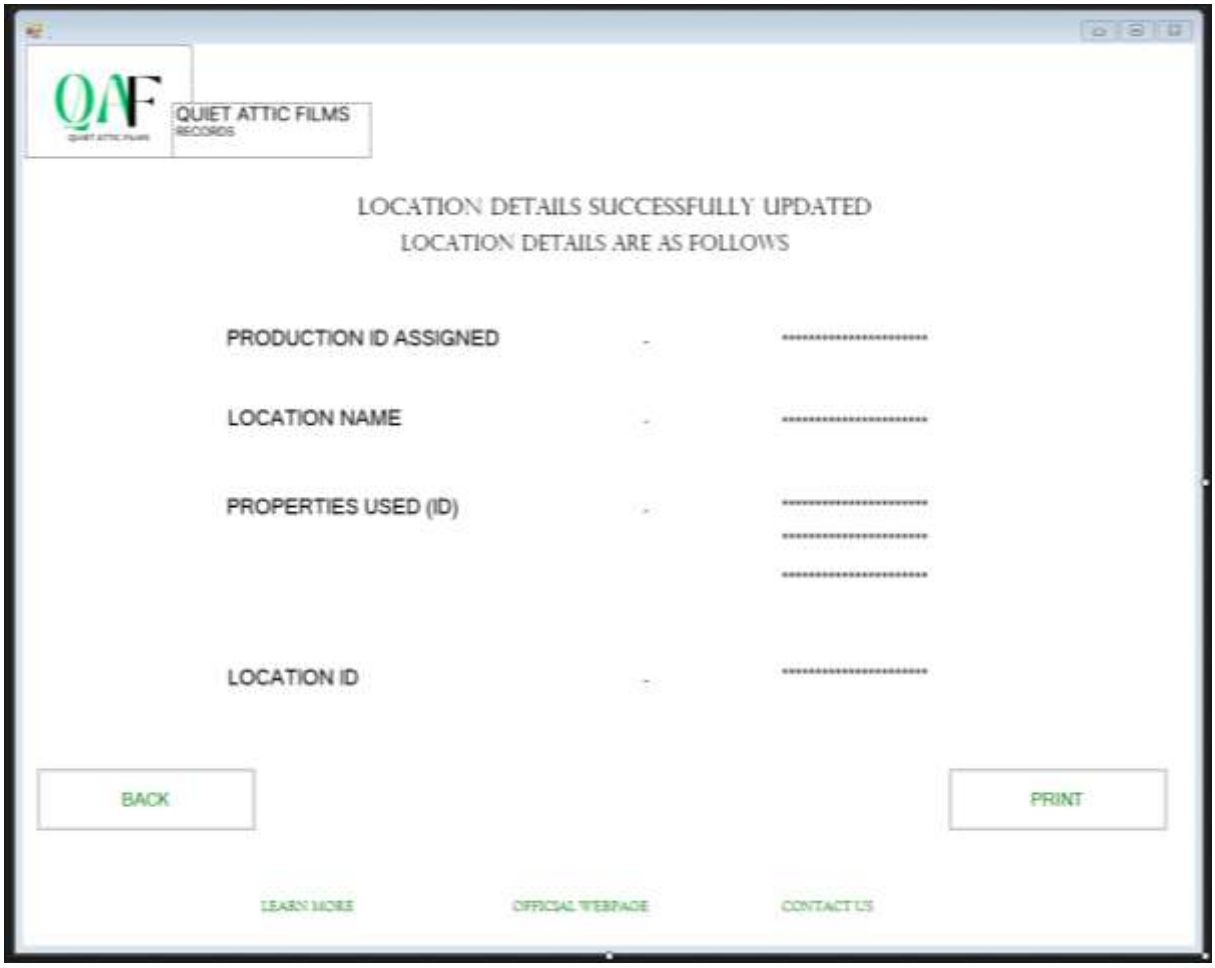

Figure 91- Location Successfully Updated Page

7.3 Delete Location Confirmation page

Figure 92 - Delete Location Confirmation page

7.4 Location successfully deleted page

Figure 93 - Location successfully deleted page

8. Manage Property Page

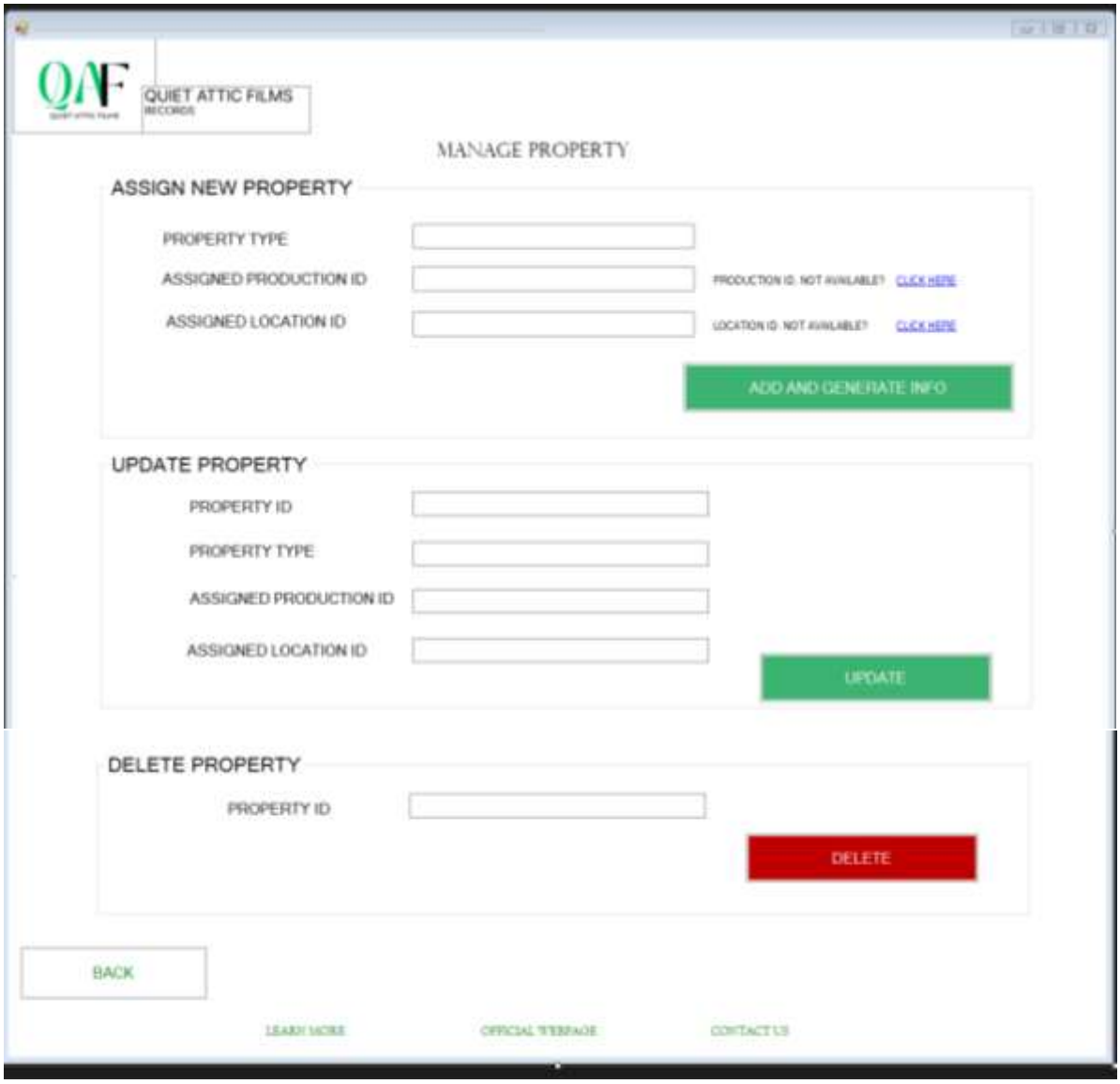

Figure 94 - Manage Property Page

Here the user can manage property data in the database.

8.1 Property Successfully Added Page

Figure 95 - Property Successfully Added Page

8.2 Property Successfully Updated Page

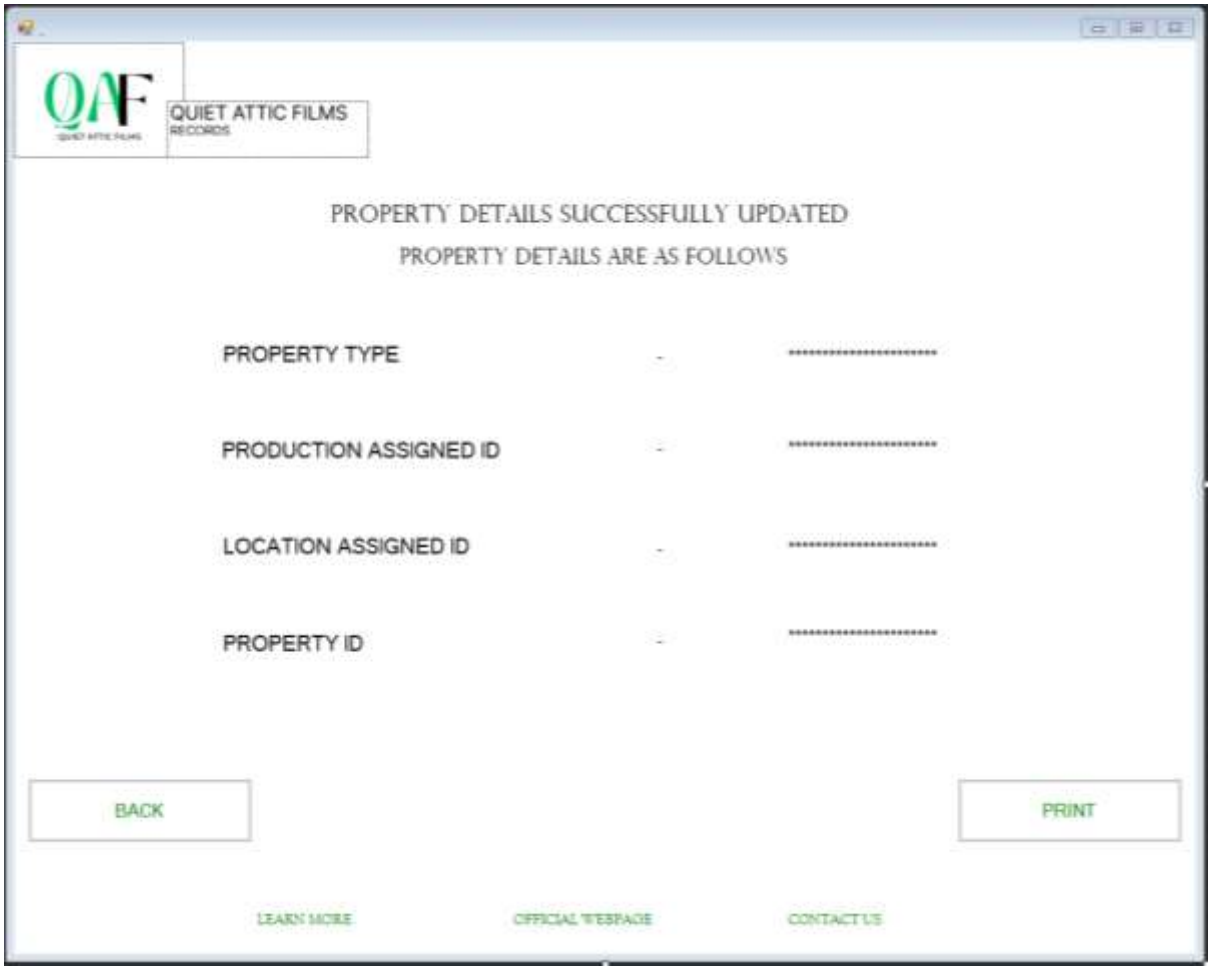

Figure 96 - Property Successfully Updated Page

8.3 Delete Property Confirmation page

Figure 97 - Delete Property Confirmation page

8.4 Property successfully deleted page

Figure 98 - Property successfully deleted page